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Long term hormone therapy for perimenopausal and postmenopausal women (Review)

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[Intervention Review]

Long term hormone therapy for perimenopausal and postmenopausal women

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ABSTRACT

Background

Hormone therapy (HT) is widely used for controlling menopausal symptoms and has also been used for the management and prevention of cardiovascular disease, osteoporosis and dementia in older women. This is an updated version of the original Cochrane review first published in 2005.

Objectives

To assess the effect of long-term HT on mortality, cardiovascular outcomes, cancer, gallbladder disease, cognition, fractures and quality of life.

Search methods

We searched the following databases to November 2007: Trials Register of the Cochrane Menstrual Disorders and Subfertility Group, Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, Biological Abstracts. Also relevant non-indexed journals and conference abstracts.

Selection criteria

Randomised double-blind trials of HT versus placebo, taken for at least one year by perimenopausal or postmenopausal women. HT included oestrogens, with or without progestogens, via oral, transdermal, subcutaneous or transnasal routes.

Data collection and analysis

Two authors independently assessed trial quality and extracted data.

Main results

Nineteen trials involving 41,904 women were included. In relatively healthy women, combined continuous HT significantly increased the risk of venous thrombo-embolism or coronary event (after one year's use), stroke (after three years), breast cancer and gallbladder disease. Long-term oestrogen-only HT significantly increased the risk of venous thrombo-embolism, stroke and gallbladder disease (after one to two years, three years and seven years' use respectively), but did not significantly increase the risk of breast cancer. The only statistically significant benefits of HT were a decreased incidence of fractures and (for combined HT) colon cancer, with long-term use.

Among women aged over 65 who were relatively healthy (i.e. generally fit, without overt disease) and taking continuous combined HT, there was a statistically significant increase in the incidence of dementia. Among women with cardiovascular disease, long-term use of combined continuous HT significantly increased the risk of venous thrombo-embolism.

One trial analysed subgroups of 2839 relatively healthy 50 to 59 year old women taking combined continuous HT and 1637 taking oestrogen-only HT, versus similar-sized placebo groups. The only significantly increased risk reported was for venous thrombo-embolism in women taking combined continuous HT: their absolute risk remained low, at less than 1/500. However, this study was not powered to detect differences between groups of younger women.

Authors' conclusions

HT is not indicated for the routine management of chronic disease. We need more evidence on the safety of HT for menopausal symptom control, though short-term use appears to be relatively safe for healthy younger women.

PLAIN LANGUAGE SUMMARY

Long term hormone therapy for perimenopausal and postmenopausal women

Hormone therapy (HT) is widely used for controlling menopausal symptoms. It has also been used for the management and prevention of chronic diseases such as cardiovascular disease, osteoporosis and dementia in older women. The present review set out to assess the long term clinical effects of using HT. Nineteen randomised double-blind trials (involving 41,904 women aged 26 to 91 years) compared HT (all oestrogens, with or without progestogens, administered by oral, transdermal, subcutaneous or intranasal routes) with placebo when taken for at least one year. In healthy women (11 studies), combined continuous HT significantly increased the risk of obstruction of a vein by a blood clot (venous thrombo-embolism), fatal or nonfatal heart attack (after one year's use), stroke (after three years), breast cancer, gallbladder disease and (in women over 65 years) dementia. Long-term oestrogen alone significantly increased the risk of venous thromboembolism, stroke and gallbladder disease. Among women with cardiovascular disease (six studies) long-term use of combined HT significantly increased the risk of venous thromboembolism, particularly in the first two years of use, and gallbladder disease. HT offered the benefit of a significant reduction in the risk of fracture (no greater in women at high risk of fractures) or colorectal cancer but only after four or five years' treatment with HT, while the highest risk of cardiovascular events with combined HT occurred in the first year of use.

BACKGROUND

Most women experience the menopause (the last menstrual period) in their early fifties, after a phase of changing ovarian function (the perimenopause) which may last several years and which is characterised by irregular menstrual cycles (Greendale 1999). Women are said to be postmenopausal when menstruation has ceased for 12 months. Many perimenopausal and postmenopausal women (though not all) report a variety of symptoms including hot flushes and vaginal dryness, which probably relate to the natural decline of oestrogen levels. Symptoms tend to fluctuate and their severity varies greatly between individuals, with some reporting intense discomfort and a substantial reduction in quality of life. Most research has focused on white women, but the experience of menopause differs between different races and ethnicities, as well as by menopausal stage (Avis 2001). The duration of regular hot flushes is very variable: most women report that they last for between six months and two years (Kronenberg 1994) but longitudinal research suggests that the time from onset to resolution of symptoms is often considerably longer than this (Guthrie 2005).

Hormone therapy (HT) includes either oestrogen alone (oestrogen-only HT) or oestrogen combined with a progestogen (combined HT) It is used in a variety of formulations and doses which can be taken orally, vaginally, trans nasally or as an implant, skin patch, cream or gel. Clinical effects vary according to the type of HT and its duration of use.

The addition of a progestogen reduces the risk of endometrial hyperplasia associated with the use of oestrogen alone in women with a uterus (Lethaby 2004), but the issue is problematic because progestogens have adverse effects on blood lipids and may have the potential to cause symptoms such as headaches, bloating and breast tenderness (McKinney 1998). Progestogens used for HT include synthetic derivatives of progesterone, synthetic derivatives of testosterone and natural progesterones derived from plants. These differ in their metabolic action and potential for adverse effects and it is currently unclear which type of progestogen has the best risk/benefit profile for use in HT. In combined HT, progestogen can be taken either continuously (every day) or sequentially (for part of each month) or less frequently. It appears that continuous therapy may be more protective than sequential therapy in the long term prevention of endometrial hyperplasia (Lethaby 2004).

Hormone therapy (HT) has been utilised for the treatment of hot flushes and other menopausal symptoms for over 50 years and its efficacy has been well established, as evidenced by a Cochrane systematic review of 24 randomised controlled trials of hot flushes published between 1971 and 2000 (MacLennan 2004).

During the past twenty-five years HT has also been used for the management or prevention of chronic disease. Oestrogens and progestogens affect most body systems and have been proposed as potentially causal or preventative of a wide range of conditions. Recommendations for use have varied over time, but through the 1990's commonly-held expert opinion was that most postmenopausal women could benefit from HT (Hemminki 2000a). This view was based on strong and consistent observational evidence that HT reduced the risk of coronary heart disease (CHD) by 30% or more. A meta-analysis of 25 cohort, case-control and angiographic studies published up to 1997 reported a risk ratio of 0.70 (95% CI 0.65 to 0.75) for CHD among oestrogen users compared to never-users. Moreover the cardioprotective benefits of oestrogen were biologically plausible, given that oestrogen has a favourable effect on many bio markers, in particular HDL and LDL cholesterol's. Other benefits reported in observational studies of HT were strong evidence of a reduction in osteoporotic fractures, a possible preventative or delaying effect on cognitive decline and/or dementia and even a reduction in overall mortality for current users (Barrett-Connor 1998).

Observational studies also indicated a range of *adverse* effects of HT, including a doubling or tripling of the risk of thromboembolic events, a large increase in endometrial cancer risk in women taking oestrogen without progestogen, an increased incidence of gallbladder disease and a possible link between HT and breast cancer. The suggestion that HT increased the risk of breast cancer was supported by evidence of an increase in breast density in a high proportion of women taking oestrogen, but findings were inconsistent and controversial (Barrett-Connor 1998). The results of a very large observational study in the UK (Beral 2003) raised concerns that current users of both combined and oestrogen-only HT were at increased risk of both incident and fatal breast cancer after relatively short periods of use. The increase in risk was greatest for users of combined HT, with no large variations between the effects of specific oestrogens or specific progestogens. However these findings have been challenged: in particular, the duration of HT use prior to the occurrence of breast cancer was based on data recorded at study entry, without allowing for subsequent use up to the time of diagnosis (Lancet 2003).

Because CHD is the most common cause of death and morbidity in older women, it was held that a significant reduction in CHD risk from HT would outweigh any potential adverse effects. However there was strong potential for selection and/or compliance bias or both in these uncontrolled studies, with oestrogen-takers more likely to be healthy, well-educated and compliant women with a lower baseline risk of cardiovascular disease and the need for randomised controlled trials was recognised (Barrett-Connor 2001; Hemminki 2000a). It has been suggested that the wide prescribing of HT in the 1990s despite the lack of randomised evidence of its efficacy and safety might reflect a conflict between commercial/professional interest groups and good public policy (Hemminki 2000).

Randomised controlled trials (RCTs) have failed to demonstrate the marked CHD benefits of HT seen in observational studies and have raised questions about its overall risk/benefit profile. A systematic review of the available randomised evidence was conducted in 1997 and updated with unpublished evidence in 2000 (Hemminki 1997; Hemminki 2000). The authors reported a conservatively-estimated odds ratio for cardiovascular events of 1.34 (95% CI 0.55 to 3.30) among those taking HT. However this result was based on only fifteen secondary events in those allocated HT and seven in the control groups and provided insufficient evidence to exclude a potential benefit from HT.

Beral et al. (Beral 2002) pooled the results of four RCTs of HT published between 1998 and 2002 (EVTET 2000; HERS 1998; WEST 2001; WHI 1998). No significant excess or reduction in the relative risk of CHD was reported and the findings negated the large beneficial effect of HT reported for cardiovascular outcomes in the earlier observational studies. Moreover excess risks of stroke, pulmonary embolus and breast cancer were reported. The risk of colorectal cancer or fractured neck of femur were found to be significantly reduced for the HT group, while the findings for endometrial cancer risk were inconclusive. In this review the authors pooled the results of studies which used different types of HT over differing time frames.

Salpeter et al (Salpeter 2004) meta-analysed 17 RCTs of HT that reported at least one death and concluded that there was a significantly reduced risk of death in women with a mean age of under 60 years taking HT compared to a placebo group, though no difference was found when older women were compared. This meta-analysis pooled studies which differed widely with respect to the type of HT used and the clinical status of the participants and in several studies death was not a prespecified outcome. Moreover women with poor prognosis ovarian cancer accounted for 60% of the events in the meta-analysis of studies of younger women.

A systematic review of studies of hormone therapy and ovarian cancer (Greiser 2006) included 30 case control studies, seven cohort studies, one randomised controlled trial and four cancer registry studies. This review found an increased risk of ovarian cancer associated with the use of either oestrogen-only hormone therapy (risk ratio (RR) 1.28, 95% confidence interval (CI) 1.18 to 1.40) or combined hormone therapy (RR 1.11, 95% CI 1.02 to 1.21). The risk applied to a range of common histological subtypes of ovarian cancer. The authors noted that their review was limited by a reliance on observational data; however, heterogeneity was low or moderate in most analyses.

Another recent systematic review (Bath 2005) meta-analysed 28 RCTs of HT that reported stroke events. HT was associated with an statistically significant excess risk of stroke, particularly ischaemic stroke. Moreover among participants who had a stroke the HT group seemed to have a worse outcome. This review had very broad inclusion criteria and pooled a wide range of trials which used different types of HT for a range of indications, some with male participants and some without placebo control. It is unclear to what extent the findings apply to perimenopausal and postmenopausal women.

Previous Cochrane systematic reviews have considered HT and others are in preparation, as follows:

Bone mineral density (Wells 2002)
Cardiovascular disease in postmenopausal women (Sanchez 2005)
Dementia and cognitive function (Hogervorst 2002a; Hogervorst 2002b; Lethaby 2008)
Endometrial hyperplasia and carcinoma (Lethaby 2004).
Hot flushes and other menopausal symptoms (MacLennan 2004)
Weight and body fat distribution (Kongnyuy 1999)
Vaginal atrophy (Suckling 2006)

In view of the large number of reviews on individual aspects of HT it was recognised that there was a need for a systematic review giving an overview of all relevant long-term clinical outcomes which might help women and their clinicians make informed judgements about the use of HT. An a priori decision was made to exclude trials of shorter duration than one year and not to include as outcomes menopausal symptom control, early onset side-effects of HT or surrogate measures such as endometrial hyperplasia and bone mineral density. This review is not intended to replace other Cochrane reviews on HT, such as those listed above. These will remain an important source of evidence on individual aspects of HT and will continue to be updated regularly.

This is an updated version of the original Cochrane review first published in 2005.

OBJECTIVES

Our objectives are to compare the effect of long-term hormone therapy (over one year's duration) versus placebo in perimenopausal or postmenopausal women on mortality, heart disease, venous thromboembolism, stroke, transient ischaemic attacks, breast cancer, colorectal cancer, ovarian cancer, endometrial cancer, gall bladder disease, cognitive function, dementia, incidence of fractures and quality of life.

HT is defined as oestrogen therapy alone or oestrogen therapy with combined, continuous or sequential progestogen therapy, delivered by oral, transdermal, subcutaneous or intranasal routes.

METHODS

Criteria for considering studies for this review

Types of studies

Only randomised, double-blind trials that include at least one HT therapy group and one placebo group that report at least one of the above outcomes will be considered for inclusion in this review. Double-blinding required blinding of the participants and all researchers/outcome assessors.

Types of participants

Suitable participants will be perimenopausal or postmenopausal women recruited from any health care setting or a population-based sample .

Perimenopausal women will be defined as women with menopausal symptoms who have not yet had their final menstrual period.

Postmenopausal women will be defined as women with surgical menopause (removal of both ovaries) or women with spontaneous menopause and amenorrhoea for more than 12 months.

Women both with and without prior history of disease (e.g. cardiovascular disease, fracture, osteoporosis etc.) will be included.

Types of interventions

All oestrogens, with and without progestogens, administered by oral, transdermal, subcutaneous or intranasal routes and given as perimenopausal or postmenopausal therapy for any reason for twelve months or more, compared with placebo.

Exclusion Criteria

Co-interventions that may potentially affect the outcomes being measured.
Topical vaginal HT creams, topical tablets and rings will not be considered. These interventions are covered in another Cochrane review (Suckling 2006).

Types of outcome measures

Total mortality

Cause specific mortality
Coronary events (myocardial infarction or coronary death)
Venous thromboembolism (pulmonary embolism or deep vein thrombosis)
Stroke (ischaemic or haemorrhagic)
Transient ischaemic attack (TIA)
Breast cancer
Colorectal cancer
Ovarian cancer
Endometrial cancer
Gallbladder disease
Cognitive function, as measured in the included trials
Dementia (including Alzheimer's disease), as measured in the included trials
Incidence of hip fractures, clinically diagnosed vertebral fractures and total clinically diagnosed fractures
Quality of life

We plan to restrict our focus to long term clinical outcomes and not to include menopausal symptom control or early onset side-effects of HT as outcomes. HT for the control of hot flushes is the subject of another review (MacLennan 2004).

Search methods for identification of studies

We obtained publications that described randomised double-blind trials of HT therapy, with a placebo group and a minimum of 12 months' duration of therapy, adapting the strategy developed by the Menstrual Disorders and Subfertility Group (*see* Review Group details for more information). We performed electronic searches of the Cochrane Menstrual Disorders and Subfertility Group Trials Register (November 2007), the Cochrane Central Register of Controlled Trials (CENTRAL) on *The Cochrane Library* (November 2007), MEDLINE (1966 to November 2007), EMBASE (1980 to November 2007), and Biological Abstracts (1969 to November 2007). The search was not restricted by language. See appendices for the specific search string used Appendix 1.Appendix 2; Appendix 3

We checked the reference lists of relevant publications returned by the above searches. We also searched the National Research Register (NRR), a register of ongoing and recently completed research projects funded by, or of interest to, the United Kingdom's National Health Service, as well as entries from the metaRegister of Controlled Trials and details on reviews in progress collected by the NHS Centre for Reviews and Dissemination (September 2007).

We contacted the following pharmaceutical companies in December 2003, via their web sites or by letter, to request data from any published or unpublished randomised controlled trials of HT in their files: Schering AG, Novartis, NovoNordisk, Paines and Byrnes/NZMS, 3M Pharmaceuticals, Organon, Wyeth. Reprints of published trials were received from one company (NovoNordisk), one company had no unpublished trials with completed study reports available (Wyeth) and our request was acknowledged by a further two companies (3M Pharmaceuticals, Organon).

Data collection and analysis

Selection of trials:

One review author screened the titles or abstracts or both of all publications obtained by the search strategy for eligible trials. If the screened abstract suggested the trial was potentially eligible for inclusion, we obtained the full article. One of the review authors checked each study against the inclusion criteria. This assessment was performed unblinded. If there was uncertainty regarding eligibility, a second review author also assessed the study and a decision was reached through discussion. If necessary, additional information was sought from the corresponding author of the study.

Quality assessment and data extraction

See Risk of Bias tables in Characterstics of included studies Two review authors independently used a ProForma to assess the quality of the trials, record methodological details, and extract outcome data. This was checked for agreement and any disagreements were resolved by discussion. The following were assessed:

Trial characteristics

1. Method of randomisation.
2. Method of allocation concealment
3. Use of stratification
4. Adequacy of double blinding (i.e. an explicit statement that therapies could not be distinguished by appearance and/or administration route)
5. Number of participants screened for eligibility, randomised, analysed, excluded, lost to follow-up or dropped out (i.e. withdrew from the trial but were followed up)
6. Level of adherence to therapy
7. Whether an "intention-to-treat" analysis was done
8.The use of a power calculation to estimate sample size
9. Duration, timing and location of the study
10. Study design (e.g. parallel or crossover, single centre or multi-centre)
11. Source of funding

Characteristics of the study participants

1. Inclusion/exclusion criteria
2. Age and any other recorded characteristics of women in the study
3. Menopausal status (i.e. peri- or postmenopausal and how status was defined, surgical or natural menopause) of the women in the study
4. Baseline equality of treatment groups
5. Means of recruitment

Interventions used

1. Type of HT (oestrogen-only or combined oestrogen/progestogen), dosage and administration route of HT/placebo
2. Duration of therapy (minimum of one year)

Outcomes

1. Which outcomes relevant to review were measured
2. How relevant outcomes were measured and defined

For cross-over trials, it was intended to use only results from the end of the first phase (before the treatment cross-over) because of the potential carry-over effect of HT therapy from the first treatment phase. However, no cross-over trials were included.

Analysis

Statistical analysis was undertaken following the guidelines of the Handbook of the Cochrane Collaboration (Higgins 2008). Analysis of treatment effects compared outcomes for each therapy group measured at follow up or at the end of therapy or both.

For dichotomous data, two by two tables were generated for each trial and expressed as a risk ratio (RR) with 95% confidence intervals (CI). This data was combined for meta-analysis with RevMan software, using the Peto-modified Mantel-Haenszel method. Since there is no consensus about whether fixed or random-effects models should be used for meta-analysis, both types of analysis were performed: this might be viewed as a sensitivity analysis to assess the impact of the choice of model on the results of the analysis; unless the results were robust to both models, they would need to be treated with caution. Published graphs display the results of the fixed-effects approach.

Continuous data was expressed as a weighted mean difference (WMD) and 95% confidence interval. We planned to combine continuous data for meta-analysis had any such data been available for pooling. We considered quality of life scores, although measured as ordinal variables, to be drawn from an underlying continuous distribution, and they were analysed as continuous outcomes. Meta-analytic methods for continuous data assume that the underlying distribution of the measurements is normal. The ratio of the mean to its standard deviation gives a crude method of assessing skew: if this ratio was less than 1.65 for any group in a trial, unless the original data were available for log transformation the results were not included in analysis tables but were reported in 'Other Data' tables. We also planned to report results in the 'Other Data' section if data were clearly skewed and results were reported in the publication as median and range with non-parametric tests of significance; however no such data were reported.

If trials reported the number of events occurring in each comparison group at a mean follow-up time (i.e. not all women had been followed up to for that duration of time, while others had been followed up for longer), the simplifying assumption was made that the risk was constant across the follow-up period and the data were reported as dichotomous data at a fixed time point. If a risk varied significantly across the follow-up period, this variation has been noted in the text in the 'Results' section.

For outcomes where studies reported no events in either the HT group or the placebo group, results were not entered in the 'Data and analyses'.

Heterogeneity

We planned to pool the results of individual studies (meta-analyse) only where they were clinically similar with respect to the study population, intervention and outcome of interest. If an individual study pooled the results of study arms that used different types of HT, the pooled results were not included in this review.

We planned to assess statistical heterogeneity (variation) between the results of different studies included in meta-analyses by inspecting the scatter in the data points on the graphs and the overlap in their confidence intervals, and by checking the I^2 quantity (Higgins 2003). This quantity describes the percentage of total variation across studies that is due to heterogeneity rather than chance. Interpretation of a given degree of heterogeneity will differ according to whether the estimates show the same direction of effect, but we planned to tentatively assign adjectives of low, moderate and high heterogeneity to I^2 values of 25%, 26 to 74% and 75% respectively.

We planned to conduct sensitivity analyses to examine the effect of methodological differences between the trials provided there were sufficient trials (> 5). These might help to explain any moderate or high statistical heterogeneity that might be detected. Specific differences we planned to explore were as follows:
1. Trials with adequate methodology versus those of poor methodology: adequate methodology is defined for this purpose as adequate allocation concealment, analysis by intention-to-treat and losses to follow-up < 10%.
2. Trials which might differ from the others with respect to their participants, interventions or clinical criteria for defining outcomes - although it was planned not to combine trials which were obviously dissimilar in these respects.

It was planned to conduct further sub group or sensitivity analyses if other possible sources of heterogeneity became evident during the preparation of the review; however the results of any such post-hoc analyses would need to be interpreted with great caution.

RESULTS

Description of studies

Forty-nine studies were retrieved by the original and updated searches and considered for inclusion, of which 30 were excluded. The primary reasons for exclusion are listed below:

- 23 studies reported no outcomes of interest to this review
- 3 studies used an intervention of less than one year's duration
- 1 study did not include a placebo group
- 3 studies were not double blinded

Nineteen studies met our inclusion criteria, including one very large study (WHI 1998). WHI 1998 incorporated randomised comparisons of two different HT regimes versus placebo which published their results separately. One (WHI 2002) compared combined oestrogen and progesterone versus placebo and is referred to in this review as WHI 1998 (*combined HTarm*) and the other compared oestrogen-only HT with placebo and is referred to in this review as WHI 1998 (*oestrogen-only HT arm*). WHI 1998 also included a subgroup study known as the Women's Health Initiative Memory Study (Espeland 2004; Rapp 2003; Shumaker 2003; Shumaker 2004;). This measured cognitive outcomes among women aged 65 to79 at trial entry from both arms of WHI 1998, and is referred to in this review as WHI 1998 *(WHIMS)*. An additional ancillary study, WHI 1998 *(WHISCA),* enrolled women from WHI 1998 *(WHIMS)* who were free of dementia. WHISCA investigated the effects of hormone therapy on specific cognitive functions in older women (Resnick 2006.)

The nineteen trials included 41,904 randomised women; 21,763 randomised to receive some form of HT and 20,125 to receive placebo (treatment allocation is unknown for 16 women on one trial (Ferenczy 2002). WISDOM 2007 also included a further 1307 women who were randomised to a comparison of two active hormone therapies and who are not included in this review. Results for over 99% of these women were analysed by intention-to-treat. Although some of the trials had biological measures as their primary outcome (for example, lumen of carotid artery) they were included because they also reported clinical endpoints relevant to this review as pre-specified secondary outcomes.

One of the included studies (HERS 1998) conducted prolonged outcome surveillance after trial completion, in the form of an open (unblinded) continuation of the study in which 93% of the original participants agreed to participate. Data from this unblinded part of the study has also been reported in this review as it is provides useful data but it has not been considered as a separate study, nor is it included in any meta-analysis.

The trials varied dramatically in size. The largest was WHI 1998 which randomised 27,347 participants, while the other studies varied in sample size from 151 to 5692 participants (Obel 1993 and WISDOM 2007 respectively). There were over 8000 women in each group in WHI 1998 *(combined HT arm)* and over 5000 in each group in WHI 1998 *(oestrogen-only HT arm).* These included over 1400 in each group on the oestrogen-only HT arm of WHI 1998 *(WHIMS)* and over 2200 in each group on the combined arm of WHI 1998 *(WHIMS)*. The Estonian trial (EPHT 2006) included around 400 women in each group. Otherwise none of the trials included more than 210 women in each comparison group. Four of the smaller trials were single centred (EPAT 2001; Haines 2003; Nachtigall 1979; Obel 1993) and it is unclear whether one trial (EVTET 2000) had more than one trial centre. The other ten trials involved between seven and 40 trial centres.

Eleven of the trials were conducted in the USA, one trial was conducted in each of the following countries: Estonia, the UK, Norway, Denmark and Hong Kong and three trials were international (one in the USA and Canada, one in Canada and the Netherlands and one in the UK, Australia and New Zealand). Two of these studies (EPHT 2006; WISDOM 2007) were originally planned as part of a larger international project. However the planning was beset with delays and in the meantime WHI 1998 began in the USA and other countries were no longer prepared to commit funds to a second trial with similar objectives. Both of these studies were prematurely closed as a result of the publication of early WHI 1998 findings.

Participants

The women included in these studies were all peri or postmenopausal, either spontaneously or surgically. The age of participants ranged from 26 to 91, with the mean or median age of each study ranging from 48 to 72 (no age was stated in Obel 1993). Inclusion criteria varied according to the primary objectives of individual trials. Some were designed to investigate the use of HT for treatment of menopausal symptoms or for disease prevention and thus enrolled women in reasonably good health. Others were designed to assess whether HT had a beneficial effect on women with a history of cancer or established diseases, including heart disease, thrombo-embolic disease, stroke, Alzheimer's disease and long-term medical conditions requiring hospitalisation; these trials restricted entry to women diagnosed with the condition of interest.

Studies of women without established medical conditions

Nine of the studies enrolled relatively healthy women (EPAT 2001; EPHT 2006; Ferenczy 2002; Haines 2003; Notelovitz 2002; Obel 1993; PEPI 1995; WHI 1998; WISDOM 2007; Yaffe 2006). Women on some of these nine trials had risk factors (such as raised cholesterol) and a small minority within individual trials had a history of cardiovascular disease, but predominantly participants were fit women without overt disease. Six of the nine trials were interested in the use of HT for disease prevention.

Three of these nine studies were large trials investigating the use of HT to prevent cardiovascular disease but also reporting a wide range of other endpoints; they had very detailed lists of inclusion and exclusion criteria (PEPI 1995; WHI 1998, WISDOM 2007). In WHI 1998, enrolment was targeted to establish set fractions for baseline age categories and to achieve representation of racial and ethnic groups in the proportion recorded in the US census for the 50 to 79 age group.

The WHI 1998 *(combined HT arm)* investigators noted that prevalence of prior cardiovascular disease in participants was low: 4.4% had a history of myocardial infarction, coronary revascularisation, stroke or transient ischaemic attack. They also commented that levels of cardiovascular risk factors were consistent with a generally healthy population of postmenopausal women: 2.9% reported a history of angina, 36% were hypertensive (or being treated for hypertension), 13% were being treated for high cholesterol, 4.4% were being treated for diabetes and 10.5% were current smokers (Manson 2003). Similarly, in WHI 1998 *(oestrogen-only HT arm),* participants were in general considered healthy, although 4.1% had a history of myocardial infarction or coronary revascularisation, 5.8% had a history of angina, 1.4% had a history of stroke,1.6% had a history of venous thrombosis, 48% were hypertensive (or being treated for hypertension), 15% were receiving treatment for high cholesterol, 7.7% were being treated for diabetes and 10.5% were current smokers (Stefanick 2003).

PEPI 1995 compared the characteristics of their cohort with values returned in large US surveys and concluded that although the PEPI 1995 cohort were generally in better health than the wider US population, they were not so markedly different as to limit the generalisability of study results. The other two "prevention" trials were much smaller and had more narrowly defined outcomes, namely the possible beneficial effect of HT on arterial wall density and bone density respectively (EPAT 2001; Notelovitz 2002). Four much smaller studies also enrolled women without stated health problems, who were either in early menopause (Obel 1993) or postmenopausal (Ferenczy 2002; Haines 2003) they gave relatively little information about inclusion and exclusion criteria (*see t*able 'Characteristics of included studies' for details).

The WISDOM 2007 study recruited women with no known major health problems from general practice registers in countries with free or low fee health care systems. Recruitment was designed to target older women first and as a result the median participant age was 63 and there were few women in the younger age group when the study closed prematurely.

Studies of women with established medical conditions or a history of cancer

Six of the studies included women with established cardiovascular disease (ESPRIT 2002; ERA 2000; EVTET 2000; HERS 1998; WAVE 2002; WEST 2001). ERA 2000 and WAVE 2002 included women who had coronary artery stenosis seen on angiogram. HERS 1998 and ESPRIT 2002 randomised women who had had a myocardial infarction or (in the case of HERS 1998) coronary artery surgery. EVTET 2000 and WEST 2001 were for women who had suffered a thrombo-embolic event (PE or DVT) or cerebrovascular event (stroke or TIA). The largest of these six studies (HERS 1998) compared their cohort of women with a similar group of women presumed to have coronary heart disease who were participants in a survey designed to produce nationally representative data:

the HERS 1998 cohort had significantly fewer smokers, women with hypertension and diabetics than the comparison group but were comparable with respect to blood pressure, body mass index, physical activity and cholesterol levels.

One study (Mulnard 2000) included women with Alzheimer's disease, while an older study (Nachtigall 1979) included women with a range of medical conditions including diabetes, a need for custodial care, arteriosclerosis and chronic neurological disorders: all participants in this study were hospitalised for the duration of the ten year study.

One trial enrolled women after surgery (including bilateral salpingo-oophorectomy) for early stage endometrial cancer (Barakat 2006).

Interventions

A wide variety of oestrogen alone or oestrogen and progestogen combinations were used as interventions in the included trials; some had more than one intervention arm, each with a different dose, formulation or route of HT. Most of the comparisons used a moderate dose of oestrogen (e.g. oestradiol 1 mg, Conjugated equine oestrogen (CEE) 0.625 mg daily or transdermal oestradiol 0.05 mg twice weekly). Nachtigall 1979 used a much higher dose than other included trials, reflecting the fact that it was conducted many years earlier than the others.

The range of interventions used was as follows:

Oestrogen-only HTs included the following:
Oestradiol (17-B oestradiol) an oestrogen derived from derived from Mexican wild yam. Doses used were 1 mg (EPAT 2001; Haines 2003; WEST 2001) and 2 mg (Haines 2003)
Oestradiol valerate, which is a pro-drug for oestradiol (meaning that it is converted in the body into the active form). The dose used was 2 mg (ESPRIT 2002)
Transdermal oestradiol skin patches. The doses used were 0.014 mg (Yaffe 2006), 0.025 mg, 0.05 mg or 0.075 mg once or twice weekly (Notelovitz 2002).
Conjugated equine oestrogen (CEE), a blend of equine oestrogens. The doses used were 0.625 mg daily (Barakat 2006; ERA 2000; Mulnard 2000; PEPI 1995; WAVE 2002; WHI 1998 *(oestrogen-only HT arm))* and 1.25 mg daily (Mulnard 2000). One study (Barakat 2006) allowed a doubling of dose for women who were symptomatic. WISDOM 2007 also included an oestrogen-only arm but the comparison group was taking combined therapy and this comparison is not relevant to this review.
Several trials using oestrogen-only HT did not randomise women to this comparison unless they had had a hysterectomy (Haines 2003; Mulnard 2000; Nachtigall 1979; Notelovitz 2002; WAVE 2002; WEST 2001; WHI 1998 *(oestrogen-only HT arm)*.

Combined HT regimens

Combined regimens included one of the above types of oestrogen in combination with one of the following progestogens:

- Medroxyprogesterone acetate (MPA): a synthetic progestogen structurally related to progesterone
- Dydrogesterone: a synthetic progestogen structurally related to progesterone
- Norethisterone (norethindrone): a synthetic progestogen structurally related to testosterone
- Micronised progesterone**:** a natural progestogen synthesized from plant sources and finely ground to improve its absorption

Continuous combined regimens included the following.

- CEE 0.625 mg with MPA 2.5 mg daily (EPHT 2006; ERA 2000; HERS 1998; PEPI 1995; WAVE 2002; WHI 1998*(combined arm)*; WISDOM 2007)
- CEE 2.5 mg with MPA 10 mg daily (Nachtigall 1979)
- Oestradiol 2 mg with 1 mg norethisterone daily (EVTET 2000)

Combined sequential regimens included the following.

- Oestradiol 1 mg daily with MPA 5 mg for 12 days once a year (WEST 2001)
- Oestradiol 2 mg days 1 to 22, 1 mg days 22 to 28, with norethisterone 1 mg days 13 to 22 (Obel 1993)
- Oestradiol 1 mg daily with dydrogesterone 5 mg or 10 mg days 14 to 28 (Ferenczy 2002)
- Oestradiol 2 mg daily with 10 to 20 mg dydrogesterone days 14 to 28 (Ferenczy 2002)
- CEE 0.625 with MPA 10 mg days 1 to 12 (PEPI 1995)
- CEE 0.625 mg with micronised progesterone 200 mg days 1 to 12 (PEPI 1995)

The control arm on each study received placebo tablets or patches or both, as appropriate.

The duration of HT use varied, with the longest trial lasting ten years (Nachtigall 1979). Four trials reported outcomes after HT use for around one year (EVTET 2000; Haines 2003; Mulnard 2000; WISDOM 2007); five measured outcomes after two years (EPAT 2001; ESPRIT 2002; Ferenczy 2002; Notelovitz 2002; Obel 1993; Yaffe 2006) and six at around three years (Barakat 2006; EPHT 2006; ERA 2000; PEPI 1995; WAVE 2002; WEST 2001). HERS 1998 measured outcomes after 4.1 years, and continued the study unblinded for a further 2.7 years.

The interventions in the WHI trial were planned to continue for 8.5 years, but both components of the trial were terminated early. WHI 1998 *(combined HT arm)* was stopped early due to net harm. Outcomes were reported at 5.2 years and subsequently for a further four months of follow-up for primary and selected outcomes, incorporating events up to the date that participants were instructed to stop their study pills. WHI 1998 *(oestrogen-only HT arm)* was also stopped early when it was decided that the prospect of obtaining more precise evidence about the effects of the intervention was unlikely to outweigh potential harms, although no predefined safety boundaries had been crossed. Results have been reported for an average follow-up of 7.1 years for primary outcomes. As noted above, two other studies also closed prematurely in response to WHI 1998 findings (EPHT 2006; WISDOM 2007).

(*See* table 'Characteristics of included studies' for more details of interventions in individual trials)

Outcomes

The outcomes measured by individual trials varied according to the trial objectives. Major clinical events were not primary outcomes for several of these studies but were measured as adverse effects. Thus although their primary outcomes were not relevant to this review, they also measured pre-specified *secondary* outcomes which included clinical endpoints of interest, for example cardiovascular events or the incidence of cancer and fractures in the trial population or both. Six trials had biological measures as their primary outcome (EPAT 2001; ERA 2000; PEPI 1995; WAVE 2002; Notelovitz 2002; Yaffe 2006).

The largest trial in the review (WHI 1998) was concerned mainly with the cardio-protective role of HT in relatively healthy women, and reported cardiovascular clinical endpoints as the primary outcome. Invasive breast cancer was designated a primary *adverse* outcome and secondary outcomes were the incidence of other cancers, fractures, gallbladder disease and death. Two other trials (EPHT 2006; WISDOM 2007) measured similar outcomes.

WHI 1998 *(WHIMS)* comprised a large subset of older women from this trial who were evaluated for probable dementia (the planned primary outcome) and for mild cognitive impairment (as a planned secondary outcome). Global cognitive function was also reported, though was not a formally pre-planned endpoint. WHI 1998 *(WHIMS)* reported separate results for the two trial arms and also pooled their results: however these are not included in this review (*see* 'Methods'). WHI 1998 (WHISCA) reported annual rates of change in specific cognitive functions in women free of dementia who were enrolled from 14 of the 39 WHI 1998 (WHIMS) sites. Outcomes derived from a battery of very specific tests and the results are summarised in the text below. Changes from baseline were not reported because pretreatment levels of specific cognitive function were unavailable. WHI 1998 (WHISCA) was intended to run for four years, but only 43% of WHI 1998 (WHISCA) participants had completed three annual follow-up assessments before WHI 1998 was terminated and the mean follow up was only 1.3 years.

Both arms of WHI 1998 measured quality of life in the whole cohort at one year and for a smaller subset at three years. As noted above, this study excluded women with menopausal symptoms severe enough to preclude randomisation. Two very much smaller trials also reported quality of life in healthy women (in one case Chinese women) (Haines 2003; Obel 1993). The latter trial (Obel 1993) reported endometrial cancer as an outcome as well.

One small trial reported only one outcome of interest to this review, namely endometrial cancer (Ferenczy 2002). Another reported on the recurrence rate of endometrial cancer, also on mortality due to endometrial cancer, CHD or any cause (Barakat 2006)

Five other trials were concerned with the effect of HT on established clinical disease. Four reported cardiovascular outcomes: their primary outcomes were myocardial infarction or death (ESPRIT 2002; HERS 1998), thrombo-embolism (EVTET 2000) and stroke (WEST 2001). The larger studies also measured a range of other major clinical outcomes such as the incidence of cancers, fractures and gallbladder disease (ESPRIT 2002; HERS 1998). One trial reported the effect of HT on the progression of symptoms in women with Alzheimer's disease (Mulnard 2000) and another measured a wide range of clinical outcomes over a period of ten years' treatment with HT, in women who were in long term hospital care for a range of medical conditions (Nachtigall 1979).

Risk of bias in included studies

Randomisation and allocation concealment

Of the nineteen included studies, sixteen described a satisfactory method of randomisation, which in all cases was computer-generated. Fourteen of these trials scored A for allocation concealment: in these trials allocation to treatment was either generated by the computer once information about an eligible participant had been entered, or was accomplished by remote contact between the recruiting centre and the study co-coordinating centre or pharmacy. One of these 13 trials (EPHT 2006) randomised women who expressed an interest in participating but did not open the randomisation envelope until their eligibility had been checked and they had consented. Two studies described using computer-generated randomisation but did not give details of the procedure for allocation to treatment (EVTET 2000; Mulnard 2000). Three studies gave no detailed information about either randomisation or allocation concealment (Ferenczy 2002; Nachtigall 1979; Notelovitz 2002).

Stratification

Thirteen studies described using stratification. NIne studies stratified according to study centre. Some also (or instead) stratified according to cardiovascular risk factors, hysterectomy status, previous use of HT, intended use of HT, cancer stage and/or age or both. In the case of the WHI 1998 study, stratification was designed to meet target numbers of women within preset age bands in order to increase the power of the study with respect to cardiovascular disease, which is commoner in older women, while still providing useful information on intermediate outcomes for younger women (*see* table 'Characteristics of included studies for more details). Five studies did not mention stratification (Ferenczy 2002; Mulnard 2000; Nachtigall 1979; Notelovitz 2002; Obel 1993). One study stated that it did not use stratification (Haines 2003).

Blinding

All the included trials described themselves as (at least) double-blinded. Thirteen trials explicitly stated that all participants, clinical staff and outcome assessors were blinded to treatment allocation, though in the WHI trial, 331 women randomised to receive active treatment were unblinded and changed arms from WHI 1998 *(oestrogen-only HT arm)* to WHI 1998 *(combined HT arm)* following a change in protocol. Nachtigall 1979 stated that the trial was double-blinded and specified that this applied to the research physicians. The other five trials were not explicit about who was blinded (EVTET 2000; Ferenczy 2002; Haines 2003; Mulnard 2000; Notelovitz 2002).

The larger trials described an unblinded mechanism to be used when required for the management of adverse effects. PEPI 1995 unblinded 39 women (4%) during the course of the trial, 32 of whom were taking oestrogen-only HT. WHI 1998 *(combined HT arm)* reported that during 5.2 years of follow-up 3444 women in the combined HT group (40%) and 548 women in the placebo group (6%) were unblinded, whereas in WHI 1998 *(oestrogen-only HT arm)*, only 100 women in the active group (< 2%) and 83 in the placebo group (< 2%) were unblinded. Nachtigall 1979 reported that 13 women in the HT group and 17 in the control group were unblinded.Two women were unblinded in WISDOM 2007. The other trials did not report such information.

One randomised blinded study (HERS 1998) completed 4.1 years of follow-up and was then extended for a further 2.7 years' duration *unblinded* .

Intention-to-treat analysis, drop-outs, adherence to treatment and losses to follow up

For the purpose of this review, *intention-to-treat* was defined as the analysis of all randomised participants in the groups to which they were randomised. Drop-outs were defined as participants who stopped their allocated treatment (and in some cases changed to a different off-trial treatment) but have known clinical outcomes and are included in analysis. *Losses to follow up* were defined as participants for whom the outcomes of interest were unknown (and who may or may not have had outcomes imputed in statistical analysis). *Adherence to treatment* refers to the number of tablets actually taken, which is often assessed by pill counts (*see* 'Additional table on Adherence to treatment').

Twelve of the included studies analysed all participants by intention-to-treat, at least for all the outcomes of interest to this review (EPHT 2006; EPAT 2001; ERA 2000; ESPRIT 2002; Haines 2003; HERS 1998; Mulnard 2000; Nachtigall 1979; Notelovitz 2002; WEST 2001; WHI 1998; WISDOM 2007; Yaffe 2006) and a further two studies analysed over 97% of participants by intention-to-treat (PEPI 1995; WAVE 2002). Three studies did not include all participants in an intention-to-treat analysis for the outcomes of interest (EVTET 2000; Ferenczy 2002; Obel 1993). It was unclear whether one study used ITT analysis because there was no description of participants other than those that were "eligible and assessable" (Barakat 2006). *See* 'Characteristics of included studies' table for details

Drop-out rates were generally high particularly in the active treatment groups, and they increased over time. In WHI 1998 *(Combined HT arm)* 42% of the active treatment group and 38% of the placebo group were no longer taking their allocated treatment at five years, and a further 10.7% of the placebo group had crossed to active therapy. In WHI 1998 (*oestrogen-only HT arm*) 53% of participants overall were no longer taking their allocated treatment at 6.8 years and a further 5.7% had initiated hormone use outside the study. *See* 'Characteristics of included studies' table and Table 1 for details of drop outs and non-adherence in other studies.

Losses to follow up were low in most of the studies, with no women lost to follow up in seven studies (EPAT 2001; ERA 2000; ESPRIT 2002; EVTET 2000; Mulnard 2000; Nachtigall 1979; WEST 2001) and 1 to 5.2% lost in five other studies, which were all large and of long duration (3 to 6.8 years) (HERS 1998; PEPI 1995; WAVE 2002; WHI 1998). Only five women (0.01%) were lost to follow up in WISDOM 2007. The Estonian trial monitored outcomes by means of the linkages to a national health insurance database and national cancer registry and the authors stated that the probability of missing data in these databases was small (EPHT 2006). In four smaller studies of one to two year duration a higher proportion of women (8.5% to 14.5%) were lost to follow up (Haines 2003; Notelovitz 2002; Obel 1993; Yaffe 2006) and in Ferenczy 2002 results were unavailable for 34% of participants for the outcome of interest to this review. It was unclear whether any women were lost to follow up in one study (Barakat 2006). *See* 'Characteristics of included studies' table for more details

Power calculations

Fourteen studies gave details of their power calculations. One of these (ESPRIT 2002) did not achieve the sample size required to achieve its target power, as accrual was lower than expected, and three studies (EPHT 2006; WHI 1998; ; WISDOM 2007) failed to achieve the required sample size or the length of follow up or both required by their power calculations because they were terminated early. Barakat 2006 failed to reach the intended power because of a combination of a low event rate and early termination. Four studies gave no information about power calculations.

We attempted to contact investigators on the following trials for more information: Barakat 2006; EPAT 2001; EVTET 2000; Ferenczy 2002; Haines 2003; HERS 1998; Mulnard 2000; Notelovitz 2002; Obel 1993; PEPI 1995; WAVE 2002; WEST 2001; WHI 1998, WISDOM 2007. The investigators from the following trials kindly supplied clarification or additional unpublished data or both: Barakat 2006, ERA 2000; EPHT 2006; Haines 2003; HERS 1998; Obel 1993; PEPI 1995; WAVE 2002; WISDOM 2007. *See* 'Characteristics of included studies' table for methodological details about all included studies and Table 1 for an overview of study quality.

Effects of interventions

Results are described below. In most cases effect measures are reported in the text only where results were statistically significant: for relative risks and confidence intervals for *all* comparisons please *see* the 'Tables of Comparison'.

Results are grouped as follows:
a) By outcome:

- Outcomes such as death, cardiovascular events, cognitive measures and quality of life are grouped according to the clinical status of participant groups, in the following order: relatively healthy women, women with a history of cardiovascular disease, women hospitalised with chronic illness and women with dementia.
- For outcomes such as cancer, fractures and gallbladder disease group all participants are grouped together as "all women".

B) By intervention

- Oestrogen-only HT
- Combined continuous HT regimens
- Combined sequential regimens

Within these categories, interventions have been grouped according to the oestrogen dose used, the equivalence between doses being based on expert sources (Ansbacher 2001; MacLennan 1998; Pharmac 2003).

Meta-analysis

Although comparisons with similar oestrogen doses are grouped together, comparisons have only been pooled (meta-analysed) if they used the same combination of oestrogen and progestagen for the same (or similar) length of time. WHI 1998 and PEPI 1995 both used the same HT regimen and reported several of the same clinical outcomes at three years, but in most cases there were no events on either arm in PEPI 1995. Three studies (ERA 2000; HERS 1998; WAVE 2002) were combined for some three year (2.8 to 3.2) outcomes, but otherwise meta-analysis was inappropriate for most outcomes because the trials used different types or doses of oestrogen or progestagen or both which do not necessarily have the same metabolic effects, or else they used different durations of HT which might differ in effect due to trends over time.

Because there were very few results suitable for pooling, statistical heterogeneity was not a major issue in this review. One meta-analysis displayed statistically significant heterogeneity ($I^2 = 66.2\%$) but it involved only two small trials with few events and we attributed the heterogeneity to chance (*see* 'Table of comparisons 2:25).

Time points for reporting results

WHI 1998 *(oestrogen-only HT arm)* reported results after a *mean* follow up of 7.1 years and WHI 1998 *(combined HT arm)* reported results after a *mean* follow up of 5.2 or 5.6 years. As mentioned above (*see* Methods), we have analysed these results as if all women had an equal length of follow up. In addition, WHI 1998 *(combined HT arm)* reported selected clinical outcomes for each year of follow up. Since all women in this arm had been enrolled for at least 3.5 years at the time of the trial publication, we have used these data to calculate outcomes on an intention-to-treat basis after one, two and three years' use of HT, using all randomised participants as the denominator.

EPHT 2006 reported results at a mean follow up of 3.43 years, with a range of 2-5 years. Results have been reported in our tables as if all women had 3 years of follow up.

WISDOM 2007 reported results after a median follow-up of 11.9 months (range 7.1-19.6). Results have been reported in our tables as if all women had one year of follow up.

Barakat 2006 reported results after a median follow up of 35.7 months. Results have been reported in our tables as if all women had 3 years of follow up.

HERS 1998 reported results from the blinded part of the trial after a mean follow up of 4.1 years, which as mentioned above (see Methods), we have reported as dichotomous data. In addition, selected clinical outcomes were reported for each year of follow up. Since all women had been enrolled for at least three years at the time of the report, in this review these data have been used to calculate outcomes on an intention-to-treat basis after one, two and three years' use of HT, using all randomised participants as the denominator.

Results for outcomes of interest

All of the statistically significant findings of this review derived from the two biggest trials, HERS 1998 and WHI 1998, both of which scored A for allocation concealment, analysed all participants by intention-to-treat and had low losses to follow up (1%-5.2%).

Death from any cause

Relevant comparisons

This outcome was measured in healthy women in five trials (EPHT 2006; EPAT 2001; PEPI 1995; WHI 1998; WISDOM 2007) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to nearly seven years.

Death from any cause was also measured in five trials of women with cardiovascular disease (EPAT 2001; ESPRIT 2002; Haines 2003; HERS 1998; WAVE 2002; WEST 2001) with a total of four different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from two to four years, with unblinded follow up to 6.8 years (HERS 1998).

In addition, the trial comparing oestrogen only HT versus placebo in women who had undergone surgery for stage I or II endometrial cancer measured this outcome (Barakat 2006).

Results

No statistically significant difference was found between HT and placebo for this outcome in any population group.

Death from coronary heart disease

Relevant comparisons

This outcome was measured in relatively healthy women in four trials (EPAT 2001; PEPI 1995; WHI 1998; WISDOM 2007) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to over seven years.

Death from coronary heart disease was also measured in five trials of women with cardiovascular disease (ERA 2000; ESPRIT 2002; HERS 1998; WAVE 2002; WEST 2001) with a total of four different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from two to four years, with unblinded follow up to 6.8 years (HERS 1998)

In addition, the trial comparing oestrogen only HT versus placebo in women who had undergone surgery for stage I or II endometrial cancer measured this outcome (Barakat 2006).

Results

No statistically significant difference was found between HT and placebo for this outcome in any population group.

Death from stroke

Relevant comparisons

This outcome was measured in three comparisons of relatively healthy women taking combined continuous HT for one year (WISDOM 2007) and for 5.2 years (WHI 1998 *(combined HT arm)*) or taking oestrogen alone for 7.1 years (WHI 1998 *(oestrogen-only HT arm)*), and also in one trial of women with a history of stroke taking oestrogen-only HT with annual progesterone for women who had a uterus for 2.8 years (WEST 2001).

Results

No statistically significant difference was found between HT and placebo for this outcome.

Death from breast cancer

This outcome was measured in two trials of relatively healthy women taking combined continuous HT for one year (WISDOM 2007) and for 5.2 years (WHI 1998). No statistically significant difference was found between HT and placebo for this outcome.

Death from endometrial cancer

This outcome was measured in the trial comparing oestrogen only HT versus placebo in women who had undergone surgery for stage I or II endometrial cancer (Barakat 2006). No statistically significant difference was found between HT and placebo.

Death from any cancer

This outcome was measured in two trials of relatively healthy women taking continuous HT for one year (WISDOM 2007) and for 5.2 years (WHI 1998 *(combined HT arm)*) and one of women with cardiovascular disease taking combined continuous HT for 4.1 years, with unblinded follow up to 6.8 years (HERS 1998).

No statistically significant difference was found between HT and placebo for this outcome.

Coronary events (myocardial infarction or cardiac death)

Relevant comparisons

This outcome was measured in relatively healthy women in five trials (EPHT 2006; EPAT 2001; PEPI 1995; WHI 1998, WISDOM 2007) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to over seven years.

Coronary events were also measured as an outcome in women with cardiovascular disease in six trials (ERA 2000; ESPRIT 2002; EVTET 2000; HERS 1998; WAVE 2002; WEST 2001) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from two to four years, with unblinded follow up to 6.8 years (HERS 1998).

Results

In WHI 1998 *(oestrogen-only HT arm)* there was no statistically significant difference between the two groups for this outcome. However, in WHI 1998 *(combined HT arm)*, relatively healthy women taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) were at a significantly higher risk of a coronary event after taking HT for one, two, three and four years (at one year: RR 1.74 (95% CI 1.05 to 2.89), at two years: RR 1.49 (95% CI 1.05 to 2.12), at three years: RR 1.43 (95% CI 1.05 to 1.95), at four years: RR 1.37 (95% CI 1.05 to 1.79)). At a mean follow up of 5.6 years there was no statistically significant difference between the groups (RR 1.22 (95% CI 0.98 to 1.51)). Data for this outcome at one year and at three years were also reported by WISDOM 2007 and by EPHT 2006 respectively. Pooling these data with the WHI 1998(combined HT arm) data resulted in a relative risk at one year of RR 1.89 (95% CI 1.15 to 3.10) and at three years of RR1.45 (95% CI 1.07 to 1.98).

No other trials found any statistically significant difference between HT and placebo for this outcome, though HERS 1998 reported results of borderline statistical significance at one year, suggesting increased risk for women with cardiovascular disease taking combined continuous therapy (RR 1.5, 95% CI 1.00 to 2.25). Although initial analysis of time trends in HERS 1998 suggested a trend towards increased risk in the HT group diminishing over time, subsequent analysis based on the entire 6.8 years of follow up (blinded and unblinded) showed no statistically significant variation in risk over time.

Stroke

Relevant comparisons

This outcome was reported in relatively healthy women in four trials (EPHT 2006; EPAT 2001; PEPI 1995; WHI 1998) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to nearly seven years.

This outcome was also measured in women with cardiovascular disease in five trials (ESPRIT 2002; EVTET 2000; HERS 1998; WAVE 2002; WEST 2001) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to four years, with unblinded follow up to 6.8 years (HERS 1998).

Results

In WHI 1998 *(oestrogen-only HT arm)* there was a statistically significant increase in the incidence of strokes at 7.1 years' follow up (RR 1.35 (95% CI 1.08 to 1.70)).The authors noted that the excess in the intervention arm was due to an increased risk of ischaemic rather than haemorrhagic stroke and that the excess risk became apparent after four years of follow up (Hendrix 2006). In WHI 1998 *(combined HT arm)*, although there was no statistically significant difference between the groups in the incidence of stroke during the first two years of the trial, women taking combined continuous HT were at a significantly higher risk of stroke after taking it for three or more years (at three years: RR 1.47 (95% CI 1.02 to 2.11), at a mean of 5.2 years: RR 1.42 (95% CI 1.08 to 1.87). Data for this outcome at three years were also reported by EPHT 2006; pooling these data with the WHI 1998 (oestrogen-only HT arm) data resulted in a relative risk at three years of 1.46 (95% CI 1.02 to 2.09s).

None of the other trials found any statistically significant difference between HT and placebo for this outcome. As noted above, most of the relevant trials were small.

Transient ischaemic attack (TIA)

Relevant comparisons

This outcome was measured in relatively healthy women in two trials (EPAT 2001; PEPI 1995) with a total of four different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for two or three years.

This outcome was also measured in three trials (ESPRIT 2002; HERS 1998; WEST 2001) of women with cardiovascular disease, with a total of three different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from two to four years, with unblinded follow up to 6.8 years (HERS 1998).

Results

No statistically significant difference was found between HT and placebo for this outcome.

Stroke or transient ischaemic attack

One trial of relatively healthy women (WISDOM 2007) reported stroke or TIA as a combined outcome. At a median of one year follow-up there was no statistically significant difference for this outcome between women taking combined continuous HT and women taking placebo.

One trial (ERA 2000) of women with known coronary disease also reported stroke or TIA as a combined outcome. Oestrogen-only HT and combined continuous therapy were compared with placebo. No statistically significant difference was found between HT and placebo
for this outcome at 3.2 year's mean follow up.

Venous thrombo-embolism (pulmonary embolus or deep vein thrombosis)

Relevant comparisons

This outcome was measured in relatively healthy women in four trials (EPAT 2001; PEPI 1995; WHI 1998, WISDOM 2007) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to over seven years.

Venous thrombo-embolism was also measured in five trials of women with cardiovascular disease (ERA 2000; ESPRIT 2002; EVTET 2000; HERS 1998; WAVE 2002) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one to four years, with unblinded follow up to 6.8 years (HERS 1998).

Results

In WHI 1998 (*oestrogen-only HT arm*), relatively healthy women taking oestrogen-only HT (CEE 0.625 mg) were at a higher risk of a thrombo-embolic event than women taking placebo. The risk was highest within the first two years and was statistically significant in this time period (RR 2.22 (95% CI 1.12 to 4.39)) At a mean follow up of seven years, the risk was lower but the intervention group was still at a higher risk that bordered on statistical significance (RR 1.32 (95% CI 1.00 to 1.74)).

In WHI 1998 (*combined HT arm)*, relatively healthy women taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) were at a significantly higher risk of a thrombo-embolic event than women taking placebo; this applied at one to over five years' follow up (at one year: RR 3.59 (95% CI 1.95 to 6.61), at two years: RR 2.98 (95% CI 1.88 to 4.71), at three years: RR 2.54 (95% CI 1.73 to 3.72), at four years: RR 2.34 (95% CI 1.69 to 3.25)), at a mean follow up of 5.6 years: RR 2.09 (95% CI 1.60 to 2.74)). Analysis of time trends in this comparison found a statistically significant time trend for a diminishing risk of venous thrombo-embolism over time. Data for this outcome at one year were also reported by WISDOM 2007; pooling these with the WHI 1998(oestrogen-only HT arm) data resulted in a relative risk at one year of RR 4.28 (95% CI 2.49 to 7.34).

Similarly, in HERS 1998, women with cardiovascular disease taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) for one to four years were significantly more likely to experience a venous thrombo-embolism than women on placebo (at one year: RR 3.26 (95% CI 1.06 to 9.96), at two years: RR 3.51 (95% CI 1.42 to 8.66), at three years: RR 3.01 (95% CI 1.50 to 6.04), at mean of 4.1 years: RR 2.62 (95% CI 1.39 to 4.94)).

None of the other trials found any statistically significant difference between HT and placebo for this outcome, though WHI 1998 *(oestrogen-only HT arm)* reported a non-statistically-significant increased risk in the HT group at 6.8 years follow up (RR 1.32 (95% CI 0.99 to 1.77)).

Breast cancer

Relevant comparisons

This outcome was measured in relatively healthy women in six trials (EPHT 2006; EPAT 2001; Notelovitz 2002; PEPI 1995; WHI 1998; WISDOM 2007) with a total of eight different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to over seven years.

This outcome was also measured in women with cardiovascular disease in four trials (ERA 2000; ESPRIT 2002; HERS 1998; WAVE 2002) with a total of four different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from two to four years, with unblinded follow up to 7.1 years (HERS 1998).

Results

In WHI 1998 *(oestrogen-only HT arm*), among relatively healthy women taking oestrogen-only HT (CEE 0.625 mg) there was a non-statistically-significant decrease in the risk of breast cancer at 7.1 years' follow up, compared with women taking placebo (RR 0.83 (95% CI 0.65 to 1.03)). There were no statistically significant differences between the groups.
WHI 1998 *(combined HT arm)* reported this outcome at yearly intervals and found no statistically significant difference between the groups in the incidence of breast cancer during the first four years of follow-up but the HT group were at a significantly higher risk of breast cancer after taking HT for five or more years (at mean of 5.6 years: RR 1.26 (95% CI 1.02 to 1.56). Analysis of time trends in this arm of WHI 1998 found a statistically significant trend for increasing breast cancer risk over time in the group taking HT. Data for this outcome at a median follow up of one year were also reported by WISDOM 2007. Pooling these data with the WHI 1998(*combined HT arm*) data resulted in a significantly reduced risk of breast cancer in the intervention arm (RR 0.53 (95% CI 0.28 to 0.96)).

No statistically significant difference was shown between any other type of HT and placebo for this outcome, though (as noted above) the relevant trials were small.

Colorectal cancer

Relevant comparisons

This outcome was measured in five trials (EPAT 2001; HERS 1998; PEPI 1995; WHI 1998, WISDOM 2007) with a total of five different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to almost seven years.

Results

WHI 1998 (*combined HT arm*) reported that among relatively healthy women taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) there was no statistically significant difference in the incidence of colorectal cancer, compared with women taking placebo, at one to four years' follow-up. However, women taking combined continuous HT had a significantly lower incidence of colon cancer after five or more years (at mean of 5.6 years: RR 0.62 (95% CI 0.43 to 0.89)).

No statistically significant difference was shown between any other type of HT and placebo for this outcome. However the relevant trials were small.

Endometrial cancer

Relevant comparisons

This outcome was measured in seven trials (EPAT 2001; ESPRIT 2002; Ferenczy 2002; HERS 1998; Obel 1993; PEPI 1995; WHI 1998) with a total of eleven different interventions, comprising comparisons of oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to over five years.

Results

No statistically significant difference was found between HT and placebo for this outcome. This finding included comparisons of oestrogen-only HT versus placebo (EPAT 2001; ESPRIT 2002; PEPI 1995); however all women with a uterus in these trials had close monitoring for endometrial hyperplasia and two trials specified that study medications were withdrawn if atypical hyperplasia wasdetected (ESPRIT 2002; PEPI 1995).

Recurrent endometrial cancer

This outcome was measured in the trial of oestrogen-only HT versus placebo in women who had undergone surgery for stage I or II endometrial cancer (Barakat 2006). No statistically significant difference was found between the groups.

Ovarian cancer

Ovarian cancer incidence was reported only in WHI 1998 *(combined HT arm)*, which used combined continuous CEE 0.625 mg + MPA 2.5 mg and has 5.6 years' follow up for this outcome. No statistically significant difference was shown between the groups.

Gallbladder disease requiring surgery

Relevant comparisons

This outcome was reported in four trials (ERA 2000; HERS 1998; PEPI 1995; WHI 1998) which compared oestrogen-only HT, combined continuous HT and sequential combined HT with placebo for three to seven years. For this outcome, the two largest studies stated that they excluded from analysis women who had had their gallbladder removed (HERS 1998) or had a history of gallbladder disease or both (WHI 1998).

Results

Meta-analysis of the three trials comparing oestrogen-only HT with placebo for this outcome (ERA 2000; PEPI 1995; WHI 1998) showed a statistically significant increase in risk in the HT group (RR 1.72, 95% CI 1.40-2.19). Meta-analysis of the four trials comparing combined continuous HT with placebo (ERA 2000; HERS 1998; PEPI 1995; WHI 1998) also showed significantly increased risk in the HT group (RR 1.55, 95% CI 1.29 to 1.86). Although these trials had differing lengths of follow up no statistical heterogeneity was observed. Similarly, the unblinded follow up of HERS 1998 reported an increase in events in the HT group which reached the borderline statistical significance (RR 1.63, 95% CI 1.00 to 2.70).

WHI 1998 investigators reported that the hazard estimates for risk in the active and placebo groups started to diverge in the first year of follow-up, with the oestrogen group separating earlier than the combined continuous HT group.

Hip fractures

Relevant comparisons

The incidence of hip fractures was reported in four trials (HERS 1998; WEST 2001; WHI 1998; WISDOM 2007). These compared combined continuous HT (HERS 1998; WHI 1998; WISDOM 2007)and oestrogen-only HT (WEST 2001; WHI 1998) with placebo for between one and 7.1 years.

Results

Both arms of WHI 1998 found a statistically significant reduction in the risk of hip fracture for women taking HT. WHI 1998 *(oestrogen-only HT arm)* reported a statistically significant reduction in the risk of hip fracture for women taking HT (CEE 0.625 mg) at 7.1 years' mean follow up (RR 0.64 (95% CI 0.45 to 0.93). WHI 1998 *(combined HT arm)* reported this outcome at yearly intervals and found no statistically significant difference in the incidence of hip fractures during the first four years of follow up, but at 5.6 years' mean follow up there was a statistically significant reduction in the risk of hip fracture for women taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) (RR 0.68 (95% CI 0.48 to 0.97)).

However, HERS 1998 found no statistically significant difference between combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) and placebo for this outcome, and the unblinded extension of this trial showed an statistically significant *increased* risk to the group taking HT from years 4.1 to 6.8 (post randomisation) (RR 2.10 (95% CI 1.06 to 4.16)).

Clinical vertebral fractures

The incidence of vertebral fractures was reported only in WHI 1998. In WHI 1998 *(oestrogen-only HT arm)*, at a mean of 6.8 years' follow-up there were significantly fewer fractures in the group taking oestrogen-only HT (CEE 0.625 mg) than in the group taking placebo (RR 0.62 (95% CI 0.42 to 0.93)). Similarly, in WHI 1998 *(combined HT arm)*, at a mean of 5.6 years' follow-up there were significantly fewer fractures in the group taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) than in the group taking placebo (RR 0.68 (95% CI 0.48 to 0.97).

Any fracture

Relevant comparisons

The incidence of any fracture was reported in six trials (EPHT 2006; ERA 2000; ESPRIT 2002; HERS 1998; WHI 1998; WISDOM 2007) comprising comparisons of oestrogen-only HT and combined continuous HT versus placebo for one to over seven years.

Results

Both arms of WHI 1998 found a statistically significant reduction in the risk of any fracture for women taking HT. This was reported at 5.6 years' mean follow up in women taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) (RR 0.78 (95% CI 0.71 to 0.85)) and at 7.1 years' mean follow -up in women taking oestrogen-only HT

(CEE 0.625 mg) (RR 0.73 (95% CI 0.65 to 0.80)). None of the other trials found any statistically significant difference between HT and placebo for this outcome.

Global cognitive function

Relevant comparisons

This outcome was reported by two studies (WHI 1998 *(WHIMS)*, Yaffe 2006), which compared low dose oestrogen patches versus placebo for two years, oestrogen-only HT versus placebo for a mean of 5.2 years and combined continuous CEE 0.625 mg + MPA 2.5 mg versus placebo for a mean of 4.2 years. Global cognitive function was measured using a cognitive screening test known as the Modified Mini-Mental State Examination (3MSE), in which a higher score reflects better cognitive functioning. WHI 1998 (*WHIMS*) included only women over 65 years of age and Yaffe 2006 included only women aged over 60 years.

Results

Over two years of follow up, there was no difference in cognitive function between women using low dose oestrogen patches (oestradiol 0.014 mg) and women using placebo patches (Yaffe 2006). Nor was there any difference in the effect of treatment when women in this study were stratified according to their cognitive status at baseline (Minimental State Examination (3MSE) ≤ 90 or > 90).

In both treatment groups and in both placebo groups of WHI 1998 (*WHIMS*), mean 3MSE scores increased from baseline and continued to increase for three to five years before starting to decline. There was a pattern of higher increases from baseline in 3MSE scores in the placebo groups which emerged after one to two years and was maintained throughout the study. The mean difference between the groups in 3MSE score changes was of borderline statistical significance in both arms of the study, with results favouring the placebo group: however in both cases the lower boundary of the confidence interval was zero (*oestrogen-only HT arm*: WMD -0.25 (95% CI -0.52 to 0.00); *combined HT arm*: WMD -0.18 (95% CI -0.35 to 0.00)).

In the WHI 1998 *(WHIMS)* combined HT arm, a decline of 10 points or more in 3MSE scores (which represents > 2 standard deviations from the baseline mean scores) was significantly more likely to occur among women in the active treatment group (RR 1.57 (95% CI 1.10 to 2.24)). The same trend was observed in the oestrogen-only HT group, but was not of statistical significance.

Specific cognitive functions

Relevant comparisons

This outcome was reported by WHI 1998 (WHISCA), which included only women aged over 65 and free of probable dementia. WHI 1998 (WHISCA) reported annual rates of change in specific cognitive functions and compared combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) versus placebo, adjusted for the time since randomisation into WHI 1998.

Results:

Among women in the combined continuous HT group (CEE 0.625 mg + MPA 2.5 mg), there was evidence that HT had a negative impact on verbal memory (P ≤ 0.01) and a non-significant trend to a positive impact on figural memory (P = 0.012) over time, compared to placebo. Both these effects were evidence only after long term therapy. There was no significant difference between the groups on other cognitive domains.

Mild cognitive impairment

This outcome was reported by WHI 1998 *(WHIMS 2003)*, which included only women over 65 years of age and compared oestrogen-only HT versus placebo for a mean of 5.2 years, and combined continuous CEE 0.625 mg + MPA 2.5 mg versus placebo for a mean of 4.2 years. No statistically significant difference was found between the groups in either arm.

Probable dementia

Relevant comparisons

This outcome was reported by WHI 1998 *(WHIMS)*, which included only women over 65 years of age and compared oestrogen-only HT (CEE 0.625 mg) versus placebo for a mean of 5.2 years, and combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) versus placebo for a mean of 4.2 years.

Results

In the oestrogen-only HT arm, no statistically significant difference was found between the groups. In the combined HT arm, the incidence of probable dementia was significantly higher in the group taking combined continuous HT than in the placebo group (RR 1.97 (95% CI 1.16 to 3.33)).

Mild cognitive impairment or dementia

Relevant comparisons

This composite endpoint was reported by WHI 1998 *(WHIMS)*, which included only women over 65 years of age and compared oestrogen-only HT (CEE 0.625 mg) versus placebo for a mean of 5.2 years, and combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) versus placebo for a mean of 4.2 years.

Results

In the oestrogen-only HT arm, the incidence of mild cognitive impairment or dementia was significantly higher in the group taking oestrogen-only HT than in the placebo group (RR 1.36 (95% CI 1.01 to 1.84). In the combined HT arm, there was no statistically significant difference between the two groups.

Quality of life

Relevant comparisons

This outcome was reported by four studies (Haines 2003; Obel 1993; WHI 1998; Yaffe 2006) which compared oestrogen patches, oestrogen-only HT, combined continuous HT and combined sequential HT versus placebo for varying durations from one year to three years. The two smallest studies (Haines 2003; Obel 1993) had only 30 to 50 women in each active treatment group. The largest (WHI 1998) did not include women with menopausal symptoms sufficiently severe to preclude randomisation, for whom this outcome may be considered most relevant.

Results

Over two years of follow up, there was no difference in health-related quality of life between women using low dose oestrogen patches (oestradiol 0.014 mg) and women using placebo patches (Yaffe 2006)

At one year, women taking oestrogen-only HT (CEE 0.625 mg) reported a slightly greater improvement in sleep disturbance than women in the placebo group which was statistically significant (mean benefit, 0.4 points on a 20 point scale). A subgroup (N = 1189) measured at three years reported no statistically significant benefit for any quality of life-related outcomes (WHI 1998 *oestrogen-only HT arm*). Haines 2003 found no statistically significant difference in overall quality of life change scores between women taking oestrogen-only HT (at moderate or moderate/high doses) and women taking placebo. The measure used was a modified version of WHO-QOL (World Health Organisation Quality of Life rating scale)

At one year, women taking combined continuous HT (CEE 0.625 mg + MPA 2.5 mg) reported a significant difference in quality of life change scores for two out of eight categories in the RAND 36 survey: these two categories were *physical functioning* (WMD 0.80 (95% CI 0.36 to 1.24)) and *role limitations due to physical problems* (RR 1.40 (95% CI 0.30 to 2.50)). However after taking HT for three years there was no statistically significant difference between the groups in any categories (WHI 1998 *(combined HT arm)*).

Obel 1993 reported overall quality of life scores but did not report change scores or their standard deviations. Results for this outcome are reported in Analysis 1.48. There was no statistically significant difference between the active and placebo groups for this outcome.

HT for women hospitalised with chronic illness

One small study (Nachtigall 1979) included women hospitalised either for chronic disease or because they required for custodial care. It comprised a comparison of combined sequential HT for ten years versus placebo. The following outcomes were measured: all cause death, myocardial infarction, venous thrombo-embolism, breast cancer, colorectal cancer and endometrial cancer. No statistically significant difference was found between the groups for any of these outcomes.

HT for women with dementia

One small study (Mulnard 2000) included women with mild to moderate Alzheimer's disease. It comprised a comparison of unopposed oestrogen for one year versus placebo and the primary outcome was change in overall status with relation to Alzheimer's disease, as measured by the Clinical Global Impression of Change scale. No statistically significant difference was found between the groups.

DISCUSSION

All of the statistically significant findings of this review derived from the two biggest trials, HERS 1998 and WHI 1998. These trials evaluated oral CEE 0.625 mg, with or without continuous methoxy progesterone (MPA 2.5 mg). Smaller trials using other types of HT reported very few or no major clinical events. We were generally unable to combine results from individual trials, either because they used different types of HT which may not be equivalent in effect or because they differed with respect to the trial population or both.

Other Cochrane reviews have found strong evidence that HT is effective in treating menopausal symptoms. One review reported a 75% reduction in the frequency of hot flushes for perimenopausal and postmenopausal women taking HT relative to placebo and also a statistically significant reduction in symptom severity for the HT group (OR 0.13, 95% CI 0.07 to 0.23) (MacLennan 2004). Another review found that local oestrogens were more effective at relieving the symptoms of vaginal atrophy in postmenopausal women than placebo or non-hormonal gel (Suckling 2006). However, women contemplating the use of HT for menopausal symptoms need to be aware of less positive findings in other areas, as discussed below.

Cardiovascular disease

There is no evidence that HT has a role in either the treatment or the prevention of cardiovascular disease. On the contrary, HT significantly increases the incidence of stroke and venous thromboembolism. Combined continuous HT also significantly increases the risk of coronary events, with the increased risk becoming evident during the first year of use. Oestrogen-only HT failed to have any statistically significant effect (either positive or negative) on coronary disease (WHI 1998 *(oestrogen-only HT arm)*).

It has been suggested, on the basis of primate experiments, that HT may reduce atherosclerosis if it is given either to those with little or no coronary artery disease, or started immediately after the menopause (Mikkola 2002). WHI 1998 *(combined HT arm)* conducted many pre-specified subgroup analyses to evaluate whether any clinical characteristics of the study population might plausibly modulate the coronary effects of HT: variables included age, time since the menopause, presence or absence of vasomotor symptoms, prior hormone use, CHD risk factor status and presence or absence of pre-existing cardiovascular disease. However none of these variables was found to significantly affect results.

The WHI 1998 investigators also conducted a nested case-control study investigating the relationship between cardiovascular bio markers, treatment assignment and risk of CHD: this included 205 women who experienced coronary events during the first three years of the study versus 513 matched controls. Overall, only one subgroup of women was found to have a significantly different relative risk of coronary heart disease from that observed for all women: those in the HT arm with a higher baseline LDL cholesterol were found to be at significantly increased excess risk. The authors commented that in view of the number of subgroup analyses performed, this may have been a chance finding. They also noted that many of the subgroups were small with limited statistical power and that their findings should thus be interpreted with caution (Manson 2003).

With respect to venous thromboembolism, subgroup analysis in WHI 1998 *(combined HT arm)* found that women in the HT arm who had factor V Leiden mutation, a blood coagulation disorder, were at higher risk than other women in the HT arm (Cushman 2004). Among women with a history of venous thromboembolism the HT group had a high event rate compared to the placebo group (7/79 versus 1/62), but there was insufficient statistical power to determine whether there was a significant excess risk in women with a history of venous thromboembolism taking HT, compared to other women in the HT arm. There was also a higher incidence of events among older and obese women taking HT, though this was related to their higher baseline risk of an event and their *relative* risk did not differ from other women in the HT arm.

In the oestrogen-only arm of WHI 1998 there was a suggestion that women in the intervention group aged 50-59 years were at lower risk of coronary heart disease, with 21 CHD events in the intervention group and 34 in the placebo group over 7.1 years (HR 0.63%, 95% CI 0.35 to 0.86). However the overall the data did not support a relationship between time since menopause and coronary risk. The trialists note that the trial may have been underpowered to demonstrate a significant difference in CHD risk in younger women, but because of the low event rate in this age group a study of over 17,000 women would be likely to be needed (Hsia 2006).

An increase in the risk of coronary events and venous thrombosis in women in the HT group was evident in the first year of treatment in women taking combined continuous HT in HERS 1998 and WHI 1998. Although there was a significant trend in both arms of WHI 1998 and in the blinded phase of HERS 1998 for cardiovascular risk in the HT group to diminish over time, subsequent analysis of HERS 1998 data which included both blinded and unblinded follow-up showed no statistically significant variation in risk over time. The WHI 1998 investigators suggest that the apparent decline in cardiovascular risk in later years may be due to an acceleration of events in earlier years among susceptible women in the HT group and point out also that with longer duration of treatment the risk of breast cancer is increased.

Breast cancer

Evidence about the effect of HT on the incidence of breast cancer is difficult to interpret. In WHI 1998 *(combined HT arm)* the breast cancer rates in the HT group were initially lower than in the placebo group and when WHI 1998 data were combined with WISDOM 2007 data at a mean or median follow up of one year the difference reached statistical significance, favouring the intervention group. However by the fourth year of use there were more events in the HT group, with a statistically significant trend for increasing risk over time. The investigators commented that breast cancers in the HT group were diagnosed at a similar grade but a more advanced stage at the time of diagnosis and suggested that combined HT may stimulate breast cancer growth but delay diagnosis, possibly by hindering mammographic detection (Chlebowski 2004). This may explain why there was a lower incidence of breast cancer among women in the HT group during the first two years of WHI 1998 *(combined HT arm)*. Subgroup analyses of prior hormone use in WHI 1998 *(combined HT arm)* found that the cumulative incidence of breast cancer over time in the intervention group increased at a higher rate than in the placebo group after about three years in prior hormone users and after about five years in women with no prior use. It was not possible to define with any reliability a timeframe for the use of combined HT (Anderson 2006). In contrast, a *decrease* in the risk of breast cancer, narrowly missing statistical significance, was found in the unopposed oestrogen arm of WHI 1998. Subgroup analyses showed significantly fewer early cancers as well as significantly fewer ductal carcinomas in the intervention group, whereas the incidence of lobular tumours did not differ significantly. The authors state that cross-study differences in the two arms of the study do not explain the differences in breast cancer incidence between the two arms and suggest that the increased risk in the combined group is due to progestin. Similar trends in other studies (HERS 1998; Beral 2003) support this theory.

The proportion of women needing repeat mammograms was significantly increased in the intervention group in both study arms. However in the combined arm the increase was in mammograms suggestive of abnormality or malignancy, whereas in the oestrogen-only arm the increase was in "recommended short interval" follow ups (Stephanick 2006). A recent review of breast biopsies conducted during the oestrogen-only arm of WHI reported a significantly increased incidence of benign proliferative breast disease in the intervention arm. The authors note that longer follow-up should clarify whether this could translate into an increased risk of breast cancer in this group (Rohan 2008).

Gynaecological cancers

None of the trials showed an increase in the incidence of endometrial cancer in the group taking HT. In three studies women with a uterus were randomised to oestrogen-only treatment (EPAT 2001; ESPRIT 2002; PEPI 1995). As endometrial cancer is well documented as an adverse effect of unopposed oestrogen (Kurman 1985), these women were closely monitored for atypical endometrial hyperplasia and received interventional treatment with discontinuation of study medications if it was detected. PEPI 1995 reported that women in the oestrogen-only HT group were significantly more likely to develop atypical endometrial hyperplasia than women in the placebo group, whereas women in the combined HT groups in the same study showed no increased risk of hyperplasia.

The trial of oestrogen-only therapy in women who had undergone surgery for stage I or II endometrial cancer was underpowered due to early discontinuation and could not conclusively refute or support the safety of the therapy with regard to the risk of recurrence. The authors note that recurrence rates were low, at 1.9% in the placebo group and 2.3% in the intervention group (Barakat 2006).

There was a trend towards an increased risk of ovarian cancer in WHI 1998 *(combined HT arm)* (RR 1.59, 95% CI 0.78 to 3.25) which did not reach statistical significance (Anderson 2003). As noted above, a systematic review of (mainly) observational studies (Greiser 2006) suggests that both oestrogen-only and combined therapy may be associated with an increased risk of ovarian cancer.

Among women with a history of ovarian cancer, a randomised study with four year follow-up of 130 women (Guidozzi 1999) found that oestrogen-only hormone therapy did not negatively affect the disease-free or overall survival time, compared with no hormone therapy. This study was not included in the present systematic review because it lacked a placebo control group.

Cognitive outcomes

With regard to cognitive outcomes, WHI 1998 *(WHIMS)* found that neither combined HT nor oestrogen-only HT conferred any benefit in global cognitive function for women aged over 65, nor did either active treatment confer any reduction in the risk of being diagnosed with mild cognitive impairment. The ancillary study WHI 1998 (*WHISCA*) noted that the effect of combined continuous HT varies across cognitive domains in this age group and reflects possible detrimental and beneficial actions on the aging brain. As noted above, this ancillary study was terminated after only 1.3 years of assessments and beneficial effects of HT did not reach statistical significance (Resnick 2006).

The rise in mean 3MSE scores (used to measure global cognitive function) that occurred in all participant groups over the first few years of WHIMS was attributed by the investigators to a learning effect known to result from repeated administration of cognitive tests (Espeland 2004). The difference in mean scores between the active therapy and placebo groups was of borderline statistical significance and consistently favoured the placebo groups, though this difference was too small to be clinically meaningful. However a *marked* decrease in 3MSE scores (defined as > 2 standard deviations from the baseline mean scores) occurred more frequently in the active treatment groups, and this trend reached statistical significance in the combined HT group. Moreover in both arms HT had a relatively greater adverse effect in women whose baseline 3MSE scores were lowest (Espeland 2004).

Similarly, for the outcome of probable dementia there was a negative trend in both active treatment groups which reached statistical significance in the combined HT group. Evidence of increased risk in this group began to appear as early as one year after randomisation and persisted over five years of follow up. The overall risk to women taking combined HT was twice that of women in the corresponding placebo group. The investigators noted however that the absolute risk of dementia remained relatively small, at 45 per 10,000 postmenopausal women aged over 65 years who took combined HT for one year (Shumaker 2004).

The WHI 1998 *(WHIMS)* results were unexpected and in striking contrast to most earlier research. The investigators suggested that this might be due to the healthy user bias in observational studies, whereby HT users had a better prognosis at baseline than the control groups. They also conjectured that it might be due to differential effects of HT on specific domains of cognition not measured individually by 3MSE, or that there might be a critical period, such as menopause, during which HT must be initiated in order to protect cognitive function at a later age. The mean age of the WHIMS population was 71 and so the study could not address this theory, although previous users of HT in WHIMS did not have higher scores (Espeland 2004). The results to date of WHI 1998 (WHISCA) suggest that the effect of combined HT on cognitive function may vary across cognitive domains in older women (Resnick 2006); the findings of the oestrogen-only arm of this study are pending.

Gallbladder disease

A statistically significant association between HT and gallbladder disease was found, with excess risk related to both oestrogen-only and combined continuous HT. Although most of the statistical power for this outcome derived from WHI 1998, the findings with respect to combined continuous HT were strongly supported by data from both the blinded and the unblinded follow-up in HERS 1998. The WHI 1998 investigators noted that the risk started to increase in the active group in the first year, and appeared to increase over time. They calculated that for one excess occurrence of gallbladder disease to happen, 323 women would need to take oestrogen-only HT or 500 women would need to take combined continuous HT for a year.

Quality of life

With respect to quality of life, WHI 1998 found that neither oestrogen-only nor combined continuous HT gave any clinically meaningful benefit at either one or three years. In both arms, subgroup analyses were conducted at one year of women who reported moderate to severe vasomotor symptoms at baseline: in each group one analysis included all women who reported such symptoms (1830 in the oestrogen-only and 2046 in the combined HT arms) and one was restricted to women aged 50 to 54. Although combined continuous HT significantly improved the severity of hot flushes and night sweats compared to placebo, results for general quality of life measures did not have a clinically meaningful effect on health-related quality of life among these subgroups. As mentioned above, these findings are unlikely to be applicable to women taking HT specifically for vasomotor or other menopausal symptoms affecting their quality of life since women with severe menopausal symptoms precluding randomisation were excluded from WHI 1998.

Possible benefits

HT offers the benefit of a significant reduction in the risk of fracture or colorectal cancer, as discussed below. However these risk reductions only become statistically significant after four or five years' treatment with HT, while the highest risk of cardiovascular events with combined HT occurs in the first year of use.

Fractures

Most women who require treatment for low bone mineral density require it lifelong and the benefits of HT will generally be outweighed by the ongoing and cumulative risk of cardiovascular disease or breast cancer. HT should only be considered for the prevention of fractures if other treatments are contraindicated and if the cardiovascular risk is low. WHI 1998 *(combined HT arm)* investigators tested the hypothesis that the beneficial effect of HT on fracture incidence differed according to fracture risk factors. They found that the reduction in risk given by HT was no greater in women at high risk of fracture (Cauley 2003). However, women with severe osteoporosis were excluded from WHI 1998 and bone mineral density was not routinely collected: thus the benefits of HT may outweigh the risks for some women with severe osteoporosis.

However the evidence on HT and fractures is not consistent. WHI 1998 found a significantly reduced risk of fractures in women taking combined continuous HT or oestrogen-only HT for five years or more but HERS 1998 reported no benefit for women on continuous combined HT. Moreover the unblinded continuation of HERS 1998 found a significantly *increased* risk of hip fracture for such women. The authors attributed this finding to chance, noting that the effect was considerably smaller in the as-treated analysis and that such a finding lacks biological plausibility (Hulley 2004). Overall, although HT is considered effective for the prevention of postmenopausal osteoporosis, other treatment options may be safer for some women and HT is generally recommended as an option only for women at significant risk for whom non-oestrogen therapies are unsuitable (Cranney 2002; NIH 2004).

Colorectal cancer

With respect to colorectal cancer, the significantly reduced incidence in women taking combined continuous HT in WHI 1998 (*combined HT arm)* was offset by the finding that colorectal cancers diagnosed in such women tended to be more advanced with more likelihood of lymphatic or metastatic involvement. The investigators suggested that women taking combined HT might benefit from routine bowel screening, despite their reduction in overall risk of colorectal cancer (Chlebowski 2004).

HT for younger women

It is important to consider any increased risk to health in absolute rather than relative terms. This review is inevitably dominated by the findings of WHI 1998, which was designed to evaluate the efficacy of HT in preventing the major causes of morbidity and mortality in older women (Matthews 1997). It was *not* designed to evaluate the risks and benefits of hormone therapy for the treatment of menopausal symptoms, and specifically excluded any woman who reported menopausal symptoms severe enough to preclude her being assigned to placebo treatment (Anderson 2003). It should also be emphasised that WHI 1998 did not include women under 50 and its findings may not apply to young surgically menopausal women - for example a woman who has had both ovaries removed in her forties (Kaunitz 2002).

A woman considering the use of HT for vasomotor symptoms is likely to be in her early fifties. For most women in their fifties the absolute risk of a life-threatening event is low (it has been estimated that absolute risk for many diseases approximately doubles with each decade of age (Hulley 2004)). Subgroup analyses of women aged 50 to 59 in WHI 1998 *(combined HT arm)* found that for relatively healthy women taking combined continuous HT the only increase in risk which reached statistical significance was for venous thrombosis. The risk in the HT group increased from eight venous thromboses per 10,000 women per year to 19 per 10,000 women per year. The increase in risk was highest in the first year of therapy, but continued over five years of treatment, and was particularly high in obese women (that is, women with a body mass index of over 30), who had a five year risk of 1.4% compared to 0.5% in normal weight women. Preliminary subgroup analyses of women in their fifties in WHI 1998 *(oestrogen-only HT arm)* found no statistically significant difference in risk for any outcome. There was even a suggestion of benefit from oestrogen-only HT for some outcomes such as coronary heart disease and breast cancer, though the authors recommended caution in interpreting this finding as they could not exclude the role of chance or limited study power (WHI 2004). However it is important to note that oestrogen-only HT is contra-indicated for women with an intact uterus, as use from one to five years has been estimated to increase the risk of endometrial cancer threefold (from a baseline lifetime risk of about 3% for a woman of 50), with effects persisting for several years after oestrogen is stopped (Grady 1995). A recent analysis which combined data from both arms of WHI 1998 found a statistically non-significant trend for women who initiated HT closer to menopause to have reduced CHD risk and also reduced mortality risk, but the elevated risk of stroke did not vary significantly by age or by length of time since menopause (Rossouw 2007).

A clinical decision model (Col 2004) has evaluated the effect of two years' combined continuous HT on life expectancy and *quality-adjusted* life expectancy for a hypothetical 50 year old menopausal woman with an intact uterus. The authors used findings from WHI 1998 and from reported utility scores for menopausal symptoms - meaning the duration of lifespan that symptomatic women would be prepared to trade off for an assurance of a shorter lifespan without menopausal symptoms. Taking into account the impact of HT on chronic disease outcomes as well as its utility value associated with menopausal symptoms, asymptomatic women taking HT for two years experienced a net loss in quality-adjusted life expectancy of one to three months, depending on their underlying risk of cardiovascular disease, while women with severe menopausal symptoms gained seven to eight months. Among women at low risk of cardiovascular disease, two years' HT was associated with a three month gain in quality-adjusted life expectancy even when menopausal symptoms were mild. However this model did not include cognitive outcomes, as WHI 1998 findings on this outcome pertain only to women over 65 and there are no equivalent data for younger women. In clinical practice a decision analysis needs to take into account individual baseline risks and the utility value that a woman ascribes to her own menopausal symptoms (Minelli 2004).There is currently a trend towards the use of low dose HT taken for the shortest possible time required to achieve treatment goals such as the relief of hot flushes, with doses individually tailored and reviewed at least annually (Grady 2003; Kaunitz 2002; MHRA 2003; NIH 2004).

Health benefits and risks after stopping combined HT

WHI 1998 (*combined HT arm*) recently reported health outcomes at a mean of 2.4 years' follow up after the intervention was stopped (Heiss 2008). Follow-up data for this period were available for 95% of women, of whom only a small minority were using hormone therapy during post-intervention follow up (4.3% in the intervention arm and 1.2% in the placebo arm at one year after the trial was stopped). Over the course of a mean 2.4 years follow up the risk of coronary events, stroke and venous-thromboembolism decreased in the group that had been randomised to combined HT and reached a level comparable with the placebo group. Similarly, there was no significant difference between the groups in the risk of fractures or of colorectal cancer by the end of post-intervention follow-up. However, in the group that had been randomised to combined HT, the hazard ratio (HR) for the outcome "all cancer" increased from 1.03 (95% CI 0.92 to 1.15) during the intervention phase to 1.24 (95% CI 1.04 to 1.48) in the post-intervention period. The increase in risk was attributable partly to the disappearance of the previous apparent protection from colorectal cancer, with some continued excess risk of breast cancer, and also some added risk of lung cancer in the HT group. A downward inflection in the temporal trend in cumulative HRs for breast cancer was observed over time, but the observed change in the HR after the intervention was not statistically significant. The authors note that clinical vigilance appears to be warranted with regard to the sustained risk of higher malignancies following termination of combined HT therapy.

Questions about the evidence

There is controversy over the degree to which the findings of WHI 1998 apply to any type of HT other than continuous combined oral CEE 0.625 mg with or without MPA 2.5 mg. There is also controversy as to the population to which the available evidence applies. For cardiovascular outcomes, the results of HERS 1998 largely support the results of WHI 1998 *(combined HT arm)*, suggesting that their findings can be generalised to older women taking combined continuous HT whether or not they have known cardiovascular risk factors (although their findings differ with respect to fracture risk).

However there is little evidence on the long term effects of HT on the healthy younger women who are most likely to use it for menopausal symptoms and nor is much known about factors that may modulate the risks involved, such as clinical characteristics or bio markers affecting individual women, different oestrogens and progestagen's, different time frames for the use of HRT, and different doses and routes of administration (e.g. unopposed oestrogen and intrauterine progestogen). There is observational evidence that transdermal oestrogen differs from oral oestrogen in that it is not associated with an increased risk of venous thromboembolism, and there is also a suggestion that some types of progestogen are thrombogenic while others are safe in this respect (Canonico 2007).

HT appears to carry an increased risk of recurrence for women with a history of breast cancer. Two unblinded trials have been conducted in Sweden which randomised breast cancer survivors with menopausal symptoms to HT or non-hormonal treatment. Both were terminated early due to a statistically significant increase in the incidence of recurrent breast cancer in the hormonal group in one of the trials (RR 3.5, 95% CI 1.5 to 8.1) (Chlebowski 2004; Holmberg 2004). After a median of four years' follow up in this trial, there was still a clinically and statistically significant increased risk of a new breast cancer event in the HT arm (RR 2.4, 95% CI 1.3 to 4.2)(Holmberg 2008). A similar trial initiated in the UK terminated recruitment prematurely in January 2004 (ICR 2001).

The results for some of the outcomes measured in this review have been influenced by safety considerations. WHI 1998 was halted more than three years early when an excess incidence of breast cancer in the HT group crossed a relatively conservative prespecified safety boundary and a "global index" balancing overall harms and benefits was also indicative of overall harm. EVTET 2000 was also terminated early due to emerging evidence of possible thrombo-embolic risk with HT, as well as a non-significant clustering of events in one group (which on un blinding was the HT group).

A high proportion of women in these studies did not receive the treatment to which they were randomised. In general the number of women who discontinued their medication or took less than 80% was disproportionately high in the HT groups, presumably because of a higher incidence of adverse effects such as vaginal bleeding. The authors of WHI 1998 note that if discontinuation of treatment and initiation of non-study treatment occurred independently of risk factors for clinical outcomes their intention-to-treat analysis underestimates both the harms and the benefits of HT among women who adhere to treatment (WHI 2002). In this trial there were also a disproportionate number of women unblinded in the HT group compared to the placebo group (40% versus 6%), primarily to manage persistent vaginal bleeding, and it has been suggested that this differential un blinding may have resulted in higher detection rates of otherwise undetectable myocardial infarction in the HT group (Shapiro 2003). However it has been suggested that detection bias on a scale to explain the differences between the groups for coronary heart disease could not have occurred - and that any bias was more likely to have been in the opposite direction, mitigating *against* the detection of effects (Tucker 2003).

The findings of this review differ somewhat from those of previous systematic reviews of HT for peri or postmenopausal women. The most notable difference from Beral 2002 was that the earlier review found no statistically significant increase in the risk of coronary heart disease among women taking HT, in contrast to the current review which found a significant increase, particularly in the first year, among women taking combined continuous HT. Unlike the current review, Beral 2002 pooled results from studies of differing participant groups and types of HT, and this appears to be the reason why the overall findings differ. In contrast to the review by Salpeter et al. (Salpeter 2004), no survival advantage was evident for women taking HT in the current review, though only one of the included trials (WHI 1998 (*oestrogen-only HT*)) analysed younger women as a subgroup for this outcome. However, of the 17 trials which Salpeter et al included in their meta-analysis of younger women, only two met the inclusion criteria for the present review: the other 15 in the earlier review were not blinded, did not report mortality as a primary or secondary outcome and/or were of less than one year's duration.

It is likely that the divergence between the findings of the randomised controlled studies included in this review and previous observational studies is primarily due to the methodological limitations of observational studies, including the "healthy woman effect", whereby women prescribed HT had more favourable prognostic factors at baseline than the comparison non-HT group.

AUTHORS' CONCLUSIONS

Implications for practice

HT for women with menopausal symptoms

Women who find menopausal symptoms intolerable and who are at low risk of cardiovascular disease or venous thrombo-embolism may wish to weigh the benefits of symptom relief against the small absolute risks of harm arising from short-term use. Although none of the trials included in this review focused specifically on women in the age group most likely to require menopausal symptom relief, WHI 1998 analysed a subgroup of 2839 relatively healthy 50 to 59 year old women taking combined continuous HT and 1637 taking oestrogen-only HT and their findings were relatively reassuring. The only significantly increased risk found in this subgroup analysis was an increased incidence of venous thrombo-embolism in those taking combined continuous HT. The absolute risk of venous thromboembolism in these women was low, at 0.5% overall for women taking HT for five years. However the risk increased to 1.4% for obese women (Cushman 2004). Additional risk factors such as a history of venous thrombosis or known increased risk of thromboembolic disease must also be considered when deciding whether likely symptom relief from HT outweighs potential harm. Moreover the risk of endometrial cancer for women with a uterus taking oestrogen-only HT is well documented.

HT for other indications

Hormone therapy may have a limited role in the treatment of osteoporosis for some women, but there is no evidence that any form of HT is beneficial for other clinical indications (except menopausal symptom relief) nor is it appropriate for the prevention of chronic disease. There is strong evidence that both oestrogen-only HT and combined therapy significantly increase the risk of stroke and gallbladder disease and that long-term use of combined continuous therapy also increases the risk of breast cancer and, in women over 65, the risk of dementia.

HT for women with previous disease

HT is not recommended for use in women with cardiovascular disease (HERS 1998) or with a history of venous thrombosis (EVTET 2000) or breast cancer (Chlebowski 2004; Holmberg 2008; ICR 2001). However, the randomised evidence provides no specific contraindications for its use in women with a history of endometrial cancer (Barakat 2006) or ovarian cancer (Guidozzi 1999), though data are scanty.

Implications for research

No studies have adequately assessed the safety of HT for symptom relief for younger women. And not much is about factors that may modulate the risks involved, such as clinical characteristics or bio markers affecting individual women, different oestrogens and progestogens, different time-frames for the use of HRT, and different doses and routes of administration (e.g. unopposed oestrogen and intrauterine progestogen). There is also a pressing need for reliable evidence on the efficacy and safety of alternatives to HT for the control of menopausal symptoms for those women who may wish to avoid using HT or for whom it is unsuitable.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Barakat 2006

Methods	Stated purpose: To determine the effect of ET on recurrence rate and survival in women who have undergone surgery for stage I or II endometrial cancer Randomisation method: Remotely generated Allocation method: Remotely dispensed Blinding: Patients and physicians blinded Stratification: Stratified by stage No of women screened for eligibility: No randomised: 1236 (see Notes) No analysed: 1236 Losses to follow up: None stated Dropouts: None stated Adherence to treatment: 41% in HT group, 50% in placebo group at trial end Analysis by intention to treat: Yes No of centres: Not stated Years of recruitment: June 1997 to January 2003 Design: Parallel Funding: National Cancer Institute grant
Participants	Included: Women post total hysterectomy and bilateral salpingo-oophorectomy (at least) for surgically staged stage I or II endometrial cancer within 20 weeks of entering study, with indication for use of ERT including hot flushes, vaginal atrophy, increased risk of CHD or increased risk of osteoporosis. Had to have undergone clinical exam with history, pelvic exam and chest X-Ray before study entry. Normal hepatic function and normal mammogram or negative breast biopsy within previous year. Excluded: Women with history or suspicion of breast cancer or other malignancies with exception of non-melanoma skin cancer. within past 5 years or with history of acute liver disease or thromboembolic disease Median age: 57 Age range: 26-91 Means of recruitment: Not stated Baseline equality of treatment groups: Well balanced Country: USA
Interventions	HT arm: 0.625 mg CEE (Unopposed oestrogen) Control arm: Placebo Duration: Planned for 3 years with 2 years' additional follow up. Closed early with median follow up 35.7 months
Outcomes	Total deaths CHD deaths Coronary event deaths Endometrial cancer deaths Endometrial cancer (recurrence)
Notes	Enrolment decreased after WHI published in July 2002. Study closed prematurely due to poor accrual. In addition, preponderance of participants had low risk profile so low event rate meant power unlikely to be reached with original power calculation. This study planned to enrol 2108 women. Numbers randomised not entirely clear: study refers to 1236 "eligible and assessable women".

Risk of bias

Barakat 2006 *(Continued)*

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Remotely generated
Allocation concealment?	Low risk	Remotely dispensed drugs
Blinding? All outcomes	Low risk	Patients and physicians blinded
Incomplete outcome data addressed? All outcomes	Low risk	No losses to follow up stated; analysed by intention to treat
Free of other bias?	Unclear risk	Planned for 3 years with 2 years' additional follow up. Closed early with median follow up 35.7 months

EPAT 2001

Methods	Stated purpose: To determine the effect of oestrogen-alone HT on the progression of subclinical atherosclerosis in healthy postmenopausal women without preexisting cardiovascular disease, as measured by changes in thickness of carotid artery wall Randomisation: Computer-generated random numbers Allocation: Blinded medication packets assigned sequentially and remotely after eligibility confirmed Stratification: By LDL cholesterol level (threshold <4.15 mmol/L), previous duration of HRT, (threshold < 5 years), and diabetes mellitus status. Blinding: Participants, gynecologists, clinical staff, and image analysts. The data monitor and data analyst were blinded to treatment assignment until analyses were completed. No of women screened for eligibility: 1161 prescreened by phone, 422 screened on site, of whom 52% randomised. No randomised: 222 No analysed: 222 for clinical outcomes Losses to follow up: 33 women were not evaluable for primary study endpoints, but clinical endpoints were reported for all Adherence to treatment in evaluable women: During the trial, mean pill adherence was 95% in the estradiol group and 92% in the placebo group (P = 0.08). Analysis by intention to treat: Yes No of centres: One Years of recruitment: 1994-8 Design: Parallel Funding: National Institute on Aging
Participants	Included: Postmenopausal women aged>45 years, no pre-existing cardiovascular disease. LDL levels>3.37 mmol/litre Excluded: Women with previous breast or gynaecological cancer, frequent hot flushes, diastolic BP >110, uncontrolled diabetes or thyroid disease, abnormal bloods, smokers Mean age: 61.15 Age range: 51.4-69.2 Means of recruitment: Not stated Baseline equality of treatment groups: No significant differences in demographics or clinical variables Country: U.S.A.
Interventions	HT arm: Unopposed micronized 17B-oestradiol 1 mg daily Control arm: Placebo

EPAT 2001 *(Continued)*

	Duration: 2 years
Outcomes	Primary outcome: Carotid artery wall thickness on ultrasound) Myocardial infarction Cerebrovascular accident Transient ischaemic attack Deep vein thrombosis, Pulmonary embolism
Notes	Power calculation: sample size of 200 required to detect a treatment effect size (the difference in carotid artery wall thickness) of 0.40 or greater with 80% power.

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Computer-generated random numbers
Allocation concealment?	Low risk	Blinded medication packets assigned sequentially and remotely after eligibility confirmed
Blinding? All outcomes	Low risk	Participants, gynaecologists, clinical staff, and image analysts. The data monitor and data analyst were blinded to treatment assignment until analyses were completed.
Incomplete outcome data addressed? All outcomes	Low risk	33 women were not able to be evaluated for primary (physiological) study endpoints, but clinical endpoints were reported for all by intention to treat analysis
Free of other bias?	Low risk	No other apparent source of bias

EPHT 2006

Methods	Stated purpose: To ascertain harms and benefits of combined CT among healthy postmenopausal Estonian women Randomisation method: Remotely randomised in permuted blocks Allocation method: Non-transparent sealed envelopes Blinding: Participants and investigators blinded Stratification: By centre No of women screened for eligibility: 39713 (whole female pop aged 50-64 of 2 areas of Estonia) No randomised and consented: 777 (see notes) No analysed: 777 Losses to follow up: None stated Dropouts: None stated Adherence to treatment: < 40% in HT group and < 30% in placebo group by 3 yrs (estimated from graph) Analysis by intention to treat: yes No of centres: 3 Years of recruitment: 1999-2001 Design: Parallel Funding: Academic and government grants
Participants	Included: Postmenopausal women >12 months since last period Excluded: Women who had used hormone therapy during the past 6 months; with untreated endometrial adenomatosis or atypical hyperplasia of the endometrium; a history of breast cancer, endometrial cancer or ovarian cancer; any other cancer treated less than 5 years ago; a history of meningioma; myocardial in-

EPHT 2006 *(Continued)*

	farction within the last 6 months; a history of hepatitis or functional liver disorders in the last 3 months; a history of deep vein thrombosis, pulmonary embolism, or cerebral infarction; porphyria; hypertension of more than 170/110 mmHg despite medication; laparoscopically or histologically confirmed endometriosis. Mean age: 59 Age range: 50-70 Means of recruitment: Invitation sent to whole female population aged 50-64 of 2 areas of Estonia Baseline equality of treatment groups: More prior use of oral contraceptive in HT group 9.2% versus 6.4%; HT group older (59 versus 58.5) Country: Estonia
Interventions	HT arm: Combined oestrogen and progesterone as one daily tablet containing conjugated equine oestrogen 0.625 mg and medroxyprogesterone acetate 2.5 mg Control arm: Matching placebo Duration: Mean follow up 3.43 years (range 2-5). Planned for 10 yr follow up but closed early.
Outcomes	Death CHD Cancers fractures CVD
Notes	Women randomised before eligibility and consent checked - envelopes only opened once these processes complete. A further 1001 women on unblinded trial arms. Designed as part of international WISDOM trial

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Remotely randomised in permuted blocks
Allocation concealment?	Low risk	Non-transparent sealed envelopes
Blinding? All outcomes	Low risk	Participants and investigators blinded
Incomplete outcome data addressed? All outcomes	Low risk	No stated losses to follow up or dropouts, analysed by intention to treat
Free of other bias?	Unclear risk	Mean follow up only 3.43 years (range 2-5). Planned for 10 yr follow up but closed early.

ERA 2000

Methods	Stated purpose: To evaluate the effects of HT on the progression of coronary atherosclerosis Randomisation method: Computerised in random blocks Allocation method: Computer displayed treatment assignment after eligible participant details entered Stratification: According to lipid lowering therapy at baseline and hospital where angiogram was performed Blinding: Participants, clinic staff and all outcomes assessment blinded Unblinding: Treatment assignment available to designated member of data management staff. Questions relating to adverse effects directed to gynaecology physician and nurse not connected with study.

	No of women screened for eligibility: Not stated No randomised: 309 No analysed: 309 (for clinical events) Losses to follow up: None (for clinical events) Dropouts: 5 women in the placebo group started taking HT. It is unclear if there were any other dropouts Adherence to treatment in the 248 participants evaluated was as follows: the unopposed oestrogen group took 74.5% of their prescribed medication, while the combined HT group took 84% and the placebo group took 85.8%. Analysis by intention to treat: Although only 248 participants were available for the primary trial end point (which was biological), clinical adverse events, including outcomes of interest to this review, were reported for all participants at 3.2 years by intention to treat. No of centres: 6 Years of recruitment: Jan 1996 - Dec 1997 Design: Parallel Funding: Grants from National Heart, Lung and Blood Institute and National Center for Research Resources General Clinical Research Center, study medications from Wyeth-Ayerst Research
Participants	Included: Postmenopausal women aged 55 - 80 years (non natural menses for at least 5 years, or for one year and FSH > 40 mu/ml or oophorectomy) with at least one stenosis > 30% in any single coronary artery confirmed by coronary angiography within 4 months of randomisation, baseline gynaecological examination normal Excluded: Failure to achieve >80% compliance during 4-week placebo run-in phase, breast or endometrial cancer, History of DVT or PE, symptomatic gallstones or elevated liver enzymes, fasting plasma triglycerides > 400 mg/dL, MI within 4 weeks, renal insufficiency, dye allergy, > 70% stenosis of coronary artery, uncontrolled hypertension, uncontrolled diabetes, planned or prior CABG, revascularization of only qualifying lesion (for study), inadequate baseline angiogram for study, other non-CHD diseases likely to be fatal or prevent adequate follow-up, participation in other intervention studies, plans to leave area within 3 years Mean age: 66 Age range: 41.8 -79.9 Means of recruitment: Media announcements, contact through hospital records and admissions, screening logs from other studies Baseline equality of treatment groups: Country: USA Follow up: 3 months, 6 months, then 6 monthly clinic visits annual smear and mammograms, annual endometrial aspiration
Interventions	HT arm: One of the following: 1. 0.625 mg CEE (Unopposed oestrogen) 2. 0.625 mg CEE plus 2.5 mg MPA (Combined continuous therapy) Control arm: Placebo Duration: 3.2 years mean
Outcomes	[Primary outcome angiographic] MI Stroke Death DVT PE
Notes	Power calculation: 80% power for primary angiographic outcome

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

ERA 2000 *(Continued)*

Adequate sequence generation?	Low risk	Computerised in random blocks
Allocation concealment?	Low risk	Computer displayed treatment assignment after eligible participant details entered
Blinding? All outcomes	Low risk	Participants, clinic staff and all outcomes assessment blinded
Incomplete outcome data addressed? All outcomes	Low risk	No losses to follow up for clinical adverse events. Analysed by intention to treat
Free of other bias?	Unclear risk	More in unopposed oestrogen group using nitrates at baseline, otherwise prognostic balance between groups

ESPRIT 2002

Methods	Stated purpose: To ascertain whether unopposed oestrogen reduces the risk of further cardiac events in postmenopausal women who survive a first myocardial infarction Randomisation method: List of random numbers generated by trial statistician in blocks of four Allocation method: Women assigned consecutively to numbers kept on list accessible to statistician only. Stratification: By clinical centre Blinding: Participants, clinicians, outcome assessors. Pharmaceutical company dispensed medication/placebo in identical numbered packages Unblinding: On request of family doctor or if participant withdrew from treatment (in later states of study, only if withdrawing participant had not had a hysterectomy). Outcome assessors remained blinded throughout No of women screened for eligibility: 3121 met inclusion criteria for MI (reasons for non-participation listed in study) No randomised: 1017 No analysed: 1017 Drop outs: Drop outs included 43 women in the HT group (8%) and 57 in the placebo group (11%) who did not take any of the trial medication. Losses to follow up: None Known NON-adherence with allocated treatment was as follows: at one year 51% of participants on the HT arm and 31% on the placebo arm were not taking their allocated tablets regularly. At two years, 57% of participants on the HT arm and 37% on the placebo arm were not taking their allocated tablets regularly. Analysis by intention to treat: Yes No of centres: 35 Years of recruitment: July 1996-Feb 2000 Design: Parallel Funding: Schering AG provided medication
Participants	Included: Postmenopausal women admitted to coronary care units or general medical wards in participating centres, who met diagnostic criteria for myocardial infarction, were discharged alive within 31 days of admission Excluded: Women with a previously documented MI, who had used HT or had vaginal bleeding in the 12 months before admission, history of breast, ovarian or endometrial cancer, active thrombophlebitis, history of DVT or PE, liver disease, Rotor syndrome, Dubin-Johnson syndrome or severe renal disease Mean age: 62 years (SD 5) Means of recruitment: Research nurses checked hospital case notes, approached potentially eligible women if their family doctor agreed to collaborate. Baseline equality of treatment groups: Yes

ESPRIT 2002 *(Continued)*

	Country: England and Wales
Interventions	HT arm: Unopposed oestradiol valerate 2 mg daily Control arm: Placebo Duration: 2 years
Outcomes	Recurrent MI Cardiac death All cause death Endometrial cancer Breast cancer Stroke Thromboembolism
Notes	Power calculation: Needed 1700 participants to give 80% power to detect 33.3% decrease in incidence of nonfatal reinfarction or cardiac death (2 sided P = 0.05). Accrual lower than anticipated: study closed with only 100 participants, giving 56% power to detect the above-mentioned outcomes, assuming full compliance

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	List of random numbers generated by trial statistician in blocks of four
Allocation concealment?	Low risk	Women assigned consecutively to numbers kept on list accessible to statistician only.
Blinding? All outcomes	Low risk	Participants, clinicians, outcome assessors.
Incomplete outcome data addressed? All outcomes	Low risk	No losses to follow up, analysed by intention to treat
Free of other bias?	Unclear risk	Accrual lower than anticipated: study closed with only 100 participants, giving 56% of planned power, assuming full compliance

EVTET 2000

Methods	Stated purpose:To determine if HT alters the risk of venous thrombo-embolism in high risk women Randomisation method: computer generated 1:1 block randomisation with fixed block sizes of 10 Allocation method: Not described Stratification: By age < 60 years or > 60 years 37 (23 HT and 14 placebo) women did not attend all visits due to premature termination of the study Blinding: Double blind No of women screened for eligibility: 328 No randomised: 140 (71 HT 69 placebo) No analysed: 140 Losses to follow-up: Nil, though 37(23 HT and 14 placebo) women did not attend all visits due to premature termination of the study Adherence to treatment: Not described Dropouts: 33: 10 in HT group (2 wanted be sure of being treated with oestrogen for postmenopausal symptoms, 8 had adverse effects), 23 in the placebo group (11 wanted be sure of being treated with oestrogen for postmenopausal symptoms, 10 had adverse effects, 2 no reason stated)

EVTET 2000 *(Continued)*

	Analysis by intention to treat: the main findings were not reported by intention to treat, since dropouts from the placebo group were not included in the denominator for the rate of recurrent thromboembolism. No of centres: Not stated Years of recruitment: February 1996-March 1999 Design: Stratified double triangular sequential design Funding: Novo-Nordisk Pharmaceutical and research forum Ulleval University Hospital.
Participants	Included: postmenopausal women with history of VTE, aged < 70 years, previous VTE was verified by objective means, i.e. venography or ultrasound in cases of DVT, and lung scan, helical computed tomography, or angiography in cases of PE. Excluded: current use or use of anticoagulants within last 3 months, familial antithrombin deficiency, any type of malignant diseases including known, suspected or past history of carcinoma of the breast; acute or chronic liver disease or history of liver disease in which liver function tests had failed to return to normal; porphyria, known drug abuse or alcoholism; life expectancy less than 2 years; or women who had taken part in other clinical trials within 12 weeks before study entry. Mean age: 55.8 years Age range: 42 to 69 years Means of recruitment: letters to family doctors, gynaecologists and hospitals, health bulletins and media. Baseline equality of treatment groups: baseline characteristics were similar for HT group and placebo group with regard to previous diseases(coronary heart disease hypertension, stroke,diabetes), smoking habits, and serum lipids. All women had previously suffered at least one VTE and the total number of previous VTE were 75 in the placebo group and 77 in the HT group. Country: Norway
Interventions	HT arm: 2 mg estradiol plus 1 mg norethisterone acetate 1 mg Control arm: placebo Duration: Planned 2 years, stopped prematurely at median 1.3 years' follow up
Outcomes	Venous thrombosis Myocardial infarction, Transient ischaemic attacks Stroke.
Notes	Power calculation: At a significance level of 5% and a power of 90% the sample size was estimated to a maximum of 240 women . After publication of the results of the HERS study which showed as a secondary end-point an increased risk of VTE, recruitment of women was discontinued in September 1998, until reviewed by the safety monitoring committee. The committee was also concerned about a non-significant clustering of end-points in one study group, but without knowing treatment allocation. The committee advised on premature termination of the study even though formal boundaries showing an excess risk of VTE were not reached. The final decision on termination of the study was made in February 1999, and by the end of March 1999, all the participants had completed a final follow-up visit.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Adequate sequence generation?	Low risk	Computer generated 1:1 block randomisation with fixed block sizes of 10
Allocation concealment?	Unclear risk	Not described
Blinding? All outcomes	Low risk	Double blinded
Incomplete outcome data addressed? All outcomes	High risk	The main findings were not reported by intention to treat, since dropouts from the placebo group were not included in the denominator for the rate of recurrent thromboembolism.

EVTET 2000 *(Continued)*

Free of other bias?	Unclear risk	Study terminated early, only 140 women enrolled of 240 planned.

Ferenczy 2002

Methods	Stated purpose:To assess the endometrial safety and bleeding pasterns of 17B-oestradiol sequentially combined with dydrogesterone Randomisation method: Not described Allocation method: Not described Stratification: Not mentioned Blinding: Double blind No of women screened for eligibility: 844 No randomised: 595 [HT group 1: 117, HT group 2: 114, HT group 3: 117, HT group 4: 118, Placebo group : 113 (See Interventions)] No analysed: 442 (for endometrial cancer, which is the only outcome of interest to this review) Losses to follow up: Endometrial status was evaluated by a biopsy, which was available only to women who remained on active treatment for over a year or who received placebo and completed the two year study. This resulted in 153 losses to follow up for this outcome (87 from the active treatment groups (24%) and 50 from the placebo group (44%), plus another 16 who received no study medication) Adherence to treatment: Not reported. Analysis by intention to treat: No No of centres: multi-centre (number not stated) Years of recruitment: not stated Design: parallel Design: parallel Funding: Solvay pharmaceutical
Participants	Included: Postmenopausal women with a uterus with amenorrhoea of at least 6 months or surgically post-menopausal (following bilateral oophorectomy without hysterectomy, more than 3 months prior to enrolment), FSH within normal postmenopausal range. Excluded: abnormal (uninvestigated bleeding), vaginal bleeding, the use of estrogens and or progestogens and or androgens in the preceding 6 months or more and any previous use of estradiol pellet/implant therapy Mean age: Age range: 45-65 years Means of recruitment: Baseline equality of treatment groups: Yes Country: Canada and Netherlands
Interventions	HT arm: 1) 1 mg/d 17B estradiol/ 5 mg dydrogesterone for the last 14 days of each 28 day cycle 2)1 mg/d 17B estradiol/10 mg dydrogesterone for the last 14 days of each 28 day cycle 3) 2 mg/d 17B estradiol/10 mg dydrogesterone for the last 14 days of each 28 day cycle 4) 2 mg/d 17B estradiol/20 mg dydrogesterone the last 14 days of each 28 day cycle. Control arm: placebo Duration: 26 cycles(104 weeks)
Outcomes	Endometrial cancer
Notes	Power calculation: not stated

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Adequate sequence generation?	Unclear risk	Not described

Ferenczy 2002 *(Continued)*

Allocation concealment?	Unclear risk	Not described
Blinding? All outcomes	Unclear risk	Double blinded
Incomplete outcome data addressed? All outcomes	High risk	Endometrial status was evaluated by a biopsy, which was available only to women who remained on active treatment for over a year or who received placebo and completed the two year study. This resulted in 153 losses to follow up (26%) for this outcome
Free of other bias?	Low risk	No other apparent source of bias

Haines 2003

Methods	Stated purpose:To assess the effects of different doses of oestrogen on menopausal symptoms, mood and quality of life in postmenopausal Chinese women Randomisation method: computer generated numbers 1:1:1 ratio Allocation methods: Trial centre sent opaque sealed sequentially numbered containers for medications Stratification: No Blinding: Double blind Unblinding: Not described No of women screened for eligibility: 169 No randomised: 152 (52: 1 mg oestradiol, 50: 2 mg oestradiol, 50 placebo) No analysed: 152 (WHO-QOL analysis performed with missing data replaced by mean score of other responders) Losses to follow up: 13 (8.5%) (including one participant who was excluded for non-adherence to treatment) Dropouts: nil Adherence to treatment: > 80% for all but one woman (by pill count) Analysis by intention to treat: Yes No of centres: One Years of recruitment: Feb 1998 - Feb 2000 Design: Parallel Funding: Novo Nordisk Asia Pacific
Participants	Included: Postmenopausal ethnically Chinese women (serum oestradiol concentration within menopausal range) not received any form of hormonal therapy during previous 3 months and had undergone a hysterectomy at least 6 months before entry into the study. Able to comply with trial protocol Excluded: Any known or suspected history of breast carcinoma, severe liver or renal disease, thromboembolic history, abnormal genital bleeding, history of DVT or thromboembolic disorder, cardiac failure, diabetes, asthma, migraine, epilepsy or treatment with liver enzyme inducing medications or those that could have affected bone metabolism. Steroids within 3 months, porphyria, severe obesity, suspected oestrogen dependent neoplasia Mean age: 48 (34-65) Age range: Means of recruitment: Outpatient clinic for menopausal problems Baseline equality of treatment groups: Not stated Country: Hong Kong
Interventions	HT arm: Oestradiol 1mg or 2mg daily Control arm: Placebo Duration: 12 months
Outcomes	Primary outcomes biological. Outcome of interest for this review: Quality of Life, measured by

Haines 2003 *(Continued)*

	WHOQOL (World Health Organisation Quality of Life)
Notes	Power calculation: Recruitment of 150 women was calculated to provide 80% power to show a 3% difference in bone mineral density at one year.

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Computer generated numbers 1:1:1 ratio
Allocation concealment?	Low risk	Opaque sealed sequentially numbered containers for medications
Blinding? All outcomes	Low risk	Double blinded
Incomplete outcome data addressed? All outcomes	Low risk	13 (8.5%) losses to follow up, analysed by intention to treat.
Free of other bias?	Low risk	No other apparent source of bias

HERS 1998

Methods	Stated purpose: To determine if combined HRT alters the risk for CHD events in postmenopausal women with established coronary disease Randomisation: Computer-generated random numbers in blocks of four Allocation: Computer displayed after participant details entered Stratification: By clinical centre Blinding: Participants, clinical centre staff, outcome assessors, data analysts, funders blinded Unblinding: When required for safety or symptom control, participants reported directly to gynaecology staff who were located separately from clinical staff, did not communicate with them about breast or gynaecological problems and were not involved in outcome ascertainment No of women screened for eligibility: 3463 of whom 43% excluded (ineligible, declined to participate, did not return for appointment or did not comply with placebo run-in therapy) No randomised: 2763 No analysed: 2763 Losses to follow up: Vital status known for all women at end of trial. 59 women did not complete follow-up (32 in experimental arm, 27 in placebo arm). Adherence to treatment in women evaluated: By self-report: At 1 year: 82% HT arm; 91% control arm; At 3 years: 75% HT arm; 81% control arm. By pill count in HT arm: at 1 yr: 79%; at 3 years: 70% HT arm. Analysis by intention to treat: Yes (also analysed by treatment received, with inclusion limited to women with > 80% compliance) No of centres: 20 Years of recruitment: Feb 1993 - Sep 1994 Design: Parallel Funding: Pharmaceutical (Wyeth-Ayerst) UNBLINDED CONTINUATION OF HERS 1998: N.B. Follow up continued unblinded, as an open-label observational study: 2321 women (93% of 2510 surviving HERS participants) followed up for a further 2.7 years - originally planned for additional 4 years but executive committee decided no further useful information likely to emerge. No analysed: 2311 for vital status Losses to follow up: 10 women (1%) not contacted at final follow up (2 in HT arm; 8 in control arm) of these, vital status known for 5.

HERS 1998 *(Continued)*

	Adherence to treatment: among women originally assigned to the HT group, 45% reported at least 80% compliance during the sixth year of follow up. Among women originally assigned to placebo, 8% reported taking HT at six years.
Participants	Included: Postmenopausal women aged under 80, with a uterus, with coronary disease (myocardial infarction, coronary artery bypass surgery, percutaneous coronary revascularization, or angiographic evidence of at least 50% narrowing of one or more major arteries, as documented by baseline ECG or hospital discharge summary), likely to be available for follow up for at least 4 years Excluded: Women whose coronary event occurred within 6 months of randomisation, use of hormone therapy within 3 months of randomisation, serum triglycerides ≥ 300 mg/dL, history or baseline findings suggestive of venous thromboembolism, breast cancer, endometrial cancer, cervical cancer, uncontrolled hypertension, uncontrolled diabetes, severe congestive heart failure,other life threatening disease, alcoholism, drug abuse, history of intolerance of HT, any pre-existing condition indicating unsuitability for long term HT or placebo therapy, > 80% compliance with placebo medication during run-in phase Mean age: 67 years (SD 7) Age range: 44 - 79 Means of recruitment: Lists of cardiac patients, mass mailing, direct advertising Baseline equality of treatment groups: More women in control arm on statins at randomisation (67% versus 54%). When adjusted in analyses - made no statistically significant difference Country: USA
Interventions	HT arm: Conjugated equine oestrogen 0.625 mg with medroxyprogesterone acetate 2.5 mg Control arm: Placebo identical in appearance Continuous oral regimen Adherence to treatment defined as >80% compliance with medication or placebo Duration: 4.2 years mean FOR UNBLINDED CONTINUATION OF HERS 1998: Continuation planned for an additional 4 years but stopped after mean of additional 2.7 years as no more useful data anticipated.
Outcomes	Coronary events (MI or coronary death) Venous thromboembolism Fracture Gallbladder disease Endometrial, breast or ovarian cancer Death
Notes	Power calculation: 90% power to observe 24% reduction in coronary events at an average of 4.2 years (P = 0.05) follow up. Further unblinded follow up 2.7 years (HERS II) - see below

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Adequate sequence generation?	Low risk	Computer-generated random numbers in blocks of four
Allocation concealment?	Low risk	Computer displayed after participant details entered
Blinding? All outcomes	Low risk	Participants, clinical centre staff, outcome assessors, data analysts, funders blinded
Incomplete outcome data addressed? All outcomes	Low risk	Vital status known for all women at end of trial. 59 women did not complete follow-up (32 in experimental arm, 27 in placebo arm). Analysed by intention to treat.

HERS 1998 *(Continued)*

Free of other bias?	Low risk	More women in control arm on statins at randomisation (67% versus 54%). When adjusted in analyses - made no statistically significant difference

Mulnard 2000

Methods	Stated purpose: To determine whether oestrogen-only HT affects global, cognitive or functional decline in women with mild to moderate Alzheimer's disease Randomisation method: Computer generated in blocks of 6 Allocation method: Not described Blinding: Double blind Stratification: Not mentioned No of women screened for eligibility: 153 No randomised: 120 (CEE 0.625 mg:42, CEE 1.25 mg:39, placebo:39) No analysed: 120 Losses to follow up: Nil Dropouts: 23 (7 in placebo group, 7 in CEE 0.625 mg group, 9 in CEE 1.25mg/d). Adherence to treatment: This was measured and was defined as the proportion of individuals who ingested at least 80% of the study medication, but was not reported in the trial publication. Analysis by intention to treat: yes No of centres: 32 Years of recruitment: Not stated Design: parallel placebo controlled Funding: National Institute on Aging, Wyeth Ayerst
Participants	Included: Women with a diagnosis of probable Alzheimer disease according to National Institute of Neurological and Communicative Disorders and Stroke-Alzheimer Disease and Related Disorders Association Criteria in the mild or moderate stage(the study protocol specified a Mini-Mental State Examination[MMSE} score of 14-28; several exceptions were made by the project director to allow for participants with MMSE scores as low as 12); female sex; previous hysterectomy(oophorectomy not required);age older than 60 years; absence of major clinical depressive disorder(as measured by score of < 17 on the Hamilton Depression Rating Scale [Ham D]; and normal gynaecological, breast, and mammography results Excluded: myocardial infarction within one year, history of thromboembolic disease or hypercoagulable state, hyperlipidaemia, or use of excluded medications (i.e. estrogens used within 3 months; current use of antipsychotics, anticonvulsants, anticoagulants, beta-blockers, narcotics, methyldopa, clonidine, or prescription cognitive-enhancing or antiparkinson medications, including experimental medications within 60 days prior to baseline. Stable doses of neuroleptics, antidepressants, anxiolytics, sedatives, and hypnotics were allowed). At the initiation of the protocol, individuals treated with donepezil or tacrine were excluded, but a protocol amendment after 20 months of enrolment allowed the stable use (minimum of 4 weeks) of these medications before screening for the study. Mean age: 75 Age range: 56-91 Means of recruitment: not stated Baseline equality of treatment groups: Baseline equality of treatment groups: there were no significant differences between the three groups in terms of baseline and demographic characteristics. Country:USA
Interventions	HT arm: (1) CEE oral 0.625 mg/day (2) CEE 1.25 mg/day Control:placebo Duration: 1 year
Outcomes	Primary outcome: Progression of Alzheimer's disease (Alzheimer's Disease Cop-operative Study version of the Clinical Global Impression of Change scale)

Mulnard 2000 *(Continued)*

Notes	Power Calculation: 81% to detect a 29% difference in the proportion of subjects who worsen in the two groups (60% worse in the placebo group versus 31% worse in the oestrogen group) using a 2-tailed[alpha] =.05 (based on data from a similar trial, with 40 subjects receiving placebo and 80 subjects receiving oestrogen. * Inclusion criteria state > 60 years but age range at baseline is 56-91

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Computer generated in blocks of 6
Allocation concealment?	Unclear risk	Not described
Blinding? All outcomes	Low risk	Double blinded
Incomplete outcome data addressed? All outcomes	Low risk	No losses to follow up stated. Analysed by intention to treat.
Free of other bias?	Unclear risk	Inclusion criteria state > 60 years but age range at baseline is 56-91

Nachtigall 1979

Methods	Stated purpose: To evaluate the effects of HT Randomisation: Women matched for diagnosis of chronic disease. From matched pairs, research nurse randomly selected which member would be assigned to which group. Method not described. Allocation concealment: Method not stated. Stratification: Not mentioned Blinding: Patients and research physicians blinded Unblinding: Code broken if a major medical complication or death occurred (13 times in HT group, 17 times in control group) No of women screened for eligibility: 403 (235 excluded: 74 ineligible, 31 refused, 130 no match for pair found) No randomised: 168 No analysed: 168 Losses to follow up: None Adherence to treatment: Not mentioned Analysis by intention to treat: Yes, though any events occurring after unblinding were not recorded. No of centres: One Years of recruitment: Unclear - Study lasted 10 years and was complete by 1976 Design: Parallel Funding: Not stated
Participants	Included: Postmenopausal inpatients with chronic disease (last menstrual period >2 years previously, GFSH >105.5 mU, total urinary oestrogen <10 micrograms/dl), never taken HT. All hospitalised for entire study period.. Screened with history, physical examination, medical record review Women matched on the basis of chronic disease diagnosis, as follows: diabetes mellitus (14 pairs), custodial care (20 pairs), arteriosclerosis (9 pairs) Other paris matched on the basis of chronic neurologic disorders. Excluded: Women with acute heart disease, hypertension (B/P >160/94), apparent malignancy, hysterectomy Mean age: 55 Age range:

Nachtigall 1979 *(Continued)*

	Means of recruitment: Baseline equality of treatment groups: Correlation for diagnosis was identical. Correlation for some other risk factors was low between individual pairs but group means were similar. Country: New York hospital for chronic diseases
Interventions	HT arm: CEE 2.5 mg daily, plus MPA 10 mg for 7 days in each month Control arm: Placebo Duration: 10 years
Outcomes	Death, myocardial infarction, "serious embolism" (pulmonary embolus), breast cancer, colon cancer, endometrial cancer , gall stones
Notes	Power calculation: Not mentioned Re generalisability: Authors point out that almost all women had long term chronic disease, were hospitalised for the entire study period, had much lower than normal overall parity and more prolonged bed rest than average woman.

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Unclear risk	Women matched for diagnosis of chronic disease. From matched pairs, research nurse randomly selected which member would be assigned to which group. Method not described.
Allocation concealment?	Unclear risk	Not described
Blinding? All outcomes	Low risk	Patients and research physicians blinded
Incomplete outcome data addressed? All outcomes	Unclear risk	No losses to follow up described. Analysed by intention to treat, but any events occurring after unblinding were not recorded.
Free of other bias?	Unclear risk	Correlation for some baseline prognostic factors was low between individual pairs but group means were similar.

Notelovitz 2002

Methods	Stated purpose: To determine the lowest effective dose of an oestradiol transdermal delivery system for preventing bone loss in postmenopausal women Randomisation and allocation methods: Not described Stratification: Not described Blinding: Double blind, double dummy Unblinding: Not described No of women screened for eligibility: Not stated No randomised: 355 (0.025 mg dose: 89, 0.05 mg dose: 90, 0.075 mg dose: 89, placebo: 87) No analysed: 355 (data imputed for losses to follow up) Losses to follow up: 34 (9.6%) Dropouts: 125 (35%) did not complete 2 years' treatment (88 in active treatment arms, 37 in placebo arm) Adherence to treatment: One participant was withdrawn for failure to adhere to the treatment schedule. Overall level of adherence to treatment in women who continued with their allocated treatment is not described. Analysis by intention to treat: Yes No of centres: 22 Years of recruitment: Not stated

Notelovitz 2002 *(Continued)*

	Design: Parallel Funding: Proctor and Gamble Pharmaceuticals
Participants	Included: Postmenopausal, non-osteoporotic and ambulatory women under 70 years of age who had had a hysterectomy, with or with out bilateral oophorectomy, at least 12 months earlier. Postmenopausal status documented by serum E2 < 23 pg/ml and FSH serum levels > 40 mlU/ml . Non-osteoporotic status defined by dual energy x-ray absorptiometry (DXA) minimum T-score of -2.5. Excluded: Participants who had received oral estrogens within 2 months of enrolment or who had contra-indications to oestrogen therapy or a history of oestrogen intolerance, women with clinically significant systemic or psychiatric disorders; history of cancer (other than basal cell carcinoma in remission or uterine cancer treated by hysterectomy); history of osteomalacia, hyperparathyroidism, or untreated hyperthyroidism, abnormal serum lipids, creatinine, or liver enzymes; or use of medications within 3 months of enrolment that could modify BMD, participants with radiographic abnormalities of the lumbar spine on anterior/posterior or lateral view, which would preclude precise DXA measurements Mean age: Not stated Age range: Not stated Means of recruitment: Not stated Baseline equality of treatment groups: Yes Country: USA
Interventions	HT arm: 2 patches, delivering total dose of oestradiol: 0.025 mg, 0.05 mg or 0.075 mg Control arm: 2 placebo patches Duration: 2 years (26 cycles)
Outcomes	Breast cancer (regular mammograms) Fractures
Notes	Power calculation: Not mentioned

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Unclear risk	Not described
Allocation concealment?	Unclear risk	Not described
Blinding? All outcomes	Low risk	Double blinded, double dummy
Incomplete outcome data addressed? All outcomes	Low risk	34 (9.6%) losses to follow-up. Analysed by intention to treat
Free of other bias?	Low risk	No other apparent source of bias

Obel 1993

Methods	Stated purpose: To compare combined and sequential therapy with respect to relief of climacteric symptoms, effects on the endometrium and on vaginal cellular maturation, steroid metabolism and side effects. Randomisation and allocation methods: Not described Stratification: Not mentioned Blinding: Double blind Unblinding: Not described

Obel 1993 *(Continued)*

	No of women screened for eligibility: 176, of whom 21 unwilling to take placebo, 2 found not post-menopausal, 2 excluded for private reasons. No randomised: 151 (combined HT:50, sequential HT:50, placebo:51) No analysed: 129 (in the groups to which they were allocated) Losses to follow up: 22 (11 from combined group, 5 from sequential group, 6 from placebo group) Adherence to treatment: Not described Analysis by intention to treat: No No of centres: One Years of recruitment: Not stated Design: Parallel Funding: Pharmaceutical Division, Novo Nordisk
Participants	Included: Women in early menopause (last spontaneous vaginal bleeding > 6 and < 24 months earlier), no HT within preceding 24 months Excluded: Women with previous or current oestrogen-dependant neoplasia, thrombo-embolic disease, liver or pancreatic disease, diabetes mellitus, severe obesity, diseases with high or low bone turnover and medication known to influence bone metabolism or provoke induction of liver enzymes. Mean age: Not stated Age range: Not stated Means of recruitment: All 5800 women born between 1930 and 1933 in Frederiksborg County, Denmark invited to participate Baseline equality of treatment groups: Yes Country: Denmark
Interventions	HT arm: 1) Oral oestradiol 2 mg + norethisterone 1mg 2) Oral oestradiol 2 mg days 1-22 + norethisterone acetate days 13-22, then oestradiol 1 mg days 22-28 Control arm: Placebo Duration: 2 years
Outcomes	Only outcomes of interest to this review: Endometrial cancer, quality of life
Notes	Power calculation: Not mentioned

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Adequate sequence generation?	Unclear risk	Not described
Allocation concealment?	Unclear risk	Not described
Blinding? All outcomes	Low risk	Double blinded
Incomplete outcome data addressed? All outcomes	Low risk	22 (15%) losses to follow up (11 from combined group, 5 from sequential group, 6 from placebo group); analysed by intention to treat
Free of other bias?	Unclear risk	Baseline quality of life scores on several measures appear substantially lower for placebo group

PEPI 1995

Methods	Stated purpose: To investigate the effects of oestrogen-only and combined therapies on cardiovascular disease risk factors, as well as on endometrial status, breast changes , bone density, menopausal symptoms and quality of life factors. Randomisation: Computer-generated variable length blocks Allocation: Allocation assignments on encrypted file loaded on computer at clinical centre and issued once eligibility confirmed (or by phone to co-ordinating centre in case of computer failure) Stratification: By clinical centre and hysterectomy status Blinding: Participants, clinical and laboratory personnel blinded, medication packages visually indistinguishable Unblinding: Unblinding officer at each trial centre or by phone call to co-ordinating centre; referral gynaecologist at each centre not directly involved with data collection or patient care able to access treatment assignment for management of safety issues No of women screened for eligibility: Approx. 1460 (states that 60% of women screened were randomised) No randomised: 875 No analysed: 847 (97%) Losses to follow up: 28 (CEE only group: 5/170, CEE + MPA sequential group 5/174, CEE + MPA continuous group 4/174, CEE + MP sequential group 5/178, Placebo group 9/174) Dropouts: Dropout rate disproportionately high in women with a uterus assigned unopposed oestrogen: 55% had to discontinue the assigned therapy, largely due to endometrial hyperplasia. Adherence to treatment: Of the 847 women who attended the 3 year follow-up, 75% of women with a uterus and 80% of women without had at least 80% adherence to treatment. [Note: 55% of women with a uterus assigned unopposed oestrogen were required to discontinue the assigned therapy due to endometrial hyperplasia]. Analysis by intention to treat: No - but 97% women analysed by ITT No of centres: 7 Years of recruitment: Dec 1989 - Feb 1990 Design: Parallel Funding: Research grants from National Heart, Lung and Blood Institute, National Institute of Child Health and Human Development, National Institute of Health and Human Development, National Institute of Arthritis and Musculoskeletal and Skin Diseases, National Institute of Diabetes and Digestive and Kidney Diseases, National Institute on Aging, USA
Participants	Included: Healthy postmenopausal women aged 45-65, with or without a uterus. Ceased menstruation 12 months prior to entry or had hysterectomy at least 2 months prior to entry and FSH levels <40 mU/ml Excluded: Women who had used hormones within past 3 months, women treated with thyroid hormone unless stabilised on treatment, serious illness including heart or thromboembolic disease, previous endometrial or breast cancer, contraindications to oestrogen Mean age: 56 years (SD 4) Age range: 45-64 Means of recruitment: Through mass media and community efforts Baseline equality of treatment groups: Women assigned to placebo had higher mean levels of fibrinogen and LDL-C at baseline Country: USA
Interventions	HT arm: One of the following 4 regimens: 1. CEE 0.625 mg daily (Unopposed oestrogen) 2. CEE 0.625 mg daily plus MPA 10 mg daily for first 10 days (Combined sequential treatment) 3. CEE 0.625 mg plus MPA 2.5 mg daily (Combined continuous treatment) 4. CEE 0.625 mg plus MP 200 mg daily for first 12 days (Combined sequential treatment) Control arm: Placebo Duration: 3 years
Outcomes	Primary endpoints biological markers, not relevant to this review; however the following pre-specified outcomes were also measured: Breast cancer Endometrial cancer Cardiovascular disease Thromboembolism Gallbladder disease.

PEPI 1995 *(Continued)*

Notes	Power calculation: Based on primary (biological) outcome: A sample of 840 women was projected to provide minimum power of 0.92 to detect differences of 5 mg/dl in HDL-cholesterol for any pairwise comparison of treatment arms at 3 years

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Computer-generated variable length blocks
Allocation concealment?	Low risk	Allocation assignments on encrypted file loaded on computer at clinical centre and issued once eligibility confirmed (or by phone to co-ordinating centre in case of computer failure)
Blinding? All outcomes	Low risk	Participants, clinical and laboratory personnel blinded, medication packages visually indistinguishable
Incomplete outcome data addressed? All outcomes	Low risk	28 (3%) lost to follow up. 97% of women analysed by intention to treat
Free of other bias?	Unclear risk	55% of women with a uterus assigned unopposed oestrogen were required to discontinue the assigned therapy due to endometrial hyperplasia. Placebo group had higher levels of fibrinogen and LDL-C at baseline, otherwise groups prognostically balanced

WAVE 2002

Methods	Stated purpose: To determine whether HT or antioxidant vitamin supplements, alone or in combination, influence the progress of coronary artery disease in postmenopausal women, as measured by angiography Randomisation: Computer randomised, permuted block design with random blocks of two and four Allocation methods: Remotely by phone call to study co-ordinating centre Stratification: Clinical centre, hysterectomy status Blinding: Participants, investigators and staff at clinical centres blinded except (when necessary) the study gynaecologist Unblinding: Adverse effects managed by gynaecologist not involved in outcome assessment who had access to treatment assignment if necessary, with permission of co-ordinating centre No of women screened for eligibility: No randomised: 211 No analysed: 206 for clinical status at end of study Losses to follow up: 5 (3 in HT group, 2 in placebo group) Adherence to treatment: Evaluated for 159/211 who had angiographic follow up: HT group took 67% of medication, placebo group took 70%; 9/108 women in placebo group crossed to open-label oestrogen Analysis by intention to treat: No - but 98% of women analysed by ITT No of centres: 7 Years of recruitment: July 1997 - August 1999 Design: Parallel Funding: National Heart, Lung and Blood Institute contract, General Clinical Research Center grant, USA
Participants	Included: Postmenopausal women with one or more 15% to 75% coronary stenoses in an artery not subjected to intervention, seen on angiogram within 4 months of study entry. Postmenopausal defined as post bilateral oophorectomy, under 55 years old with an FSH of 40 Mu/ml or higher, or older than 55 years

WAVE 2002 *(Continued)*

	Excluded: HT use within 3 months, concurrent use of over 60 mg day of vitamin C or 30 IBU daily of vitamin E and unwilling to stop taking them, suspected breast, uterine or cervical cancer, uncontrolled diabetes or hypertension, MI within 4 weeks, elevated triglycerides or creatinine levels, symptomatic gallstones, heart failure, history of haemorrhagic stroke, bleeding diathesis, PE, DVT or untreated osteoporosis. Mean age: 65 Age range: 56-74 Means of recruitment: Recruited at clinical sites in USA and Canada Baseline equality of treatment groups: Higher prevalence of diabetes and higher fasting blood glucose levels in the HT group Country: USA and Canada
Interventions	HT arm: One of the following regimens: 1) CEE 0.625 (oestrogen only therapy) - for women who had had a hysterectomy 2) CEE 0.625 and MPA 2.5 mg daily (continuous combined therapy) - for women who had not had a hysterectomy Control arm: Placebo Duration: 3 years In addition, this study included women on this study were prescribed a regimen of vitamins E and C or placebo vitamins. The only comparison considered in this review is HT/placebo vitamins versus placebo HT/placebo vitamins
Outcomes	Primary outcome biological: change in minimum lumen diameter of qualifying coronary lesions Outcomes of interest to review: All cause death Total mortality Cardiovascular events Venous thromboembolism Stroke Breast cancer Quality of life
Notes	Study publication pools results for women on unopposed and combined therapies Power calculation: Based on primary (biological) outcome: 423 women provide 90% power to detect an effect size of at least 0.33 (corresponding to a change in minimum lumen diameter of 0.1 mm and assuming 20% of women would not undergo a follow-up angiogram)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Adequate sequence generation?	Low risk	Computer randomised, permuted block design with random blocks of two and four
Allocation concealment?	Low risk	Remotely by phone call to study co-ordinating centre
Blinding? All outcomes	Low risk	Participants, investigators and staff at clinical centres blinded except (when necessary) the study gynaecologist
Incomplete outcome data addressed? All outcomes	Low risk	Losses to follow-up 5 (3 in HT group, 2 in placebo group), 98% of women analysed by intention to treat
Free of other bias?	Unclear risk	Active group had higher prevalence of diabetes and higher fasting blood glucose levels, otherwise groups prognostically balanced

WEST 2001

Methods	Stated purpose:To determine whether 17B oestradiol reduces the risk of recurrent stroke or death among postmenopausal women who have experienced a transient ischaemic attack or nondisabling ischaemic stroke Randomisation: Computer generated at pharmacy, in blocks of four Allocation methods: By remote contact with trial pharmacy Stratification: By trial centre and risk level (3 levels) Blinding: Participants, investigators and endpoint assessors blinded Unblinding: study internist unblinded in the case of overriding concern about a woman's clinical care No of women screened for eligibility: 5296 (2772 ineligible, 1843 declined to participate, 17 unable to be randomised within protocol time frame) No randomised: 664 (HT: 337, Placebo: 327) No analysed: 664 Losses to follow up: Nil Dropouts: 34% of the oestradiol group and 24% of placebo group. Non adherence to allocated treatment: Overall mean: HT group: 44%, Placebo: 36%. Among women who continued with treatment, adherence to treatment was 90% in both groups. Analysis by intention to treat: Yes No of centres: 21 (single recruitment hub) Years of recruitment: December 1993-May 1998 Design: Parallel Funding: National Institute of Neurological Disorders and Stroke grant, Medical Research Council of Canada grant, Mead Johnson laboratories provided support and study drug
Participants	Included: Postmenopausal women (i.e. amenorrhoea for at least 12 months, or having undergone hysterectomy and > 55 years of age) over 44 years of age within 90 days of a qualifying ischaemic stroke or transient ischaemic attack. Excluded: Women whose index event was disabling or if it occurred while taking oestrogen. Women with a history of breast of endometrial cancer, who had had a venous thromboembolic event while on oestrogen replacement therapy, had a neurological or psychiatric disease that could complicate the evaluation of the endpoints, or had a co-existing condition that limited their life expectancy Mean age: 71 Age range: 46-91 Means of recruitment: Admissions to 20 largest regional hospitals in Connecticut and Massachusetts; also via contact with selected neurology groups and direct referrals from physicians Baseline equality of treatment groups: Yes Country: USA
Interventions	HT arm: 17-Beta estradiol 1 mg daily plus, for women with a uterus, a course of medroxyprogesterone acetate once a year, 5 mg daily for 12 days Control arm: Placebo Duration: 2.8 years
Outcomes	Death or recurrent stroke Myocardial infarction Cognitive function
Notes	Study publication pools results for women on unopposed and combined therapies Power calculation: 652 women required to give 80% power to detect a reduction in the rate of deaths or nonfatal strokes from 25% in the placebo group to 15% in the HT group (2 tailed P = 0.05)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Adequate sequence generation?	Low risk	Computer generated at pharmacy, in blocks of four

WEST 2001 *(Continued)*

Allocation concealment?	Low risk	By remote contact with trial pharmacy
Blinding? All outcomes	Low risk	Participants, investigators and endpoint assessors blinded
Incomplete outcome data addressed? All outcomes	Low risk	No losses to follow up, analysed by intention to treat
Free of other bias?	Low risk	No apparent source of bias

WHI 1998

Methods	Stated purpose: To test the hypothesis that women taking HT will have lower rates of coronary heart disease and osteoporosis-related fractures. COMBINED HT ARM: Randomisation: Centrally randomised by permuted block algorithm Stratification: By clinical centre site and age group Allocation: By local access to remote study database Blinding: All participants, clinic staff, and outcome assessors blinded, with the exception of 331 participants who were unblinded from the unopposed oestrogen arm and reassigned to combined HT arm due to change in protocol (see Notes). A further 432 women (248 in the experimental arm and 183 in the placebo arm) had a hysterectomy after randomisation (for reasons other than cancer) and switched to unopposed oestrogen or the corresponding placebo in the unopposed oestrogen study arm. Unblinding: When required for safety or symptom management, unblinding officer unblinded clinic gynaecologist, who was not involved with outcomes assessment. At average 5.2 year follow-up, 3444 women in experimental group and 548 women in placebo group had been unblinded, mainly to manage persistent vaginal bleeding) No randomised: 16,608 (8506 to experimental group, 8102 to placebo group) No analysed: 16,608 Losses to follow-up: 583 participants (3.5%) - i.e. no outcomes data for > 18 months: [307 in HT arm (3%), 276 in control group (3.5%). Vital status known for 96.5% Dropouts/Non adherence to allocated treatment: Women with adherence to treatment of under 80% (by pill count) were counted as dropouts. Dropout rates at 5.2 years were 42% in the experimental arm, and 38% in the placebo group. In addition, 10.7% of women in the placebo group crossed to receive active treatment. Analysis by intention to treat: Yes (analysed with and without unblinded group in experimental arm) No of centres: 40 Power calculation: Sample gives 80-95% power for primary endpoint comparisons at 5% significance, assuming an intervention effect of 20% for CHD and 21% for combined fractures at 6-9 year follow-up, and an intervention effect of 22% for breast cancer at 14 year follow up (relative risk of 1.3 assumed for increased risk of breast cancer in intervention group) Years of recruitment: 1993-1998 Note: Planned 8.5 years follow up. Trial stopped after mean 5.2 years as test statistic for breast cancer exceeded predetermined stopping boundary and global risk index indicated risks exceeding benefits. WHI 1998 UNOPPOSED OESTROGEN ARM: Randomisation: as above Stratification: as above Allocation: as above Blinding: as above Unblinding: see above No randomised: 10739 (including 248 in experimental arm, 183 in placebo arm) joined this study after randomisation to corresponding arms in WHI 2002 and having subsequently had hysterectomy (for reasons other than cancer) No analysed: 10739 Losses to follow-up: 563

WHI 1998 *(Continued)*

	Drop outs/ Non adherence to allocated treatment: Women with adherence to treatment of under 80% by pill count were counted as dropouts. The dropout rate was 53.8% by the end of the study (6.8 yrs), and did not vary significantly between study arms. In addition, 9.1% of women in the placebo arm and 5.7% of women in the active treatment group initiated hormone use outside of the study through their own physician. Analysis by intention to treat: Yes No of centres: 40 Power calculation: 12,375 participants needed to detect a 21% reduction in CHD rates over projected 9 year average follow-up. Years of recruitment: 1993-1998 N.B. This arm of WHI 1998 was stopped early after a mean follow up of 6.8 years (planned for 9), when it was determined that the prospect of obtaining more precise evidence about the effects of the intervention was unlikely to outweigh potential harms, although no predefined safety boundaries had been crossed. WHIMS ANCILLARY STUDY: Enrolled 7479 WHI 1998 participants who were free of probable dementia and aged 56-79. Of these, 4532 were from the combined HT arm of WHI 1998 [WHI 1998(WHIMS:combined arm) and 2947 were from the unopposed oestrogen arm [WHI 1998(WHIMS:unopposed oestrogen arm)]. Overall, 92.4% of eligible women participated Years of recruitment: May 1996-December 1999 Analysis by intention to treat: All analysed by ITT for the planned primary and secondary outcomes. For a third outcome (global cognitive function), which was not formally pre-planned, 178 participants (3.9%) were excluded from the combined arm (151 because relevant follow-up data was missing and 27 because they consented to join WHIMS more than 6 months after WHI treatment assignment, by which time treatment effects may already have been underway) and 139 (4.7%) were excluded from the unopposed oestrogen arm (109 due to missing follow-up data and 30 due to enrolment 6 months or more after randomisation). Adherence to allocated treatment (i.e. Proportion taking > 80% of study medication): Unopposed oestrogen arm: Year 1: 77.2% in HT group versus 84.1% in placebo group. Year 6: 42%% in HT group versus 47.8% in placebo group. Combined HT arm: Year 1: 71% in HT group vs 83% in the placebo group. Year 4: 49% in HT group vs 61% in placebo group Power: designed to provide >80% power of detect an observed 40% relative reduction in the incidence rate of clinically diagnosed all-cause dementia Duration: The mean time from randomisation into WHI 1998 to the last WHIMS cognitive screening examination was 4.05 years for women on the combined HT arm and 5.21 years for women on the unopposed oestrogen arm.
Participants	COMBINED HT ARM: Included: Postmenopausal women (no vaginal bleeding for 6 months, or for 12 months for 50-54 year olds; any use of postmenopausal hormones), with a uterus, aged 50-79 at initial screening, likely to reside in area for 3 years, provision of written informed consent Excluded: Medical condition predictive of survival time <3 years, invasive cancer in past 10 years (except non-melanoma skin cancer), breast cancer at any time or suspicion of breast cancer at baseline screening, acute myocardial infarction, stroke, transient ischaemic attack in previous 6 months, known chronic active hepatitis or severe cirrhosis, blood counts indicative of disease, severe hypertension or current use of oral corticosteroids, femoral neck bone mineral density of more than 3 standard deviations below the corresponding age-specific mean, endometrial cancer or endometrial hyperplasia at baseline, malignant melanoma, pulmonary embolism or deep vein thrombosis that was nontraumatic or that had occurred in the previous six months, bleeding disorder, lipaemic serum and hypertriglyceridaemia diagnosis, current use of anticoagulants or tamoxifen, PAP smear or pelvic abnormalities, unwillingness or inability to complete baseline study requirements, alcoholism, drug dependency, mental illness, dementia, severe menopausal symptoms inconsistent with assignment to placebo, inability or unwillingness to discontinue current HT use or oral testosterone use, inadequate adherence with placebo run-in, unwillingness to have baseline or follow up endometrial aspirations, active participant in another randomised clinical trial Mean age: 63 years (SD 7) Age range: 50-79. Age ratio of 33%:45%:21% for the baseline age categories of 50-59, 60-69, 70-79 respectively (enrolment targeted to achieve ratio of 30:45:25)

WHI 1998 *(Continued)*

	Recruitment: Letter of invitation in conjunction with media awareness programme. Sampling method gave women from minority groups six-fold higher odds for selection than Caucasian women and resulted in sample with 84% racially/ethnically designated "white", 16% non-"white" Screening: Interested women screened by phone or mail for eligibility, then attended 3 screening visits for history, clinical exam and tests. Three month washout period before baseline evaluation of women using postmenopausal hormones at baseline screening. Lead-in placebo pills given for at least 4 weeks during screening process to establish compliance with pill taking. Baseline equality of treatment groups: No substantive differences between study groups at baseline Country: USA UNOPPOSED OESTROGEN ARM: Included: Postmenopausal women who had undergone hysterectomy (therefore considered postmenopausal for enrolment purposes), aged 50-79 at initial screening, likely to reside in area for 3 years, provision of written informed consent Excluded: as above Mean age: 64 Age range: 50-79. Age ratio of 33%:45%:21% for the baseline age categories of 50-59, 60-69, 70-79 respectively (enrolment targeted to achieve ratio of 30:45:25) Recruitment: as above Screening: as above Baseline equality of treatment groups: No substantive differences between study groups at baseline Country: USA WHIMS ancillary study: Included: Participants in either arm of WHI 1998, aged at least 65 and free of probable dementia
Interventions	Combined HT arm: Experimental group: Combined oestrogen and progesterone as one daily tablet containing conjugated equine oestrogen 0.625 mg and medroxyprogesterone acetate 2.5 mg Control group: Matching placebo Duration: 5.2 years Permanent discontinuation of medication: Women who developed breast cancer, endometrial hyperplasia not responsive to treatment, endometrial atypia, endometrial cancer, deep vein thrombosis, pulmonary embolus, malignant melanoma, meningioma, triglyceride level over 1000 mg/dL, prescription of oestrogen, testosterone or selective oestrogen-receptor modulators by their personal physician. Temporary discontinuation of medication: women who had acute MI, stroke, fracture, major injury involving hospitalisation, surgery involving anaesthesia, illness resulting in immobilisation for over one week, or other severe illness in which hormone use temporarily inappropriate Duration of intervention: 4.2 years N.B. The WHI 1998 (WHIMS) investigators reported outcomes according to study arm (unopposed oestrogen or combined HT therapy), and also (as per protocol) reported results pooled across the two arms. However there were significant baseline prognostic differences between the two arms (see Quality Table) and the results have not been pooled in this review. Unopposed oestrogen arm: Experimental group: 0.635 mg CEE daily Control arm: Placebo Permanent discontinuation of medication: as above WHIMS ancillary study: As for either arm of WHI 1998 above
Outcomes	Combined HT arm: Cardiovascular disease: acute MI, silent MI, coronary death, stroke, pulmonary embolus Cancer: breast, colorectal, endometrial, other cancers Fractures: Hip, vertebral, osteoporotic Unopposed oestrogen arm: as above (with the exception of endometrial cancer)

WHI 1998 *(Continued)*

	WHIMS ancillary study: Cognitive function Mild cognitive impairment Dementia For assessment of outcomes women in WHIMS underwent up to four phases of testing as follows 1. Participants underwent cognitive screening with the Modified Mini-Mental State Examination (3MSE) at baseline and annually. 2. Women who scored below an education-adjusted cut off point proceeded to a battery of psychoneurological tests, and standardised interviews, plus interviews with a designated informant (friend or relative) 3. Clinical assessments from local physicians 4. CT and blood tests to rule out reversible pathology All cases judged locally as probable dementia were independently evaluated by two adjudicators blind to the diagnosis, as were 50% of cases of mild cognitive impairment and 10% of all cases without dementia. Mild cognitive impairment defined as per current DSM IV criteria - operationally defined as follows: poor performance (< 10th percentile) in a battery of neuropsychological tests, a report of mild functional impairment from designated informant, no evidence of a psychiatric or medical explanation for the cognitive decline, and an absence of dementia. Dementia defined as per DSM-IV criteria
Notes	N.B. The original WHI protocol allowed women with a uterus to be randomised to receive unopposed oestrogen. As evidence emerged (from the PEPI trial) that this could be unsafe, 331 participants with a uterus in the intervention group in the unopposed oestrogen arm were reassigned to the intervention group in the combined HT arm.

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Centrally randomised by permuted block algorithm
Allocation concealment?	Low risk	By local access to remote study database
Blinding? All outcomes	Low risk	All participants, clinic staff, and outcome assessors blinded, with the exception of 331 participants who were unblinded from the unopposed oestrogen arm and reassigned to combined HT arm due to change in protocol
Incomplete outcome data addressed? All outcomes	Low risk	Combined HT arm : 583 participants (3.5%), lost to follow up. Vital status known for 96.5%. Oestrogen only arm: 563 (5%). Analysed by intention to treat.
Free of other bias?	Unclear risk	Both arms closed early: Combined arm at 5.2 years (8.5 planned); oestrogen-only arm at 6.8 years (9 planned). In the unopposed oestrogen arm there was greater use of aspirin at bedtime in the placebo group at baseline. In the combined HRT arm there was a lower prevalence of stroke and a higher percentage of participants using statins in the active treatment group at baseline.

WISDOM 2007

Methods	Stated purpose: To assess the log term benefits and risks of HT Randomisation method: Remote computer-generated Allocation method: Remote computer-generated

WISDOM 2007 *(Continued)*

	Blinding: All participants, clinic staff, and outcome assessors blinded except when vaginal bleeding triggered a code break Stratification: By hysterectomy status and intended use of HT: women with no uterus and unwilling to take placebo randomised to CEE or combined HT. equal probability of any treatment within each stratum. No of women screened for eligibility: 14,203 No randomised: 4385; 2196 on combined therapy, 2189 on placebo (See Notes) No analysed: 4385 Losses to follow up: 5 Dropouts: 615 (14%) had withdrawn from randomised treatment by trial closure Adherence to treatment: Trial treatment delivered 73% of time to women in combined HT arm and 86% of time to women on placebo. Analysis by intention to treat: Yes No of centres: 384 UK, 91 Australian and 24 NZ GP practices. Years of recruitment: 1999-2002 Design: Parallel Funding: Non-commercial medical research funding
Participants	Included: Postmenopausal women aged 50-69 Excluded: History of breast cancer, any other cancer in past 10 years except basal or squamous cell skin cancer, endometriosis or endometrial hyperplasia, venous thromboembolism, gall bladder disease, MI, unstable angina, CVA, subarachnoid haemorrhage, TIA, use of HT within past 6 months, unlikely to be able to give informed consent Mean age: 63 Age range: 50-69 Means of recruitment: GP practice registers Baseline equality of treatment groups: Country: UK, Australia, New Zealand
Interventions	HT arm: Daily CEE 0.625 mg plus medroxyprogesterone acetate 2.5 mg Control arm: Placebo Duration: Planned for median 10 years. but prematurely closed after median 11.9 months (range 7.1-19.6) Women with a uterus within 3 yrs of last period, those aged 50-53 and older women with unacceptable breakthrough bleeding took medroxyprogesterone acetate 5.0 mg. Women with a uterus who experienced unacceptable spotting or bleeding on the above therapy were offered open label CEE 0.625 mg plus medroxyprogesterone acetate 10.0 mg daily for the last 14 days of a 28 day cycle. All women took placebo medication during run in: those who achieved 80% compliance were randomised A further 1307 women were randomised to a comparison of oestrogen-only versus combined HT: these results not reported here
Outcomes	Major cardiovascular disease (primary) Osteoporotic fractures (primary) Breast cancer Mortality VTE CVD Dementia (No follow-up data collected) Adverse events
Notes	Powered in protocol to detect 25% reduction in CHD over 10 years - this assumed an 18,000 sample size but trial stopped early with 26% of target

WISDOM 2007 *(Continued)*

	A further 1307 women were in comparison of combined therapy vs oestrogen only and not included in this review.

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Remote computer-generated
Allocation concealment?	Low risk	Remote computer-generated
Blinding? All outcomes	Low risk	All participants, clinic staff, and outcome assessors blinded except when vaginal bleeding triggered a code break
Incomplete outcome data addressed? All outcomes	Low risk	615 (14%) had withdrawn from randomised treatment by trial closure. Analysed by intention to treat
Free of other bias?	Unclear risk	Powered in protocol to detect 25% reduction in CHD over 10 years - this assumed an 18,000 sample size but trial stopped early with 26% of target.

Yaffe 2006

Methods	Stated purpose: To investigate the effect of unopposed ultra-low-dose transdermal oestradiol on cognition and health-related quality of life in postmenopausal women. Randomisation method: Remotely by computer Allocation method: By pharmacy Blinding: Participants, investigators and outcome assessors blind Stratification: By clinical centre No of women screened for eligibility: 1509 (of whom 605 had one-week run in phase. 10/605 were non-compliant ad 1678 were found ineligible or refused to continue screening) No randomised: 417 (treatment group: 208; placebo group: 209) No analysed: analysed using a time x treatment interaction. 388 at year 1, 376 at year 2 Losses to follow up: 40 Dropouts: 41 Adherence to treatment: Among those who completed treatment, 84% used at least 75% of study drug during the entire 2 years Analysis by intention to treat: No of centres: 9 Years of recruitment: 1999-2000 Design: Parallel Funding: Industry funded
Participants	Included: Women aged 60-80 with intact uterus, at least 5 years post menopause, normal bone density Excluded: Women with unexplained uterine bleeding, endometrial hyperplasia, endometrium >mm double thickness on ultrasound, abnormal mammogram suggestive of breast cancer history of metabolic bone disease, cancer, coronary disease, stroke, TIA, VTE, uncontrolled hypertension, uncontrolled thyroid disease, liver disease, abnormal fasting triglyceride or fasting glucose, ever taken fluoride, calcitonin or bisphosphates, oestrogen or progestin within past 3 months Median age: 67 Means of recruitment: Not stated Baseline equality of treatment groups: Mean 3MS scores slightly higher in intervention group (P 0.04) Country: USA

Yaffe 2006 *(Continued)*

Interventions	HT arm: oestradiol patch delivering approx. 0.014 mg oestradiol daily, applied to abdomen weekly Control arm: identical placebo patch Duration: 2 years All participants also received 400 mg calcium twice daily and 400 iu vitamin D daily
Outcomes	Preplanned secondary outcome: Changes in cognition (MMSE) SF-36: Physical component scale and mental component scale. Bone mineral density was primary outcome (not reported here)
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Adequate sequence generation?	Low risk	Remotely by computer
Allocation concealment?	Low risk	By pharmacy
Blinding? All outcomes	Low risk	Participants, investigators and outcome assessors blind
Incomplete outcome data addressed? All outcomes	Low risk	40/417 (9.5%) women lost to follow up. Analysed by intention to treat
Free of other bias?	Low risk	No apparent source of bias

CABG: Coronary artery bypass graft
CEE: Conjugated equine oestrogen
CHD: Coronary heart disease
CVD:Cardiovascular disease
DSM IV: Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition
DVT: Deep vein thrombosis
DXA: Dual energy x-ray absorptiometry
ET: Oestrogen therapy
ERT: Oestrogen replacement therapy
FSH: Follicle stimulating hormone
MI: Myocardial infarction
MMSE: Mini Mental State Examination
MP: Micronised progesterone
MPA: Medroxyprogesterone acetate
MI: Myocardial infarction
PE: Pulmonary embolism
TIA: Transient ischaemic attack
VTE: Venous thrombo-embolism
Adherence to treatment refers to the number of tablets actually taken, which is often assessed by pill counts (see Additional Table 2)
Drop-outs: Participants who stopped their allocated treatment (and in some cases changed to a different off-trial treatment) but have known clinical outcomes and were included in analysis.
Intention to treat: the analysis of all randomised participants in the groups to which they were randomised.
Losses to follow up: Participants for whom the outcomes of interest were unknown (and who may or may not have had outcomes imputed in statistical analysis).

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Aitken 1971	No outcomes of interest
Aitken 1973	No outcomes of interest
Angerer 2000	Duration under one year
Bloch Thomsen 2002	No outcomes of interest
Chen 2001	No placebo, no outcomes of interest
Christiansen 1981	No outcomes of interest
Corrado 2002	No placebo, no outcomes of interest
Corson 1999	No placebo, no outcomes of interest
de Roo 1999	No outcomes of interest
Eiken 1996	No outcomes of interest
Estratab 1977	No outcomes of interest
EWA 2000	No placebo, no outcomes of interest
Genant 1990	No outcomes of interest
Graser 2001	Duration less than one year, no outcomes of interest
HABITS 2004	Not double-blinded
Hall 1998	Not double-blinded
Jensen 1985	No outcomes of interest
Kuopio 1998	Not blinded
Lufkin 1992	No outcomes of interest
Newhouse 2000	Duration less than one year
Ng 1992	No outcomes of interest
Ory 1998	No outcomes of interest
Os 2002	No placebo, no outcomes of interest
Papworth 2002	No placebo
Post 2001	No outcomes of interest
Saitta A 2001	Duration less than one year
Steiner 2007	Combines EPAT and WELL-HART data. No outcomes of interest
Teede 2002	No outcomes of interest

Study	Reason for exclusion
ULTRA 2005	No outcomes of interest
Virtanen 1999	Duration less than one year, no outcomes of interest

Characteristics of ongoing studies *[ordered by study ID]*

PREPARE 2002

Trial name or title	Prevent Postmenopausal Alzheimer's with Replacement Estrogens
Methods	
Participants	Women over 65 with a family history of Alzheimer's disease, not currently taking estrogen
Interventions	Oestrogen for women post hysterectomy or oestrogen plus progesterone for women with a uterus, versus placebo
Outcomes	Incidence of dementia
Starting date	1998
Contact information	Dr Mary Sano, Associate Professor of Clinical Neuropsychology, Gertrude Sergievsky Center, Columbia University
Notes	?Ended August 2003

DATA AND ANALYSES

Comparison 1. Women without major health problems (Selected outcomes: death, CVD, cognition, QOL)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Death from any cause: Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.1 Oestradiol 1 mg for 2 yrs	1	222	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.10]
1.2 CEE 0.625 mg for 6.8 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.03 [0.88, 1.21]
2 Death from any cause: Combined continuous HT (moderate dose oestrogen)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr	2	20993	Risk Ratio (M-H, Fixed, 95% CI)	1.31 [0.76, 2.27]
2.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.05 [0.71, 1.56]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs	2	17385	Risk Ratio (M-H, Fixed, 95% CI)	1.06 [0.79, 1.42]
2.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.84, 1.21]
3 Death from any cause: Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs	1	338	Risk Ratio (M-H, Fixed, 95% CI)	3.18 [0.13, 77.55]
3.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	352	Risk Ratio (M-H, Fixed, 95% CI)	4.89 [0.24, 101.09]
4 Death from coronary heart disease: Oestrogen-only HT ((moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.1 Oestradiol 1 mg daily for 2 yrs	1	222	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.10]
4.2 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.71, 1.43]
5 Death from coronary heart disease: Combined continuous HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
5.1 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.09 [0.69, 1.73]
6 Death from stroke: Oestrogen-only HT (moderate dose)	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.58, 2.32]
6.1 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.58, 2.32]
7 Death from stroke: Combined continuous HT (moderate dose)	2	20993	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [0.51, 2.46]
7.1 CEE 0.625 mg + MPA 2.5 mg for median 1 yr	1	4385	Risk Ratio (M-H, Fixed, 95% CI)	2.99 [0.12, 73.37]
7.2 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.04 [0.46, 2.35]
8 Death from breast cancer: Combined continuous HT (moderate dose oestrogen)	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.24, 3.81]
8.1 CEE 0.625 mg daily + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.24, 3.81]
9 Death from cancer: Combined continuous HT (moderate dose oestrogen)	2	17385	Risk Ratio (M-H, Fixed, 95% CI)	1.17 [0.88, 1.54]
9.1 CEE O.625 mg daily + MPA 2.5 mg for 3 yrs	1	777	Risk Ratio (M-H, Fixed, 95% CI)	2.77 [0.11, 67.80]
9.2 CEE 0.625 mg daily + MPA 2.5 mg for mean 5.2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.87, 1.53]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10 Coronary events (MI or cardiac death): Oestrogen-only HT (moderate dose)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
10.1 Oestradiol 1 mg for 2 yrs	1	222	Risk Ratio (M-H, Fixed, 95% CI)	0.5 [0.05, 5.43]
10.2 CEE 0.625 mg for 3 yrs	1	349	Risk Ratio (M-H, Fixed, 95% CI)	2.98 [0.12, 72.72]
10.3 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.78, 1.14]
11 Coronary events (MI or cardiac death): Combined continuous HT (moderate dose oestrogen)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
11.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr	2	20993	Risk Ratio (M-H, Fixed, 95% CI)	1.89 [1.15, 3.10]
11.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.49 [1.05, 2.12]
11.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs	2	17385	Risk Ratio (M-H, Fixed, 95% CI)	1.45 [1.07, 1.98]
11.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 years	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.22 [0.98, 1.51]
12 Coronary events (MI or cardiac death): Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
12.1 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	352	Risk Ratio (M-H, Fixed, 95% CI)	4.89 [0.24, 101.09]
13 Stroke: Unopposed oestrogen (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
13.1 Oestradiol 1 mg for 2 yrs	1	222	Risk Ratio (M-H, Fixed, 95% CI)	3.0 [0.12, 72.86]
13.2 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.35 [1.08, 1.70]
14 Stroke: Combined continuous HT (moderate dose oestrogen)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
14.1 CEE 0.625 mg + MPA 2.5 mg for 1 yr	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.49, 1.86]
14.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.31 [0.83, 2.06]
14.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs	2	17385	Risk Ratio (M-H, Fixed, 95% CI)	1.46 [1.02, 2.09]
14.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.34 [1.05, 1.72]
15 Stroke: Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
15.1 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	352	Risk Ratio (M-H, Fixed, 95% CI)	2.93 [0.12, 71.51]
16 Stroke or transient ischaemic attack	1	4385	Odds Ratio (M-H, Fixed, 95% CI)	0.73 [0.37, 1.47]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
16.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr	1	4385	Odds Ratio (M-H, Fixed, 95% CI)	0.73 [0.37, 1.47]
17 Transient ischaemic attack: Oestrogen-only HT (moderate dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
17.1 Oestradiol 1 mg for 2 yrs	1	222	Risk Ratio (M-H, Fixed, 95% CI)	3.0 [0.12, 72.86]
18 Transient ischaemic attack: Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
18.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs	1	348	Risk Ratio (M-H, Fixed, 95% CI)	3.0 [0.12, 73.14]
19 Venous thrombo-embolism (DVT or PE): Oestrogen-only HT ((moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
19.1 CEE 0.625 mg for up to 2 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	2.22 [1.12, 4.39]
19.2 CEE 0.625 mg for 3 yrs	1	349	Risk Ratio (M-H, Fixed, 95% CI)	6.96 [0.36, 133.75]
19.3 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.32 [1.00, 1.74]
20 Venous thrombo-embolism (DVT or PE): Combined continuous HT (moderate dose oestrogen)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
20.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr	2	20993	Risk Ratio (M-H, Fixed, 95% CI)	4.28 [2.49, 7.34]
20.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	2.98 [1.88, 4.71]
20.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	2.54 [1.73, 3.72]
20.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	2.09 [1.60, 2.74]
21 Venous thrombo-embolism (DVT or PE): Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
21.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs	1	348	Risk Ratio (M-H, Fixed, 95% CI)	3.0 [0.12, 73.14]
22 Global cognitive function: change difference in 3MSE scores: Oestrogen only (low dose)	1		Change difference (Fixed, 95% CI)	-0.31 [-0.74, 0.12]
22.1 Transdermal estradiol 0.014 mg (Baseline 3MSE =/< 90)	1		Change difference (Fixed, 95% CI)	-1.21 [-5.05, 2.63]
22.2 Transdermal estradiol 0.014 mg (Baseline 3MSE > 90)	1		Change difference (Fixed, 95% CI)	-0.3 [-0.73, 0.13]
23 Global cognitive function: change difference in 3MSE scores (combined cont HT (mod dose)	1		Change difference (Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
23.1 CEE 0.625 mg for mean 5.2 years	1		Change difference (Fixed, 95% CI)	-0.26 [-0.52, 0.00]
23.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.2 yrs	1		Change difference (Fixed, 95% CI)	-0.18 [-0.36, 0.00]
24 Large decline (> 2SD) in global cognitive function	1	7152	Risk Ratio (M-H, Fixed, 95% CI)	1.38 [1.08, 1.75]
24.1 CEE 0.625 mg for mean 5.2 years	1	2808	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.89, 1.70]
24.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.2 yrs	1	4344	Risk Ratio (M-H, Fixed, 95% CI)	1.57 [1.10, 2.24]
25 Mild cognitive impairment: Combined continuous HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
25.1 CEE 0.625 mg for mean 5.2 years	1	2947	Risk Ratio (M-H, Fixed, 95% CI)	1.33 [0.95, 1.85]
25.2 Combined continuous CEE 0.625mg + MPA 2.5 mg for mean 4.05 yrs	1	4532	Risk Ratio (M-H, Fixed, 95% CI)	1.05 [0.73, 1.52]
26 Probable dementia: Combined continuous HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
26.1 CEE 0.625 mg for mean 5.2 years	1	2947	Risk Ratio (M-H, Fixed, 95% CI)	1.49 [0.84, 2.66]
26.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.05 yrs	1	4532	Risk Ratio (M-H, Fixed, 95% CI)	1.97 [1.16, 3.33]
27 Mild cognitive impairment or probable dementia	1	7474	Risk Ratio (M-H, Fixed, 95% CI)	1.35 [1.08, 1.68]
27.1 CEE 0.625 mg for mean 5.4 years	1	2942	Risk Ratio (M-H, Fixed, 95% CI)	1.36 [1.01, 1.84]
27.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.05 yrs	1	4532	Risk Ratio (M-H, Fixed, 95% CI)	1.33 [0.97, 1.83]
28 Change in quality of life: General health (RAND 36): Oestrogen-only HT	1	10383	Mean Difference (IV, Fixed, 95% CI)	0.01 [-0.54, 0.55]
28.1 CEE 0.625 mg for 1 yr	1	9421	Mean Difference (IV, Fixed, 95% CI)	0.06 [-0.50, 0.62]
28.2 CEE 0.625 mg for 3 yrs	1	962	Mean Difference (IV, Fixed, 95% CI)	-0.61 [-2.50, 1.28]
29 Change in quality of life: General health (RAND 36): Combined continuous HT (moderate dose oestrogen)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
29.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	14942	Mean Difference (IV, Fixed, 95% CI)	0.30 [-0.13, 0.73]

Outcome or subgroup title	No. of studies	No. of partici-pants	Statistical method	Effect size
29.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1272	Mean Difference (IV, Fixed, 95% CI)	-0.10 [-1.78, 1.58]
30 Change difference in quality of life: Physical health (SF-36) Oestrogen only HT (low dose)	1		Change difference (Fixed, 95% CI)	-0.37 [-1.47, 0.73]
30.1 Transdermal oestradiol 0.014 mg for 2 years	1		Change difference (Fixed, 95% CI)	-0.37 [-1.47, 0.73]
31 Change in quality of life: Physical functioning (RAND 36): Oestrogen-only HT (mod dose)	1	10197	Mean Difference (IV, Fixed, 95% CI)	0.74 [0.13, 1.36]
31.1 CEE 0.625 mg for 1 yr	1	9258	Mean Difference (IV, Fixed, 95% CI)	0.69 [0.05, 1.33]
31.2 CEE 0.625 mg for 3 yrs	1	939	Mean Difference (IV, Fixed, 95% CI)	1.54 [-0.89, 3.97]
32 Change in quality of life: Physical functioning (RAND 36): Combined continuous HT (moderate dose oestrogen)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
32.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	14663	Mean Difference (IV, Fixed, 95% CI)	0.80 [0.36, 1.24]
32.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1248	Mean Difference (IV, Fixed, 95% CI)	-1.30 [-3.10, 0.50]
33 Change in quality of life: Role limitations due to physical problems (RAND 36):Oestrogen-only HT	1	10413	Mean Difference (IV, Fixed, 95% CI)	-0.14 [-1.59, 1.30]
33.1 CEE 0.625 mg for 1 yr	1	9447	Mean Difference (IV, Fixed, 95% CI)	-0.23 [-1.78, 1.32]
33.2 CEE 0.625 mg for 3 yrs	1	966	Mean Difference (IV, Fixed, 95% CI)	0.43 [-3.60, 4.46]
34 Change in quality of life: Role limitations due to physical problems (RAND 36): Combined cont. HT (mod dose)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
34.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	14985	Mean Difference (IV, Fixed, 95% CI)	1.4 [0.30, 2.50]
34.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1269	Mean Difference (IV, Fixed, 95% CI)	-0.10 [-4.25, 4.05]
35 Change in quality of life: Bodily pain (RAND 36): Oestrogen-only HT	1	10628	Mean Difference (IV, Fixed, 95% CI)	0.76 [-0.12, 1.63]
35.1 CEE 0.625 mg for 1 yr	1	9649	Mean Difference (IV, Fixed, 95% CI)	0.76 [-0.15, 1.67]

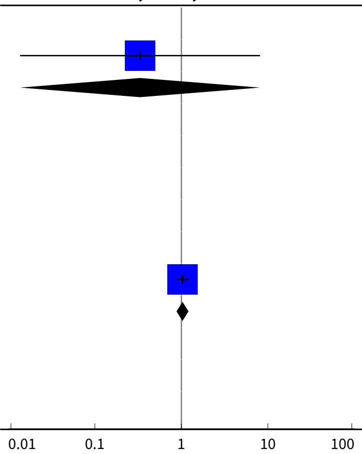
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
35.2 CEE 0.625 mg for 3 yrs	1	979	Mean Difference (IV, Fixed, 95% CI)	0.71 [-2.40, 3.82]
36 Change in quality of life: Bodily pain (RAND 36): Combined continuous HT (moderate dose oestrogen)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
36.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	15209	Mean Difference (IV, Fixed, 95% CI)	1.90 [1.23, 2.57]
36.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1292	Mean Difference (IV, Fixed, 95% CI)	1.60 [-0.86, 4.06]
37 Change in quality of life:Energy and fatigue (RAND 36): Oestrogen-only HT	1	10373	Mean Difference (IV, Fixed, 95% CI)	-0.03 [-0.66, 0.60]
37.1 CEE 0.625 mg for 1 yr	1	9419	Mean Difference (IV, Fixed, 95% CI)	-0.12 [-0.77, 0.53]
37.2 CEE 0.625 mg for 3 yrs	1	954	Mean Difference (IV, Fixed, 95% CI)	0.98 [-1.24, 3.20]
38 Change in quality of life:Energy and fatigue (RAND 36): Combined continuous HT (moderate dose oestrogen)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
38.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	14941	Mean Difference (IV, Fixed, 95% CI)	0.2 [-0.29, 0.69]
38.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1241	Mean Difference (IV, Fixed, 95% CI)	-0.3 [-2.10, 1.50]
39 Change in quality of life: Social functioning (RAND 36): Oestrogen-only HT	1	10549	Mean Difference (IV, Fixed, 95% CI)	-1.31 [-2.15, -0.47]
39.1 CEE 0.625 mg for 1 yr	1	9572	Mean Difference (IV, Fixed, 95% CI)	-1.32 [-2.20, -0.44]
39.2 CEE 0.625 mg for 3 yrs	1	977	Mean Difference (IV, Fixed, 95% CI)	-1.24 [-4.18, 1.70]
40 Change in quality of life: Social functioning (RAND 36): Combined continuous HT (moderate dose oestrogen)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
40.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	15117	Mean Difference (IV, Fixed, 95% CI)	-0.10 [-0.71, 0.51]
40.2 Combined CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1285	Mean Difference (IV, Fixed, 95% CI)	0.5 [-1.72, 2.72]
41 Change in quality of life: Role limitations due to emotional problems (RAND 36):Oestrogen-only HT	1	10457	Mean Difference (IV, Fixed, 95% CI)	-0.95 [-2.07, 0.17]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
41.1 CEE 0.625 mg for 1 yr	1	9483	Mean Difference (IV, Fixed, 95% CI)	-1.13 [-2.28, 0.02]
41.2 CEE 0.625 mg for 3 yrs	1	974	Mean Difference (IV, Fixed, 95% CI)	2.16 [-2.67, 6.99]
42 Change in quality of life: Role limitations due to emotional problems (RAND 36): Combined cont. HT (mod dose)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
42.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	14985	Mean Difference (IV, Fixed, 95% CI)	-0.20 [-1.18, 0.78]
42.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1279	Mean Difference (IV, Fixed, 95% CI)	-1.70 [-5.29, 1.89]
43 Change difference in quality of life: Mental health (SF-36) Oestrogen only HT (low dose)	1		Change difference (Fixed, 95% CI)	-0.95 [-2.17, 0.27]
43.1 Transdermal oestradiol 0.014 mg for 2 years	1		Change difference (Fixed, 95% CI)	-0.95 [-2.17, 0.27]
44 Change in quality of life: Mental health (RAND 36): Oestrogen-only HT (mod dose)	1	9403	Mean Difference (IV, Fixed, 95% CI)	0.04 [-0.50, 0.58]
44.1 CEE 0.625 mg for 1 yr	1	9401	Mean Difference (IV, Fixed, 95% CI)	0.04 [-0.50, 0.58]
44.2 CEE 0.625 mg for 3 yrs	1	2	Mean Difference (IV, Fixed, 95% CI)	0.0 [0.0, 0.0]
45 Change in quality of life: Mental health (RAND 36): Combined continuous HT (moderate dose oestrogen)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
45.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)	1	14910	Mean Difference (IV, Fixed, 95% CI)	-0.10 [-0.49, 0.29]
45.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)	1	1254	Mean Difference (IV, Fixed, 95% CI)	1.0 [-0.52, 2.52]
46 Change in quality of life overall (HQOL): Oestrogen-only HT (moderate dose)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only
46.1 Oestradiol 1 mg daily (1 yr)	1	102	Mean Difference (IV, Fixed, 95% CI)	-0.41 [-8.95, 8.13]
46.2 CEE 0.625 mg for 1 yr	0	0	Mean Difference (IV, Fixed, 95% CI)	0.0 [0.0, 0.0]
47 Change in quality of life overall (HQOL): Oestrogen-only HT (mod/high dose)	1		Mean Difference (IV, Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
47.1 Unopposed oestradiol 2 mg daily (1 yr)	1	100	Mean Difference (IV, Fixed, 95% CI)	-3.31 [-11.79, 5.17]
48 Change in quality of life score (GHQ-11 scale) (combined HT)			Other data	No numeric data
48.1 Combined continuous: oestradiol 2 mg + norethisterone 1 mg (2 yrs)			Other data	No numeric data
48.2 Combined sequential: oestradiol (2 mg days 1-12, 1 mg days 13-22) + norethisterone 1 mg days 13-22) for 2 yrs			Other data	No numeric data

Analysis 1.1. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 1 Death from any cause: Oestrogen-only HT (moderate dose).

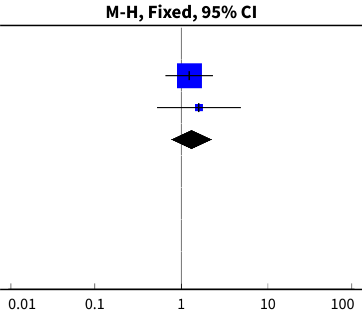
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.1.1 Oestradiol 1 mg for 2 yrs					
EPAT 2001	0/111	1/111		100%	0.33[0.01,8.1]
Subtotal (95% CI)	**111**	**111**		**100%**	**0.33[0.01,8.1]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
1.1.2 CEE 0.625 mg for 6.8 yrs					
WHI 1998	291/5310	289/5429		100%	1.03[0.88,1.21]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**1.03[0.88,1.21]**
Total events: 291 (Treatment), 289 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.36(P=0.72)					



Favours treatment | Favours control

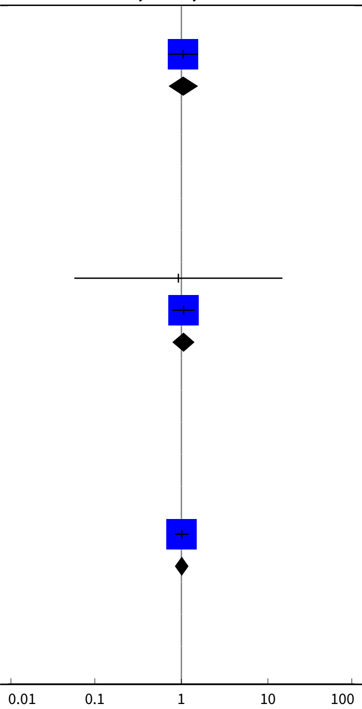
Analysis 1.2. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 2 Death from any cause: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.2.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr					
WHI 1998	22/8506	17/8102		77.66%	1.23[0.66,2.32]
WISDOM 2007	8/2196	5/2189		22.34%	1.59[0.52,4.87]
Subtotal (95% CI)	**10702**	**10291**		**100%**	**1.31[0.76,2.27]**
Total events: 30 (Treatment), 22 (Control)					
Heterogeneity: Tau²=0; Chi²=0.16, df=1(P=0.69); I²=0%					
Test for overall effect: Z=0.97(P=0.33)					



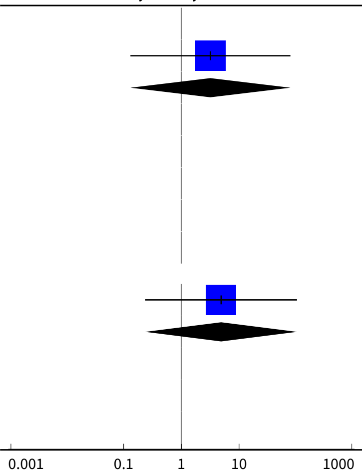
Favours treatment | Favours control

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.2.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	52/8506	47/8102		100%	1.05[0.71,1.56]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.05[0.71,1.56]**
Total events: 52 (Treatment), 47 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.26(P=0.79)					
1.2.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs					
EPHT 2006	1/404	1/373		1.22%	0.92[0.06,14.71]
WHI 1998	91/8506	82/8102		98.78%	1.06[0.79,1.42]
Subtotal (95% CI)	**8910**	**8475**		**100%**	**1.06[0.79,1.42]**
Total events: 92 (Treatment), 83 (Control)					
Heterogeneity: Tau²=0; Chi²=0.01, df=1(P=0.92); I²=0%					
Test for overall effect: Z=0.36(P=0.72)					
1.2.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.2 yrs					
WHI 1998	231/8506	218/8102		100%	1.01[0.84,1.21]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.01[0.84,1.21]**
Total events: 231 (Treatment), 218 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.1(P=0.92)					



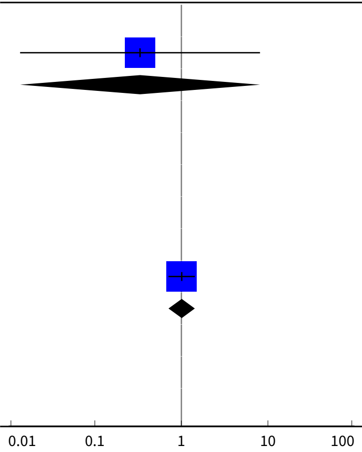
Analysis 1.3. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 3 Death from any cause: Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.3.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs					
PEPI 1995	1/164	0/174		100%	3.18[0.13,77.55]
Subtotal (95% CI)	**164**	**174**		**100%**	**3.18[0.13,77.55]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.71(P=0.48)					
1.3.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	2/178	0/174		100%	4.89[0.24,101.09]
Subtotal (95% CI)	**178**	**174**		**100%**	**4.89[0.24,101.09]**
Total events: 2 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.03(P=0.3)					



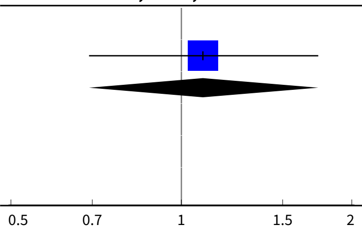
Analysis 1.4. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 4 Death from coronary heart disease: Oestrogen-only HT ((moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.4.1 Oestradiol 1 mg daily for 2 yrs					
EPAT 2001	0/111	1/111		100%	0.33[0.01,8.1]
Subtotal (95% CI)	**111**	**111**		**100%**	**0.33[0.01,8.1]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
1.4.2 CEE 0.625 mg for 7.1 yrs					
WHI 1998	62/5310	63/5429		100%	1.01[0.71,1.43]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**1.01[0.71,1.43]**
Total events: 62 (Treatment), 63 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.03(P=0.97)					



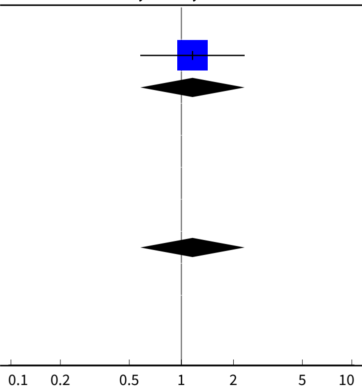
Analysis 1.5. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 5 Death from coronary heart disease: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.5.1 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	39/8506	34/8102		100%	1.09[0.69,1.73]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.09[0.69,1.73]**
Total events: 39 (Treatment), 34 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.38(P=0.71)					



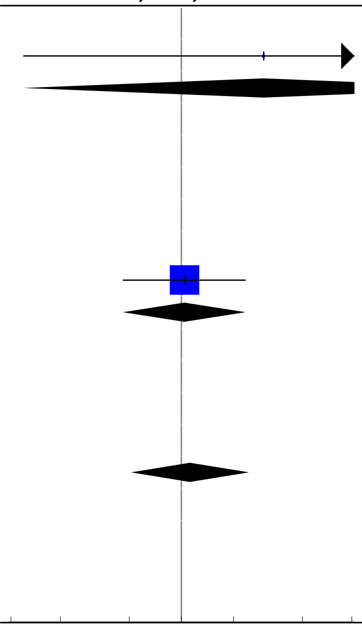
Analysis 1.6. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 6 Death from stroke: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.6.1 CEE 0.625 mg for 7.1 yrs					
WHI 1998	17/5310	15/5429		100%	1.16[0.58,2.32]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**1.16[0.58,2.32]**
Total events: 17 (Treatment), 15 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.42(P=0.68)					
Total (95% CI)	**5310**	**5429**		**100%**	**1.16[0.58,2.32]**
Total events: 17 (Treatment), 15 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.42(P=0.68)					



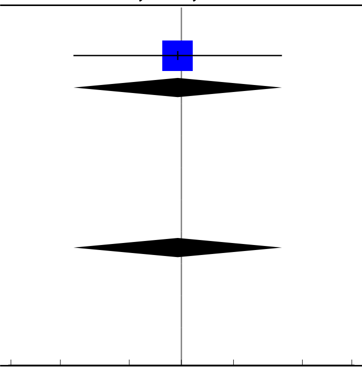
Analysis 1.7. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 7 Death from stroke: Combined continuous HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.7.1 CEE 0.625 mg + MPA 2.5 mg for median 1 yr					
WISDOM 2007	1/2196	0/2189		4.26%	2.99[0.12,73.37]
Subtotal (95% CI)	**2196**	**2189**		**4.26%**	**2.99[0.12,73.37]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.67(P=0.5)					
1.7.2 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	12/8506	11/8102		95.74%	1.04[0.46,2.35]
Subtotal (95% CI)	**8506**	**8102**		**95.74%**	**1.04[0.46,2.35]**
Total events: 12 (Treatment), 11 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.09(P=0.93)					
Total (95% CI)	**10702**	**10291**		**100%**	**1.12[0.51,2.46]**
Total events: 13 (Treatment), 11 (Control)					
Heterogeneity: $Tau^2=0$; $Chi^2=0.39$, df=1(P=0.53); $I^2=0\%$					
Test for overall effect: Z=0.29(P=0.77)					
Test for subgroup differences: Not applicable					



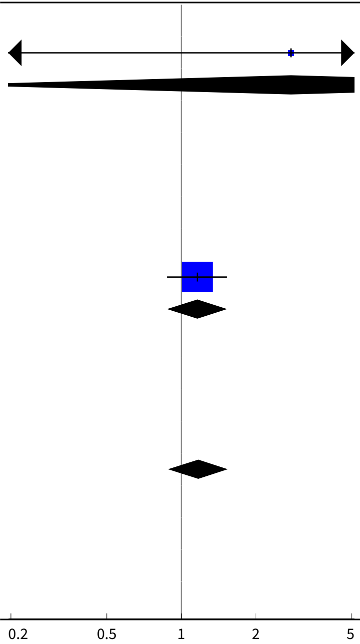
Analysis 1.8. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 8 Death from breast cancer: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.8.1 CEE 0.625 mg daily + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	4/8506	4/8102		100%	0.95[0.24,3.81]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.95[0.24,3.81]**
Total events: 4 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.07(P=0.95)					
Total (95% CI)	**8506**	**8102**		**100%**	**0.95[0.24,3.81]**
Total events: 4 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.07(P=0.95)					



Analysis 1.9. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 9 Death from cancer: Combined continuous HT (moderate dose oestrogen).

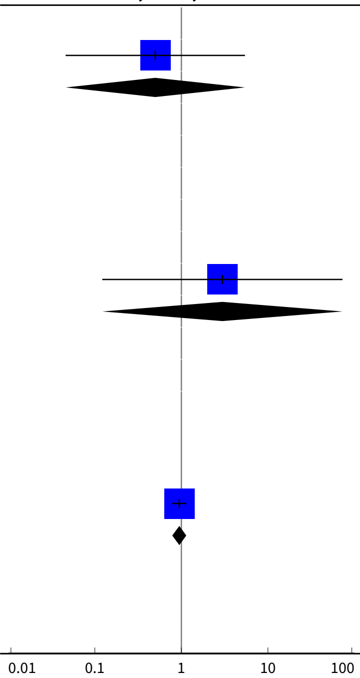
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.9.1 CEE 0.625 mg daily + MPA 2.5 mg for 3 yrs					
EPHT 2006	1/404	0/373		0.57%	2.77[0.11,67.8]
Subtotal (95% CI)	**404**	**373**		**0.57%**	**2.77[0.11,67.8]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.62(P=0.53)					
1.9.2 CEE 0.625 mg daily + MPA 2.5 mg for mean 5.2 yrs					
WHI 1998	107/8506	88/8102		99.43%	1.16[0.87,1.53]
Subtotal (95% CI)	**8506**	**8102**		**99.43%**	**1.16[0.87,1.53]**
Total events: 107 (Treatment), 88 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.03(P=0.3)					
Total (95% CI)	**8910**	**8475**		**100%**	**1.17[0.88,1.54]**
Total events: 108 (Treatment), 88 (Control)					
Heterogeneity: Tau²=0; Chi²=0.28, df=1(P=0.59); I²=0%					
Test for overall effect: Z=1.09(P=0.28)					
Test for subgroup differences: Not applicable					



Favours treatment | Favours control

Analysis 1.10. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 10 Coronary events (MI or cardiac death): Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.10.1 Oestradiol 1 mg for 2 yrs					
EPAT 2001	1/111	2/111		100%	0.5[0.05,5.43]
Subtotal (95% CI)	**111**	**111**		**100%**	**0.5[0.05,5.43]**
Total events: 1 (Treatment), 2 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.57(P=0.57)					
1.10.2 CEE 0.625 mg for 3 yrs					
PEPI 1995	1/175	0/174		100%	2.98[0.12,72.72]
Subtotal (95% CI)	**175**	**174**		**100%**	**2.98[0.12,72.72]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.67(P=0.5)					
1.10.3 CEE 0.625 mg for 7.1 yrs					
WHI 1998	201/5310	217/5429		100%	0.95[0.78,1.14]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**0.95[0.78,1.14]**
Total events: 201 (Treatment), 217 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.57(P=0.57)					



Favours treatment | Favours control

Analysis 1.11. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 11 Coronary events (MI or cardiac death): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.11.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr					
WHI 1998	42/8506	23/8102		97.92%	1.74[1.05,2.89]
WISDOM 2007	4/2196	0/2189		2.08%	8.97[0.48,166.53]
Subtotal (95% CI)	**10702**	**10291**		**100%**	**1.89[1.15,3.1]**
Total events: 46 (Treatment), 23 (Control)					
Heterogeneity: Tau²=0; Chi²=1.19, df=1(P=0.27); I²=16.31%					
Test for overall effect: Z=2.52(P=0.01)					
1.11.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	80/8506	51/8102		100%	1.49[1.05,2.12]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.49[1.05,2.12]**
Total events: 80 (Treatment), 51 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.25(P=0.02)					
1.11.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs					
EPHT 2006	2/404	0/373		0.76%	4.62[0.22,95.86]
WHI 1998	99/8506	66/8102		99.24%	1.43[1.05,1.95]
Subtotal (95% CI)	**8910**	**8475**		**100%**	**1.45[1.07,1.98]**
Total events: 101 (Treatment), 66 (Control)					
Heterogeneity: Tau²=0; Chi²=0.57, df=1(P=0.45); I²=0%					
Test for overall effect: Z=2.38(P=0.02)					
1.11.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 years					
WHI 1998	188/8506	147/8102		100%	1.22[0.98,1.51]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.22[0.98,1.51]**
Total events: 188 (Treatment), 147 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.81(P=0.07)					

Favours treatment 0.001 0.1 1 10 1000 Favours control

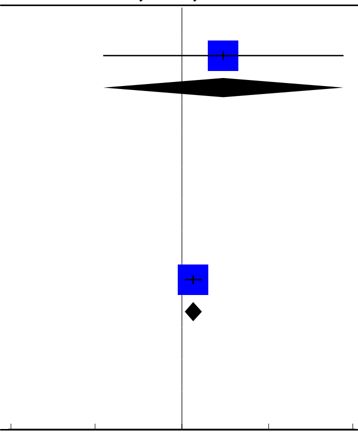
Analysis 1.12. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 12 Coronary events (MI or cardiac death): Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.12.1 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	2/178	0/174		100%	4.89[0.24,101.09]
Subtotal (95% CI)	**178**	**174**		**100%**	**4.89[0.24,101.09]**
Total events: 2 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.03(P=0.3)					

Favours treatment 0.001 0.1 1 10 1000 Favours control

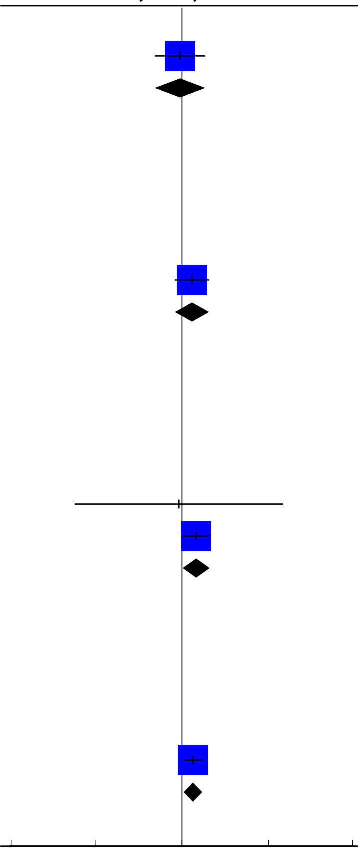
Analysis 1.13. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 13 Stroke: Unopposed oestrogen (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.13.1 Oestradiol 1 mg for 2 yrs					
EPAT 2001	1/111	0/111		100%	3[0.12,72.86]
Subtotal (95% CI)	**111**	**111**		**100%**	**3[0.12,72.86]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
1.13.2 CEE 0.625 mg for 7.1 yrs					
WHI 1998	168/5310	127/5429		100%	1.35[1.08,1.7]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**1.35[1.08,1.7]**
Total events: 168 (Treatment), 127 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.6(P=0.01)					



Analysis 1.14. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 14 Stroke: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.14.1 CEE 0.625 mg + MPA 2.5 mg for 1 yr					
WHI 1998	17/8506	17/8102		100%	0.95[0.49,1.86]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.95[0.49,1.86]**
Total events: 17 (Treatment), 17 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.14(P=0.89)					
1.14.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	44/8506	32/8102		100%	1.31[0.83,2.06]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.31[0.83,2.06]**
Total events: 44 (Treatment), 32 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.16(P=0.24)					
1.14.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs					
EPHT 2006	1/404	1/373		2.07%	0.92[0.06,14.71]
WHI 1998	74/8506	48/8102		97.93%	1.47[1.02,2.11]
Subtotal (95% CI)	**8910**	**8475**		**100%**	**1.46[1.02,2.09]**
Total events: 75 (Treatment), 49 (Control)					
Heterogeneity: $Tau^2=0$; $Chi^2=0.11$, df=1(P=0.74); $I^2=0\%$					
Test for overall effect: Z=2.06(P=0.04)					
1.14.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	151/8506	107/8102		100%	1.34[1.05,1.72]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.34[1.05,1.72]**
Total events: 151 (Treatment), 107 (Control)					



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Heterogeneity: Not applicable					
Test for overall effect: Z=2.36(P=0.02)					

Favours treatment 0.01 0.1 1 10 100 Favours control

Analysis 1.15. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 15 Stroke: Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.15.1 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	1/178	0/174		100%	2.93[0.12,71.51]
Subtotal (95% CI)	**178**	**174**		**100%**	**2.93[0.12,71.51]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.66(P=0.51)					

Favours treatment 0.01 0.1 1 10 100 Favours control

Analysis 1.16. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 16 Stroke or transient ischaemic attack.

Study or subgroup	Treatment n/N	Control n/N	Odds Ratio M-H, Fixed, 95% CI	Weight	Odds Ratio M-H, Fixed, 95% CI
1.16.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr					
WISDOM 2007	14/2196	19/2189		100%	0.73[0.37,1.47]
Subtotal (95% CI)	**2196**	**2189**		**100%**	**0.73[0.37,1.47]**
Total events: 14 (Treatment), 19 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.88(P=0.38)					
Total (95% CI)	**2196**	**2189**		**100%**	**0.73[0.37,1.47]**
Total events: 14 (Treatment), 19 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.88(P=0.38)					

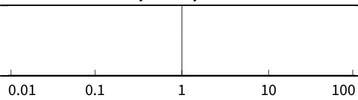
Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Analysis 1.17. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 17 Transient ischaemic attack: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.17.1 Oestradiol 1 mg for 2 yrs					
EPAT 2001	1/111	0/111		100%	3[0.12,72.86]
Subtotal (95% CI)	**111**	**111**		**100%**	**3[0.12,72.86]**
Total events: 1 (Treatment), 0 (Control)					

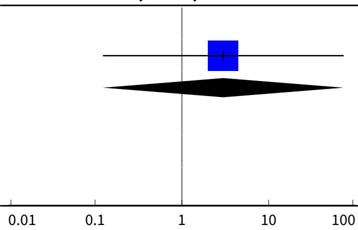
Favours treatment 0.01 0.1 1 10 100 Favours control

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	



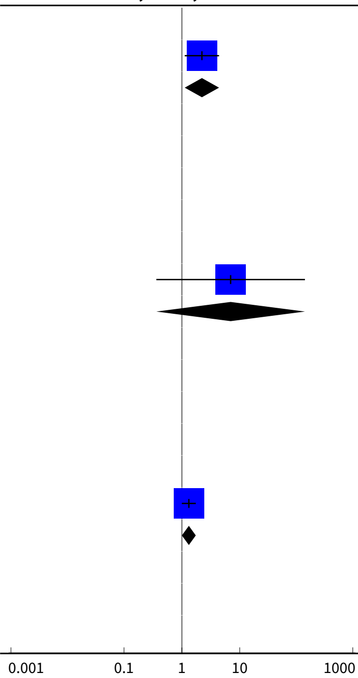
Analysis 1.18. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 18 Transient ischaemic attack: Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.18.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs					
PEPI 1995	1/174	0/174		100%	3[0.12,73.14]
Subtotal (95% CI)	**174**	**174**		**100%**	**3[0.12,73.14]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.67(P=0.5)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	



Analysis 1.19. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 19 Venous thrombo-embolism (DVT or PE): Oestrogen-only HT ((moderate dose).

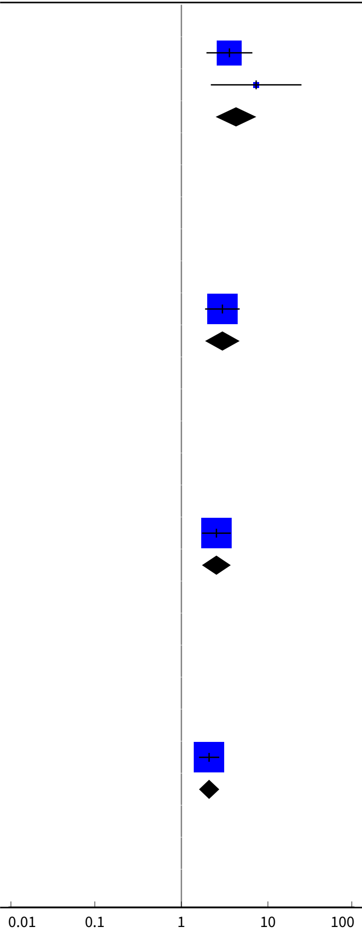
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.19.1 CEE 0.625 mg for up to 2 yrs					
WHI 1998	26/5310	12/5429		100%	2.22[1.12,4.39]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**2.22[1.12,4.39]**
Total events: 26 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.28(P=0.02)					
1.19.2 CEE 0.625 mg for 3 yrs					
PEPI 1995	3/175	0/174		100%	6.96[0.36,133.75]
Subtotal (95% CI)	**175**	**174**		**100%**	**6.96[0.36,133.75]**
Total events: 3 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.29(P=0.2)					
1.19.3 CEE 0.625 mg for 7.1 yrs					
WHI 1998	111/5310	86/5429		100%	1.32[1,1.74]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**1.32[1,1.74]**
Total events: 111 (Treatment), 86 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.95(P=0.05)					
		Favours treatment	0.001 0.1 1 10 1000	Favours control	



Analysis 1.20. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 20 Venous thrombo-embolism (DVT or PE): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
1.20.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr				
WHI 1998	49/8506	13/8102	81.59%	3.59[1.95,6.61]
WISDOM 2007	22/2196	3/2189	18.41%	7.31[2.19,24.39]
Subtotal (95% CI)	**10702**	**10291**	**100%**	**4.28[2.49,7.34]**
Total events: 71 (Treatment), 16 (Control)				
Heterogeneity: Tau²=0; Chi²=1.08, df=1(P=0.3); I²=7.02%				
Test for overall effect: Z=5.26(P<0.0001)				
1.20.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs				
WHI 1998	75/8506	24/8102	100%	2.98[1.88,4.71]
Subtotal (95% CI)	**8506**	**8102**	**100%**	**2.98[1.88,4.71]**
Total events: 75 (Treatment), 24 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=4.66(P<0.0001)				
1.20.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs				
WHI 1998	96/8506	36/8102	100%	2.54[1.73,3.72]
Subtotal (95% CI)	**8506**	**8102**	**100%**	**2.54[1.73,3.72]**
Total events: 96 (Treatment), 36 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=4.78(P<0.0001)				
1.20.4 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs				
WHI 1998	167/8506	76/8102	100%	2.09[1.6,2.74]
Subtotal (95% CI)	**8506**	**8102**	**100%**	**2.09[1.6,2.74]**
Total events: 167 (Treatment), 76 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=5.37(P<0.0001)				

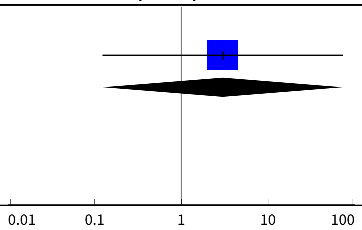
Risk Ratio M-H, Fixed, 95% CI



Analysis 1.21. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 21 Venous thrombo-embolism (DVT or PE): Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
1.21.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs				
PEPI 1995	1/174	0/174	100%	3[0.12,73.14]
Subtotal (95% CI)	**174**	**174**	**100%**	**3[0.12,73.14]**
Total events: 1 (Treatment), 0 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=0.67(P=0.5)				

Risk Ratio M-H, Fixed, 95% CI



Analysis 1.22. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 22 Global cognitive function: change difference in 3MSE scores: Oestrogen only (low dose).

Study or subgroup	Control N	Treatment N	Change difference (SE)	Change difference IV, Fixed, 95% CI	Weight	Change difference IV, Fixed, 95% CI
1.22.1 Transdermal estradiol 0.014 mg (Baseline 3MSE =/< 90)						
Yaffe 2006	1	1	-1.2 (1.96)		1.24%	-1.21[-5.05,2.63]
Subtotal (95% CI)					**1.24%**	**-1.21[-5.05,2.63]**
Heterogeneity: Not applicable						
Test for overall effect: Z=0.62(P=0.54)						
1.22.2 Transdermal estradiol 0.014 mg (Baseline 3MSE > 90)						
Yaffe 2006	1	1	-0.3 (0.22)		98.76%	-0.3[-0.73,0.13]
Subtotal (95% CI)					**98.76%**	**-0.3[-0.73,0.13]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.36(P=0.17)						
Total (95% CI)					**100%**	**-0.31[-0.74,0.12]**
Heterogeneity: Tau²=0; Chi²=0.21, df=1(P=0.64); I²=0%						
Test for overall effect: Z=1.42(P=0.15)						
Test for subgroup differences: Chi²=0.21, df=1 (P=0.64), I²=0%						

Favours control -10 -5 0 5 10 Favours treatment

Analysis 1.23. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 23 Global cognitive function: change difference in 3MSE scores (combined cont HT (mod dose).

Study or subgroup	Control N	Treatment N	Change difference (SE)	Change difference IV, Fixed, 95% CI	Weight	Change difference IV, Fixed, 95% CI
1.23.1 CEE 0.625 mg for mean 5.2 years						
WHI 1998	1	1	-0.3 (0.133)		100%	-0.26[-0.52,0]
Subtotal (95% CI)					**100%**	**-0.26[-0.52,0]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.96(P=0.05)						
1.23.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.2 yrs						
WHI 1998	1	1	-0.2 (0.094)		100%	-0.18[-0.36,0]
Subtotal (95% CI)					**100%**	**-0.18[-0.36,0]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.91(P=0.06)						
Test for subgroup differences: Chi²=0.24, df=1 (P=0.62), I²=0%						

Favours control -1 -0.5 0 0.5 1 Favours treatment

Analysis 1.24. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 24 Large decline (> 2SD) in global cognitive function.

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.24.1 CEE 0.625 mg for mean 5.2 years					

Favours treatment 0.2 0.5 1 2 5 Favours control

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
WHI 1998	77/1387	64/1421		56.81%	1.23[0.89,1.7]
Subtotal (95% CI)	**1387**	**1421**		**56.81%**	**1.23[0.89,1.7]**
Total events: 77 (Treatment), 64 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.27(P=0.2)					
1.24.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.2 yrs					
WHI 1998	74/2131	49/2213		43.19%	1.57[1.1,2.24]
Subtotal (95% CI)	**2131**	**2213**		**43.19%**	**1.57[1.1,2.24]**
Total events: 74 (Treatment), 49 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.48(P=0.01)					
Total (95% CI)	**3518**	**3634**		**100%**	**1.38[1.08,1.75]**
Total events: 151 (Treatment), 113 (Control)					
Heterogeneity: Tau²=0; Chi²=0.96, df=1(P=0.33); I²=0%					
Test for overall effect: Z=2.63(P=0.01)					
Test for subgroup differences: Not applicable					
		Favours treatment	0.2 0.5 1 2 5	Favours control	

Analysis 1.25. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 25 Mild cognitive impairment: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.25.1 CEE 0.625 mg for mean 5.2 years					
WHI 1998	76/1464	58/1483		100%	1.33[0.95,1.85]
Subtotal (95% CI)	**1464**	**1483**		**100%**	**1.33[0.95,1.85]**
Total events: 76 (Treatment), 58 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.66(P=0.1)					
1.25.2 Combined continuous CEE 0.625mg + MPA 2.5 mg for mean 4.05 yrs					
WHI 1998	56/2229	55/2303		100%	1.05[0.73,1.52]
Subtotal (95% CI)	**2229**	**2303**		**100%**	**1.05[0.73,1.52]**
Total events: 56 (Treatment), 55 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.27(P=0.79)					
		Favours treatment	0.5 0.7 1 1.5 2	Favours control	

Analysis 1.26. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 26 Probable dementia: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.26.1 CEE 0.625 mg for mean 5.2 years					
WHI 1998	28/1464	19/1483		100%	1.49[0.84,2.66]
Subtotal (95% CI)	**1464**	**1483**		**100%**	**1.49[0.84,2.66]**
Total events: 28 (Treatment), 19 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.36(P=0.17)					
1.26.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.05 yrs					
WHI 1998	40/2229	21/2303		100%	1.97[1.16,3.33]
Subtotal (95% CI)	**2229**	**2303**		**100%**	**1.97[1.16,3.33]**
Total events: 40 (Treatment), 21 (Control)					
Heterogeneity: $Tau^2=0$; $Chi^2=0$, df=0(P<0.0001); $I^2=100\%$					
Test for overall effect: Z=2.53(P=0.01)					

Favours treatment 0.2 0.5 1 2 5 Favours control

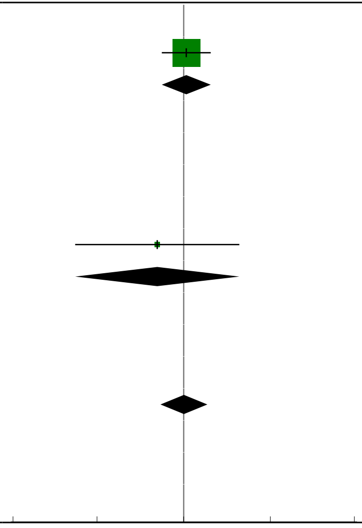
Analysis 1.27. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 27 Mild cognitive impairment or probable dementia.

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.27.1 CEE 0.625 mg for mean 5.4 years					
WHI 1998	93/1463	69/1479		51.39%	1.36[1.01,1.84]
Subtotal (95% CI)	**1463**	**1479**		**51.39%**	**1.36[1.01,1.84]**
Total events: 93 (Treatment), 69 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2(P=0.05)					
1.27.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.05 yrs					
WHI 1998	85/2229	66/2303		48.61%	1.33[0.97,1.83]
Subtotal (95% CI)	**2229**	**2303**		**48.61%**	**1.33[0.97,1.83]**
Total events: 85 (Treatment), 66 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.77(P=0.08)					
Total (95% CI)	**3692**	**3782**		**100%**	**1.35[1.08,1.68]**
Total events: 178 (Treatment), 135 (Control)					
Heterogeneity: $Tau^2=0$; $Chi^2=0.01$, df=1(P=0.92); $I^2=0\%$					
Test for overall effect: Z=2.67(P=0.01)					
Test for subgroup differences: Not applicable					

Favours treatment 0.5 0.7 1 1.5 2 Favours control

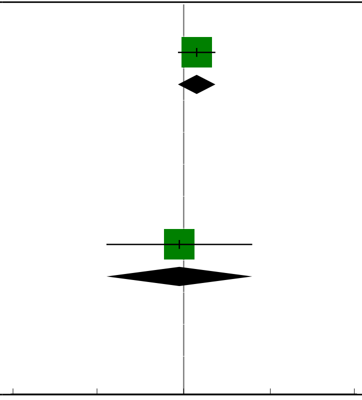
Analysis 1.28. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 28 Change in quality of life: General health (RAND 36): Oestrogen-only HT.

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Weight	Mean Difference Fixed, 95% CI
1.28.1 CEE 0.625 mg for 1 yr						
WHI 1998	4654	-0.3 (13.7)	4767	-0.4 (14.2)	91.84%	0.06[-0.5,0.62]
Subtotal ***	**4654**		**4767**		**91.84%**	**0.06[-0.5,0.62]**
Heterogeneity: Not applicable						
Test for overall effect: Z=0.21(P=0.83)						
1.28.2 CEE 0.625 mg for 3 yrs						
WHI 1998	475	-3.5 (14.9)	487	-2.9 (15.1)	8.16%	-0.61[-2.5,1.28]
Subtotal ***	**475**		**487**		**8.16%**	**-0.61[-2.5,1.28]**
Heterogeneity: Not applicable						
Test for overall effect: Z=0.63(P=0.53)						
Total ***	**5129**		**5254**		**100%**	**0.01[-0.54,0.55]**
Heterogeneity: Tau²=0; Chi²=0.44, df=1(P=0.51); I²=0%						
Test for overall effect: Z=0.02(P=0.98)						
Test for subgroup differences: Chi²=0.44, df=1 (P=0.51), I²=0%						



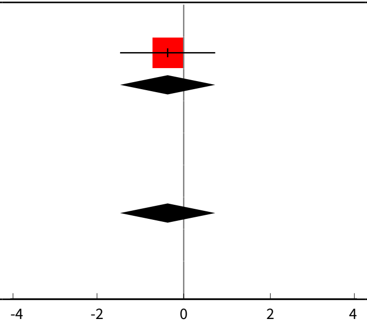
Analysis 1.29. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 29 Change in quality of life: General health (RAND 36): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Weight	Mean Difference Fixed, 95% CI
1.29.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)						
WHI 1998	7637	-0.4 (13.4)	7305	-0.7 (13.5)	100%	0.3[-0.13,0.73]
Subtotal ***	**7637**		**7305**		**100%**	**0.3[-0.13,0.73]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.36(P=0.17)						
1.29.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)						
WHI 1998	652	-2.2 (15.9)	620	-2.1 (14.7)	100%	-0.1[-1.78,1.58]
Subtotal ***	**652**		**620**		**100%**	**-0.1[-1.78,1.58]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%						
Test for overall effect: Z=0.12(P=0.91)						
Test for subgroup differences: Chi²=0.2, df=1 (P=0.65), I²=0%						



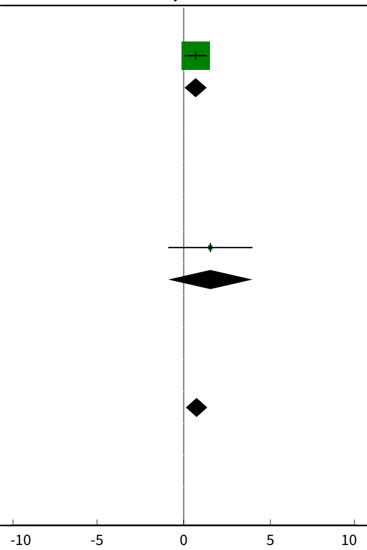
Analysis 1.30. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 30 Change difference in quality of life: Physical health (SF-36) Oestrogen only HT (low dose).

Study or subgroup	Control N	Treatment N	Change difference (SE)	Weight	Change difference IV, Fixed, 95% CI
1.30.1 Transdermal oestradiol 0.014 mg for 2 years					
Yaffe 2006	1	1	-0.4 (0.56)	100%	-0.37[-1.47,0.73]
Subtotal (95% CI)				**100%**	**-0.37[-1.47,0.73]**
Heterogeneity: Not applicable					
Test for overall effect: Z=0.66(P=0.51)					
Total (95% CI)				**100%**	**-0.37[-1.47,0.73]**
Heterogeneity: Not applicable					
Test for overall effect: Z=0.66(P=0.51)					



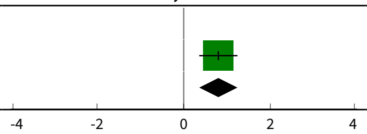
Analysis 1.31. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 31 Change in quality of life: Physical functioning (RAND 36): Oestrogen-only HT (mod dose).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Weight	Mean Difference Fixed, 95% CI
1.31.1 CEE 0.625 mg for 1 yr						
WHI 1998	4580	-1.2 (15.6)	4678	-1.9 (15.8)	93.53%	0.69[0.05,1.33]
Subtotal ***	**4580**		**4678**		**93.53%**	**0.69[0.05,1.33]**
Heterogeneity: Not applicable						
Test for overall effect: Z=2.12(P=0.03)						
1.31.2 CEE 0.625 mg for 3 yrs						
WHI 1998	456	-4.7 (18.9)	483	-6.2 (19.1)	6.47%	1.54[-0.89,3.97]
Subtotal ***	**456**		**483**		**6.47%**	**1.54[-0.89,3.97]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.24(P=0.21)						
Total ***	**5036**		**5161**		**100%**	**0.74[0.13,1.36]**
Heterogeneity: Tau^2=0; Chi^2=0.44, df=1(P=0.51); I^2=0%						
Test for overall effect: Z=2.36(P=0.02)						
Test for subgroup differences: Chi^2=0.44, df=1 (P=0.51), I^2=0%						

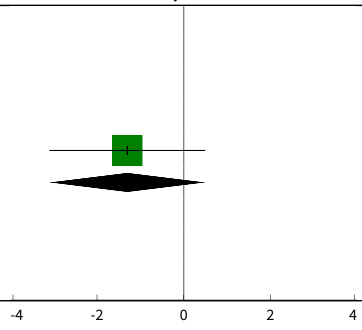


Analysis 1.32. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 32 Change in quality of life: Physical functioning (RAND 36): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Weight	Mean Difference Fixed, 95% CI
1.32.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)						
WHI 1998	7503	-0.6 (13.6)	7160	-1.4 (13.6)	100%	0.8[0.36,1.24]
Subtotal ***	**7503**		**7160**		**100%**	**0.8[0.36,1.24]**

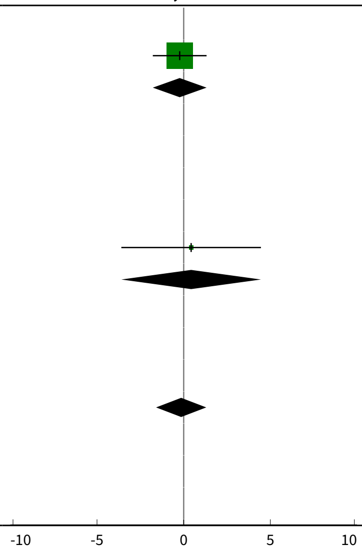


Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
Heterogeneity: Not applicable							
Test for overall effect: Z=3.56(P=0)							
1.32.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)							
WHI 1998	636	-3.2 (16.9)	612	-1.9 (15.5)		100%	-1.3[-3.1,0.5]
Subtotal ***	**636**		**612**			**100%**	**-1.3[-3.1,0.5]**
Heterogeneity: Not applicable							
Test for overall effect: Z=1.42(P=0.16)							
Test for subgroup differences: Chi^2=4.94, df=1 (P=0.03), I^2=79.77%							



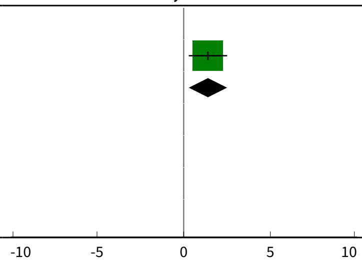
Analysis 1.33. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 33 Change in quality of life: Role limitations due to physical problems (RAND 36):Oestrogen-only HT.

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.33.1 CEE 0.625 mg for 1 yr							
WHI 1998	4664	-2 (38.3)	4783	-1.8 (38.6)		87.09%	-0.23[-1.78,1.32]
Subtotal ***	**4664**		**4783**			**87.09%**	**-0.23[-1.78,1.32]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.29(P=0.77)							
1.33.2 CEE 0.625 mg for 3 yrs							
WHI 1998	476	-5.8 (40.7)	490	-6.2 (19.1)		12.91%	0.43[-3.6,4.46]
Subtotal ***	**476**		**490**			**12.91%**	**0.43[-3.6,4.46]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.21(P=0.83)							
Total ***	**5140**		**5273**			**100%**	**-0.14[-1.59,1.3]**
Heterogeneity: Tau^2=0; Chi^2=0.09, df=1(P=0.76); I^2=0%							
Test for overall effect: Z=0.2(P=0.84)							
Test for subgroup differences: Chi^2=0.09, df=1 (P=0.76), I^2=0%							

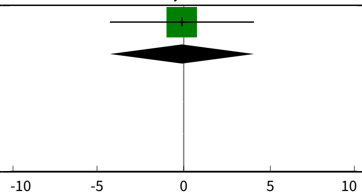


Analysis 1.34. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 34 Change in quality of life: Role limitations due to physical problems (RAND 36): Combined cont. HT (mod dose).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.34.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)							
WHI 1998	7658	-0.9 (34.8)	7327	-2.3 (34.2)		100%	1.4[0.3,2.5]
Subtotal ***	**7658**		**7327**			**100%**	**1.4[0.3,2.5]**
Heterogeneity: Not applicable							
Test for overall effect: Z=2.48(P=0.01)							
1.34.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)							

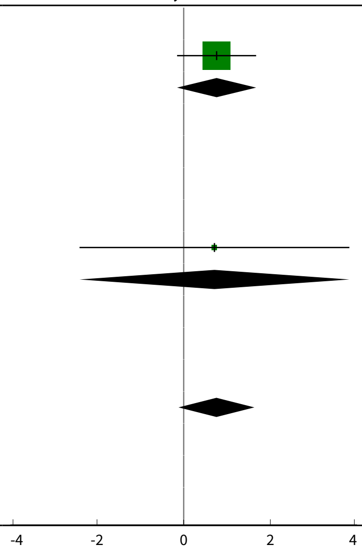


Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
WHI 1998	643	-2.9 (38.5)	626	-2.8 (37)		100%	-0.1[-4.25,4.05]
Subtotal ***	**643**		**626**			**100%**	**-0.1[-4.25,4.05]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%							
Test for overall effect: Z=0.05(P=0.96)							
Test for subgroup differences: Chi²=0.47, df=1 (P=0.49), I²=0%							



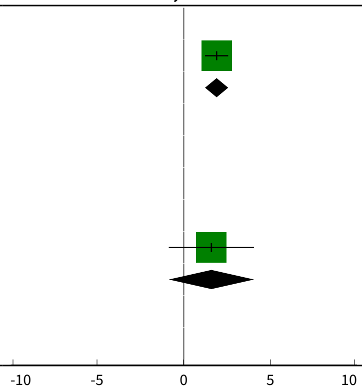
Analysis 1.35. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 35 Change in quality of life: Bodily pain (RAND 36): Oestrogen-only HT.

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.35.1 CEE 0.625 mg for 1 yr							
WHI 1998	4764	-1.2 (22.8)	4885	-2 (22.9)		92.1%	0.76[-0.15,1.67]
Subtotal ***	**4764**		**4885**			**92.1%**	**0.76[-0.15,1.67]**
Heterogeneity: Not applicable							
Test for overall effect: Z=1.63(P=0.1)							
1.35.2 CEE 0.625 mg for 3 yrs							
WHI 1998	479	-4.6 (24.3)	500	-5.3 (25.4)		7.9%	0.71[-2.4,3.82]
Subtotal ***	**479**		**500**			**7.9%**	**0.71[-2.4,3.82]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%							
Test for overall effect: Z=0.45(P=0.65)							
Total ***	**5243**		**5385**			**100%**	**0.76[-0.12,1.63]**
Heterogeneity: Tau²=0; Chi²=0, df=1(P=0.98); I²=0%							
Test for overall effect: Z=1.69(P=0.09)							
Test for subgroup differences: Chi²=0, df=1 (P=0.98), I²=0%							



Analysis 1.36. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 36 Change in quality of life: Bodily pain (RAND 36): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.36.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)							
WHI 1998	7777	0.1 (21.1)	7432	-1.8 (20.9)		100%	1.9[1.23,2.57]
Subtotal ***	**7777**		**7432**			**100%**	**1.9[1.23,2.57]**
Heterogeneity: Not applicable							
Test for overall effect: Z=5.58(P<0.0001)							
1.36.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)							
WHI 1998	659	-1.7 (22.9)	633	-3.3 (22.2)		100%	1.6[-0.86,4.06]
Subtotal ***	**659**		**633**			**100%**	**1.6[-0.86,4.06]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%							
Test for overall effect: Z=1.28(P=0.2)							



Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
Test for subgroup differences: Chi²=0.05, df=1 (P=0.82), I²=0%							

Favours control -10 -5 0 5 10 Favours treatment

Analysis 1.37. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 37 Change in quality of life:Energy and fatigue (RAND 36): Oestrogen-only HT.

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.37.1 CEE 0.625 mg for 1 yr							
WHI 1998	4662	-0.1 (16)	4757	-0 (16.4)		91.99%	-0.12[-0.77,0.53]
Subtotal ***	**4662**		**4757**			**91.99%**	**-0.12[-0.77,0.53]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%							
Test for overall effect: Z=0.36(P=0.72)							
1.37.2 CEE 0.625 mg for 3 yrs							
WHI 1998	471	-1.3 (16.4)	483	-2.3 (18.5)		8.01%	0.98[-1.24,3.2]
Subtotal ***	**471**		**483**			**8.01%**	**0.98[-1.24,3.2]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.87(P=0.39)							
Total ***	**5133**		**5240**			**100%**	**-0.03[-0.66,0.6]**
Heterogeneity: Tau²=0; Chi²=0.87, df=1(P=0.35); I²=0%							
Test for overall effect: Z=0.1(P=0.92)							
Test for subgroup differences: Chi²=0.87, df=1 (P=0.35), I²=0%							

Favours treatment -4 -2 0 2 4 Favours control

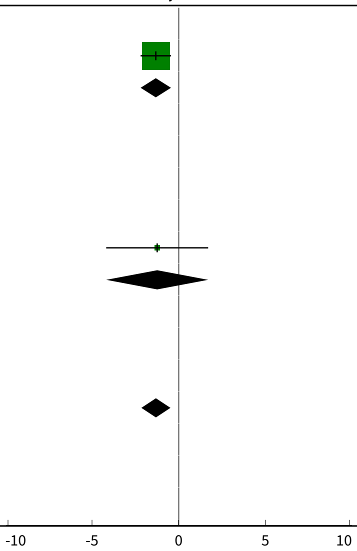
Analysis 1.38. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 38 Change in quality of life:Energy and fatigue (RAND 36): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.38.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)							
WHI 1998	7648	0.2 (15)	7293	0 (15.3)		100%	0.2[-0.29,0.69]
Subtotal ***	**7648**		**7293**			**100%**	**0.2[-0.29,0.69]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.81(P=0.42)							
1.38.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)							
WHI 1998	631	-0.6 (16.8)	610	-0.3 (15.6)		100%	-0.3[-2.1,1.5]
Subtotal ***	**631**		**610**			**100%**	**-0.3[-2.1,1.5]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.33(P=0.74)							
Test for subgroup differences: Chi²=0.28, df=1 (P=0.6), I²=0%							

Favours control -4 -2 0 2 4 Favours treatment

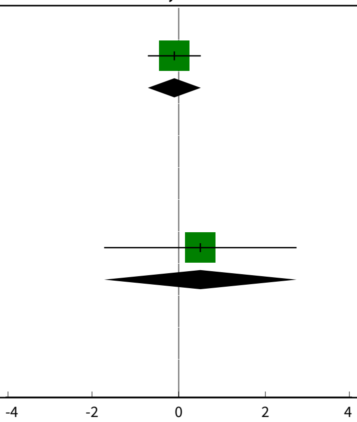
Analysis 1.39. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 39 Change in quality of life: Social functioning (RAND 36): Oestrogen-only HT.

Study or subgroup	Treatment		Control		Mean Difference	Weight	Mean Difference
	N	Mean(SD)	N	Mean(SD)	Fixed, 95% CI		Fixed, 95% CI
1.39.1 CEE 0.625 mg for 1 yr							
WHI 1998	4736	-3.1 (22)	4836	-1.8 (21.8)		91.85%	-1.32[-2.2,-0.44]
Subtotal ***	**4736**		**4836**			**91.85%**	**-1.32[-2.2,-0.44]**
Heterogeneity: Not applicable							
Test for overall effect: Z=2.95(P=0)							
1.39.2 CEE 0.625 mg for 3 yrs							
WHI 1998	476	-4.9 (23.6)	501	-3.6 (23.3)		8.15%	-1.24[-4.18,1.7]
Subtotal ***	**476**		**501**			**8.15%**	**-1.24[-4.18,1.7]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.83(P=0.41)							
Total ***	**5212**		**5337**			**100%**	**-1.31[-2.15,-0.47]**
Heterogeneity: Tau²=0; Chi²=0, df=1(P=0.96); I²=0%							
Test for overall effect: Z=3.07(P=0)							
Test for subgroup differences: Chi²=0, df=1 (P=0.96), I²=0%							



Analysis 1.40. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 40 Change in quality of life: Social functioning (RAND 36): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment		Control		Mean Difference	Weight	Mean Difference
	N	Mean(SD)	N	Mean(SD)	Fixed, 95% CI		Fixed, 95% CI
1.40.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)							
WHI 1998	7720	-1.9 (18.9)	7397	-1.8 (19.4)		100%	-0.1[-0.71,0.51]
Subtotal ***	**7720**		**7397**			**100%**	**-0.1[-0.71,0.51]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.32(P=0.75)							
1.40.2 Combined CEE 0.625 mg + MPA 2.5 mg (3 yrs)							
WHI 1998	658	-1.6 (20.5)	627	-2.1 (20.1)		100%	0.5[-1.72,2.72]
Subtotal ***	**658**		**627**			**100%**	**0.5[-1.72,2.72]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.44(P=0.66)							
Test for subgroup differences: Chi²=0.26, df=1 (P=0.61), I²=0%							



Analysis 1.41. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 41 Change in quality of life: Role limitations due to emotional problems (RAND 36):Oestrogen-only HT.

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.41.1 CEE 0.625 mg for 1 yr							
WHI 1998	4682	-1.6 (34)	4801	-0.4 (21.8)		94.62%	-1.13[-2.28,0.02]
Subtotal ***	**4682**		**4801**			**94.62%**	**-1.13[-2.28,0.02]**
Heterogeneity: Not applicable							
Test for overall effect: Z=1.92(P=0.05)							
1.41.2 CEE 0.625 mg for 3 yrs							
WHI 1998	478	-1.7 (37.8)	496	-3.9 (39.2)		5.38%	2.16[-2.67,6.99]
Subtotal ***	**478**		**496**			**5.38%**	**2.16[-2.67,6.99]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.88(P=0.38)							
Total ***	**5160**		**5297**			**100%**	**-0.95[-2.07,0.17]**
Heterogeneity: Tau²=0; Chi²=1.69, df=1(P=0.19); I²=40.7%							
Test for overall effect: Z=1.67(P=0.1)							
Test for subgroup differences: Chi²=1.69, df=1 (P=0.19), I²=40.7%							

Favours treatment -10 -5 0 5 10 Favours control

Analysis 1.42. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 42 Change in quality of life: Role limitations due to emotional problems (RAND 36): Combined cont. HT (mod dose).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.42.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)							
WHI 1998	7650	-0.7 (30.8)	7335	-0.5 (30.7)		100%	-0.2[-1.18,0.78]
Subtotal ***	**7650**		**7335**			**100%**	**-0.2[-1.18,0.78]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.4(P=0.69)							
1.42.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)							
WHI 1998	650	-2.5 (31)	629	-0.8 (34.4)		100%	-1.7[-5.29,1.89]
Subtotal ***	**650**		**629**			**100%**	**-1.7[-5.29,1.89]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%							
Test for overall effect: Z=0.93(P=0.35)							
Test for subgroup differences: Chi²=0.62, df=1 (P=0.43), I²=0%							

Favours control -10 -5 0 5 10 Favours treatment

Analysis 1.43. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 43 Change difference in quality of life: Mental health (SF-36) Oestrogen only HT (low dose).

Study or subgroup	Control N	Treatment N	Change difference (SE)	Change difference IV, Fixed, 95% CI	Weight	Change difference IV, Fixed, 95% CI
1.43.1 Transdermal oestradiol 0.014 mg for 2 years						
Yaffe 2006	1	1	-0.9 (0.62)		100%	-0.95[-2.17,0.27]
Subtotal (95% CI)					**100%**	**-0.95[-2.17,0.27]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.53(P=0.13)						
Total (95% CI)					**100%**	**-0.95[-2.17,0.27]**
Heterogeneity: Not applicable						
Test for overall effect: Z=1.53(P=0.13)						

Favours control -4 -2 0 2 4 Favours treatment

Analysis 1.44. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 44 Change in quality of life: Mental health (RAND 36): Oestrogen-only HT (mod dose).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.44.1 CEE 0.625 mg for 1 yr							
WHI 1998	4651	0.6 (13.3)	4750	0.6 (13.3)		100%	0.04[-0.5,0.58]
Subtotal ***	**4651**		**4750**			**100%**	**0.04[-0.5,0.58]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.15(P=0.88)							
1.44.2 CEE 0.625 mg for 3 yrs							
WHI 1998	1	0 (0)	1	0 (0)			Not estimable
Subtotal ***	**1**		**1**				**Not estimable**
Heterogeneity: Not applicable							
Test for overall effect: Not applicable							
Total ***	**4652**		**4751**			**100%**	**0.04[-0.5,0.58]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.15(P=0.88)							
Test for subgroup differences: Not applicable							

Favours treatment -1 -0.5 0 0.5 1 Favours control

Analysis 1.45. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 45 Change in quality of life: Mental health (RAND 36): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.45.1 Combined continuous CEE 0.625 mg + MPA 2.5 mg (1 yr)							
WHI 1998	7631	0.6 (12.1)	7279	0.7 (12.4)		100%	-0.1[-0.49,0.29]
Subtotal ***	**7631**		**7279**			**100%**	**-0.1[-0.49,0.29]**

Favours control -4 -2 0 2 4 Favours treatment

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
Heterogeneity: Not applicable							
Test for overall effect: Z=0.5(P=0.62)							
1.45.2 Combined continuous CEE 0.625 mg + MPA 2.5 mg (3 yrs)							
WHI 1998	635	1.3 (13.5)	619	0.3 (13.9)		100%	1[-0.52,2.52]
Subtotal ***	**635**		**619**			**100%**	**1[-0.52,2.52]**
Heterogeneity: Not applicable							
Test for overall effect: Z=1.29(P=0.2)							
Test for subgroup differences: Chi²=1.89, df=1 (P=0.17), I²=47.15%							

Favours control -4 -2 0 2 4 Favours treatment

Analysis 1.46. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 46 Change in quality of life overall (HQOL): Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.46.1 Oestradiol 1 mg daily (1 yr)							
Haines 2003	52	0.5 (20.9)	50	0.9 (23)		100%	-0.41[-8.95,8.13]
Subtotal ***	**52**		**50**			**100%**	**-0.41[-8.95,8.13]**
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%							
Test for overall effect: Z=0.09(P=0.93)							
1.46.2 CEE 0.625 mg for 1 yr							
Subtotal ***	**0**		**0**				**Not estimable**
Heterogeneity: Not applicable							
Test for overall effect: Not applicable							
Test for subgroup differences: Not applicable							

Favours control -10 -5 0 5 10 Favours treatment

Analysis 1.47. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 47 Change in quality of life overall (HQOL): Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment N	Treatment Mean(SD)	Control N	Control Mean(SD)	Mean Difference Fixed, 95% CI	Weight	Mean Difference Fixed, 95% CI
1.47.1 Unopposed oestradiol 2 mg daily (1 yr)							
Haines 2003	50	-2.4 (20.2)	50	0.9 (23)		100%	-3.31[-11.79,5.17]
Subtotal ***	**50**		**50**			**100%**	**-3.31[-11.79,5.17]**
Heterogeneity: Not applicable							
Test for overall effect: Z=0.76(P=0.44)							

Favours treatment -100 -50 0 50 100 Favours control

Analysis 1.48. Comparison 1 Women without major health problems (Selected outcomes: death, CVD, cognition, QOL), Outcome 48 Change in quality of life score (GHQ-11 scale) (combined HT).

Change in quality of life score (GHQ-11 scale) (combined HT)

Study	HRT no.	Change score (SD)	Placebo no.	Change score (SD)	p value
Combined continuous: oestradiol 2 mg + norethisterone 1 mg (2 yrs)					
Obel 1993	31 (of 50 randomised)	-0.26 (3.71)	37 (of 51 randomised)	1.11 (3.85)	p</= 0.10
Combined sequential: oestradiol (2 mg days 1-12, 1 mg days 13-22) + norethisterone 1 mg days 13-22) for 2 yrs					
Obel 1993	37 (of 50 randomised)	- 0.81 (3.08)	37 (of 51 randomised)	1.11 (3.85)	p = </= 0.10

Comparison 2. Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Death from any cause: Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.1 CEE 0.625 mg daily for 3 yrs (2.8-3.2)	2	327	Risk Ratio (M-H, Fixed, 95% CI)	1.31 [0.53, 3.22]
2 Death from any cause: Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	0.81 [0.51, 1.27]
3 Death from any cause: Oestrogen-only HT (mod dose) for women without uterus, plus sequential MPA if uterus	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs	1	664	Risk Ratio (M-H, Fixed, 95% CI)	1.14 [0.77, 1.67]
4 Death from any cause: Combined continuous HT (moderate dose oestrogen)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.1 CEE 0.625 mg + MPA 2.5 mg for 2.8 - 3.2 yrs	2	297	Risk Ratio (M-H, Fixed, 95% CI)	0.86 [0.28, 2.62]
4.2 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.06 [0.84, 1.34]
4.3 CEE 0.625 mg + MPA 2.5 mg for 4-7 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.14 [0.90, 1.44]
5 Death from coronary heart disease: Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
5.1 CEE 0.625 mg daily for 2.8 - 3.2 yrs	2	327	Risk Ratio (M-H, Fixed, 95% CI)	1.31 [0.36, 4.77]
6 Death from coronary heart disease: Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	0.69 [0.40, 1.18]
7 Death from CHD: Oestrogen-only HT (mod dose) for women without uterus, plus sequential MPA if uterus intact	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
7.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs	1	664	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.37, 1.81]
8 Death from coronary heart disease: Combined continuous CEE + MPA	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
8.1 CEE 0.625 mg daily + MPA 2.5 mg for 1 yr	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.55 [0.73, 3.29]
8.2 CEE 0.625 mg daily + MPA 2.5 mg for 2 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.50 [0.90, 2.51]
8.3 CEE 0.625 mg daily + MPA 2.5 mg for 3 yrs (2.8-3.2)	3	3060	Risk Ratio (M-H, Fixed, 95% CI)	1.30 [0.88, 1.90]
8.4 CEE 0.625 mg daily + MPA 2.5 mg for 4+ yrs (median 4.1)	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.19 [0.85, 1.67]
8.5 CEE 0.625 mg daily + MPA 2.5 mg for 4-6.8 yrs	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	0.99 [0.71, 1.39]
9 Death from stroke: Oestrogen-only HT (mod dose) if no uterus, plus sequential MPA if uterus intact	1	664	Risk Ratio (M-H, Fixed, 95% CI)	2.91 [0.95, 8.93]
9.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs	1	664	Risk Ratio (M-H, Fixed, 95% CI)	2.91 [0.95, 8.93]
10 Death from cancer: Combined continuous HT (moderate dose oestrogen)	1	5084	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.78, 1.74]
10.1 CEE 0.625 mg daily + MPA 2.5 mg for 4+ yrs (median 4.1)	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.88 [0.49, 1.57]
10.2 CEE 0.625 mg daily + MPA 2.5 mg for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.51 [0.86, 2.65]
11 Coronary event (MI or cardiac death): Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
11.1 CEE 0.625 daily for 2.8-3.2 yrs	2	327	Risk Ratio (M-H, Fixed, 95% CI)	1.13 [0.54, 2.40]
12 Coronary event (MI or cardiac death): Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
12.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	1.00 [0.72, 1.39]

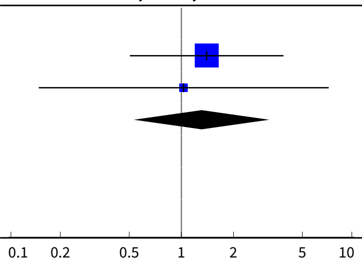
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
13 Coronary event : Oestrogen-only HT (mod dose) for women without uterus, plus sequential MPA if uterus intact	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
13.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs	1	664	Risk Ratio (M-H, Fixed, 95% CI)	0.97 [0.57, 1.65]
14 Coronary event (MI or cardiac death): Combined continuous HT (moderate dose oestrogen)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
14.1 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.20 [0.91, 1.58]
14.2 CEE 0.625 mg + MPA 2.5 mg for 1 yr	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.50 [1.00, 2.25]
14.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs (2.8 - 3.2)	3	3060	Risk Ratio (M-H, Fixed, 95% CI)	1.07 [0.86, 1.33]
14.4 CEE 0.625 mg + MPA 2.5 mg for median 4.1 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.99 [0.81, 1.19]
14.5 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UN-BLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.78, 1.29]
15 Stroke (first or recurrent): Oestrogen-only HT (moderate dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
15.1 CEE 0.625 daily for 2.8 yrs	1	122	Risk Ratio (M-H, Fixed, 95% CI)	0.69 [0.12, 3.98]
16 Stroke (first or recurrent): Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
16.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	1.64 [0.60, 4.47]
17 Stroke (first or recurrent): Oestrogen-only HT (mod dose) if no uterus, plus annual MPA if uterus intact	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
17.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs	1	664	Risk Ratio (M-H, Fixed, 95% CI)	1.09 [0.79, 1.51]
18 Stroke (first or recurrent): Combined continuous HT (moderate dose oestrogen)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
18.1 CEE 0.625 mg + MPA for 2.8 yrs	1	88	Risk Ratio (M-H, Fixed, 95% CI)	5.23 [0.26, 105.85]
18.2 CEE 0.625 mg + MPA 2.5 mg for median 4.1 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.90, 1.68]
18.3 CEE 0.625 mg + MPA for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.05 [0.71, 1.57]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
19 Stroke (first or recurrent): Combined continuous HT (mod/high dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
19.1 Continuous oestradiol 2 mg + norethisterone acetate 1 mg for 1.3 yrs	1	140	Risk Ratio (M-H, Fixed, 95% CI)	0.32 [0.01, 7.82]
20 Transient ischaemic attack: Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
20.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	1.13 [0.54, 2.36]
21 Transient ischaemic attack: Oestrogen-only HT (mod dose) if no uterus, plus sequential MPA if uterus intact	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
21.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 yrs	1	664	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.70, 1.94]
22 Transient ischaemic attack: Combined continuous HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
22.1 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.51, 1.23]
22.2 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.49, 1.84]
23 Stroke or transient ischaemic attack: Combined continuous HT (moderate dose oestrogen)	1	209	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.34, 3.03]
23.1 CEE 0.625 mg + MPA 2.5 mg for 3.2 yrs	1	209	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.34, 3.03]
24 Stroke or transient ischaemic attack : Oestrogen-only HT (moderate dose)	1	205	Risk Ratio (M-H, Fixed, 95% CI)	0.88 [0.28, 2.78]
24.1 CEE 0.625 daily for 3.2 yrs	1	205	Risk Ratio (M-H, Fixed, 95% CI)	0.88 [0.28, 2.78]
25 VTE (first or recurrent PE or DVT): Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
25.1 CEE 0.625 daily for 2.8 - 3.2 yrs	2	327	Risk Ratio (M-H, Fixed, 95% CI)	1.64 [0.44, 6.17]
26 VTE (first or recurrent PE or DVT): Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
26.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.33, 4.55]
27 VTE (first or recurrent PE or DVT): Combined continuous HT (moderate dose oestrogen)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
27.1 CEE 0.625 mg + MPA 2.5 mg for 1 yr	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	3.26 [1.06, 9.96]
27.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	3.51 [1.42, 8.66]
27.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs (2.8 - 3.2)	3	3060	Risk Ratio (M-H, Fixed, 95% CI)	3.01 [1.50, 6.04]
27.4 CEE 0.625 mg + MPA 2.5 mg for median 4.1 years	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	2.62 [1.39, 4.94]
27.5 CEE 0.625 mg + MPA 2.5 mg for 4-7 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.37 [0.63, 2.98]
28 VTE (first or recurrent PE or DVT): Combined continuous HT (mod/high dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
28.1 Continuous oestradiol 2 mg + norethisterone acetate 1 mg for 1.3 yrs	1	140	Risk Ratio (M-H, Fixed, 95% CI)	6.80 [0.86, 53.85]

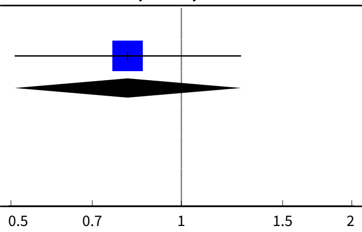
Analysis 2.1. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 1 Death from any cause: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.1.1 CEE 0.625 mg daily for 3 yrs (2.8-3.2)					
ERA 2000	8/100	6/105		74.85%	1.4[0.5,3.89]
WAVE 2002	2/60	2/62		25.15%	1.03[0.15,7.1]
Subtotal (95% CI)	**160**	**167**		**100%**	**1.31[0.53,3.22]**
Total events: 10 (Treatment), 8 (Control)					
Heterogeneity: Tau²=0; Chi²=0.07, df=1(P=0.79); I²=0%					
Test for overall effect: Z=0.58(P=0.56)					



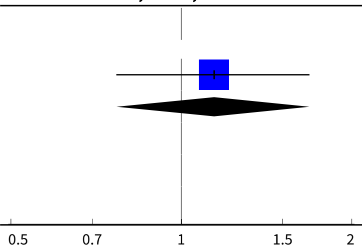
Analysis 2.2. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 2 Death from any cause: Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.2.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	32/513	39/504		100%	0.81[0.51,1.27]
Subtotal (95% CI)	**513**	**504**		**100%**	**0.81[0.51,1.27]**
Total events: 32 (Treatment), 39 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.94(P=0.35)					



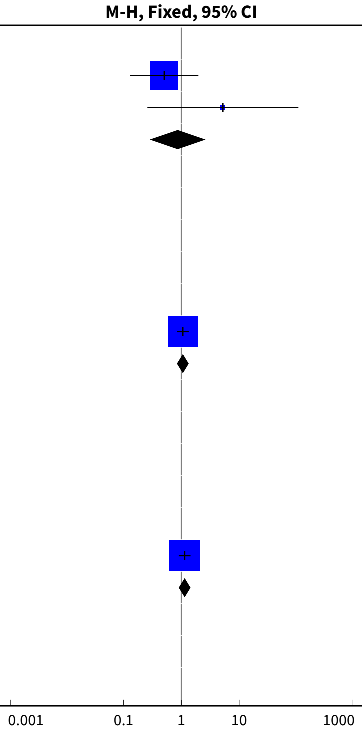
Analysis 2.3. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 3 Death from any cause: Oestrogen-only HT (mod dose) for women without uterus, plus sequential MPA if uterus.

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
2.3.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs				
WEST 2001	48/337	41/327	100%	1.14[0.77,1.67]
Subtotal (95% CI)	**337**	**327**	**100%**	**1.14[0.77,1.67]**
Total events: 48 (Treatment), 41 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=0.64(P=0.52)				



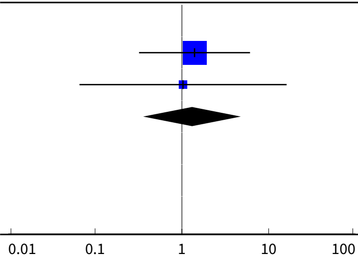
Analysis 2.4. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 4 Death from any cause: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
2.4.1 CEE 0.625 mg + MPA 2.5 mg for 2.8 - 3.2 yrs				
ERA 2000	3/104	6/105	92.43%	0.5[0.13,1.97]
WAVE 2002	2/43	0/45	7.57%	5.23[0.26,105.85]
Subtotal (95% CI)	**147**	**150**	**100%**	**0.86[0.28,2.62]**
Total events: 5 (Treatment), 6 (Control)				
Heterogeneity: Tau²=0; Chi²=1.97, df=1(P=0.16); I²=49.36%				
Test for overall effect: Z=0.26(P=0.79)				
2.4.2 CEE 0.625 mg + MPA 2.5 mg for 4 yrs				
HERS 1998	130/1380	123/1383	100%	1.06[0.84,1.34]
Subtotal (95% CI)	**1380**	**1383**	**100%**	**1.06[0.84,1.34]**
Total events: 130 (Treatment), 123 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=0.48(P=0.63)				
2.4.3 CEE 0.625 mg + MPA 2.5 mg for 4-7 yrs UNBLINDED				
HERS 1998	131/1156	116/1165	100%	1.14[0.9,1.44]
Subtotal (95% CI)	**1156**	**1165**	**100%**	**1.14[0.9,1.44]**
Total events: 131 (Treatment), 116 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=1.07(P=0.28)				



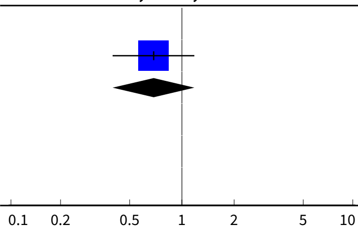
Analysis 2.5. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 5 Death from coronary heart disease: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
2.5.1 CEE 0.625 mg daily for 2.8 - 3.2 yrs				
ERA 2000	4/100	3/105	74.85%	1.4[0.32,6.1]
WAVE 2002	1/60	1/62	25.15%	1.03[0.07,16.15]
Subtotal (95% CI)	**160**	**167**	**100%**	**1.31[0.36,4.77]**
Total events: 5 (Treatment), 4 (Control)				
Heterogeneity: Tau²=0; Chi²=0.04, df=1(P=0.85); I²=0%				
Test for overall effect: Z=0.41(P=0.68)				



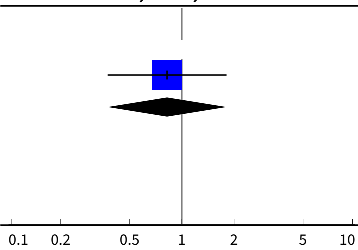
Analysis 2.6. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 6 Death from coronary heart disease: Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
2.6.1 Oestradiol valerate 2 mg for 2 yrs				
ESPRIT 2002	21/513	30/504	100%	0.69[0.4,1.18]
Subtotal (95% CI)	**513**	**504**	**100%**	**0.69[0.4,1.18]**
Total events: 21 (Treatment), 30 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=1.35(P=0.18)				



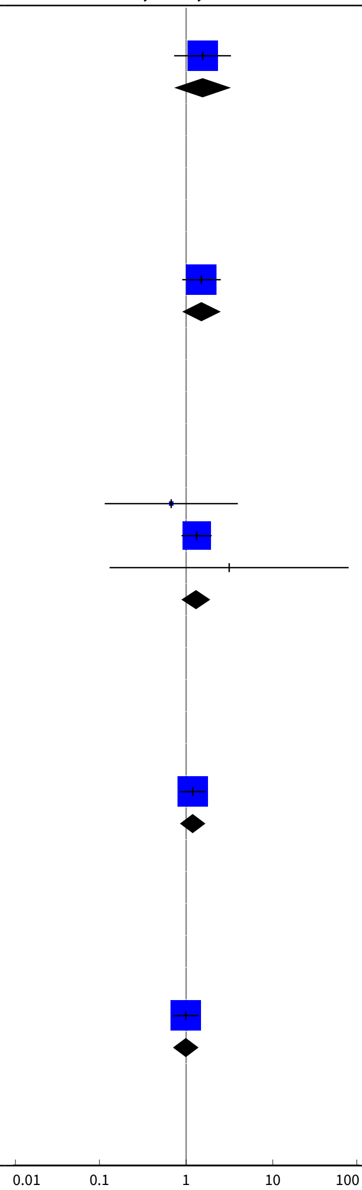
Analysis 2.7. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 7 Death from CHD: Oestrogen-only HT (mod dose) for women without uterus, plus sequential MPA if uterus intact.

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
2.7.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs				
WEST 2001	11/337	13/327	100%	0.82[0.37,1.81]
Subtotal (95% CI)	**337**	**327**	**100%**	**0.82[0.37,1.81]**
Total events: 11 (Treatment), 13 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=0.49(P=0.62)				



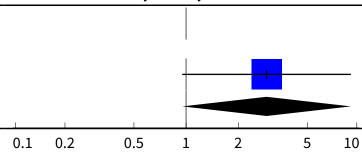
Analysis 2.8. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 8 Death from coronary heart disease: Combined continuous CEE + MPA.

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.8.1 CEE 0.625 mg daily + MPA 2.5 mg for 1 yr					
HERS 1998	17/1380	11/1383		100%	1.55[0.73,3.29]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.55[0.73,3.29]**
Total events: 17 (Treatment), 11 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.14(P=0.26)					
2.8.2 CEE 0.625 mg daily + MPA 2.5 mg for 2 yrs					
HERS 1998	36/1380	24/1383		100%	1.5[0.9,2.51]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.5[0.9,2.51]**
Total events: 36 (Treatment), 24 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.56(P=0.12)					
2.8.3 CEE 0.625 mg daily + MPA 2.5 mg for 3 yrs (2.8-3.2)					
ERA 2000	2/104	3/105		6.72%	0.67[0.11,3.95]
HERS 1998	54/1380	41/1383		92.18%	1.32[0.89,1.97]
WAVE 2002	1/43	0/45		1.1%	3.14[0.13,74.95]
Subtotal (95% CI)	**1527**	**1533**		**100%**	**1.3[0.88,1.9]**
Total events: 57 (Treatment), 44 (Control)					
Heterogeneity: Tau²=0; Chi²=0.83, df=2(P=0.66); I²=0%					
Test for overall effect: Z=1.32(P=0.19)					
2.8.4 CEE 0.625 mg daily + MPA 2.5 mg for 4+ yrs (median 4.1)					
HERS 1998	70/1380	59/1383		100%	1.19[0.85,1.67]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.19[0.85,1.67]**
Total events: 70 (Treatment), 59 (Control)					
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%					
Test for overall effect: Z=1(P=0.32)					
2.8.5 CEE 0.625 mg daily + MPA 2.5 mg for 4-6.8 yrs					
HERS 1998	62/1156	63/1165		100%	0.99[0.71,1.39]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**0.99[0.71,1.39]**
Total events: 62 (Treatment), 63 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.05(P=0.96)					



Analysis 2.9. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 9 Death from stroke: Oestrogen-only HT (mod dose) if no uterus, plus sequential MPA if uterus intact.

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.9.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs					
WEST 2001	12/337	4/327		100%	2.91[0.95,8.93]
Subtotal (95% CI)	**337**	**327**		**100%**	**2.91[0.95,8.93]**



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Total events: 12 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.87(P=0.06)					
Total (95% CI)	**337**	**327**		**100%**	**2.91[0.95,8.93]**
Total events: 12 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.87(P=0.06)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

Analysis 2.10. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 10 Death from cancer: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.10.1 CEE 0.625 mg daily + MPA 2.5 mg for 4+ yrs (median 4.1)					
HERS 1998	21/1380	24/1383		54.61%	0.88[0.49,1.57]
Subtotal (95% CI)	**1380**	**1383**		**54.61%**	**0.88[0.49,1.57]**
Total events: 21 (Treatment), 24 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.44(P=0.66)					
2.10.2 CEE 0.625 mg daily + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					
HERS 1998	30/1156	20/1165		45.39%	1.51[0.86,2.65]
Subtotal (95% CI)	**1156**	**1165**		**45.39%**	**1.51[0.86,2.65]**
Total events: 30 (Treatment), 20 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.45(P=0.15)					
Total (95% CI)	**2536**	**2548**		**100%**	**1.16[0.78,1.74]**
Total events: 51 (Treatment), 44 (Control)					
Heterogeneity: Tau²=0; Chi²=1.75, df=1(P=0.19); I²=42.87%					
Test for overall effect: Z=0.75(P=0.45)					
Test for subgroup differences: Not applicable					
		Favours treatment	0.2 0.5 1 2 5	Favours control	

Analysis 2.11. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 11 Coronary event (MI or cardiac death): Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.11.1 CEE 0.625 daily for 2.8-3.2 yrs					
ERA 2000	10/100	10/105		83.22%	1.05[0.46,2.41]
WAVE 2002	3/60	2/62		16.78%	1.55[0.27,8.95]
Subtotal (95% CI)	**160**	**167**		**100%**	**1.13[0.54,2.4]**
Total events: 13 (Treatment), 12 (Control)					
Heterogeneity: Tau²=0; Chi²=0.15, df=1(P=0.69); I²=0%					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Test for overall effect: Z=0.33(P=0.74)					

Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Analysis 2.12. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 12 Coronary event (MI or cardiac death): Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.12.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	62/513	61/504		100%	1[0.72,1.39]
Subtotal (95% CI)	**513**	**504**		**100%**	**1[0.72,1.39]**
Total events: 62 (Treatment), 61 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.01(P=0.99)					

Favours treatment 0.5 0.7 1 1.5 2 Favours control

Analysis 2.13. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 13 Coronary event : Oestrogen-only HT (mod dose) for women without uterus, plus sequential MPA if uterus intact .

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.13.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs					
WEST 2001	25/337	25/327		100%	0.97[0.57,1.65]
Subtotal (95% CI)	**337**	**327**		**100%**	**0.97[0.57,1.65]**
Total events: 25 (Treatment), 25 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.11(P=0.91)					

Favours treatment 0.5 0.7 1 1.5 2 Favours control

Analysis 2.14. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 14 Coronary event (MI or cardiac death): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.14.1 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
HERS 1998	104/1380	87/1383		100%	1.2[0.91,1.58]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.2[0.91,1.58]**
Total events: 104 (Treatment), 87 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.29(P=0.2)					
2.14.2 CEE 0.625 mg + MPA 2.5 mg for 1 yr					
HERS 1998	57/1380	38/1383		100%	1.5[1,2.25]

Favours treatment 0.01 0.1 1 10 100 Favours control

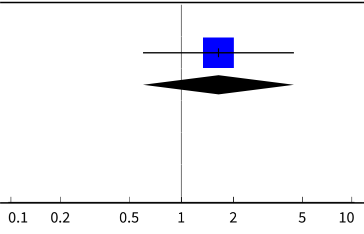
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.5[1,2.25]**
Total events: 57 (Treatment), 38 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.98(P=0.05)					
2.14.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs (2.8 - 3.2)					
ERA 2000	8/104	10/105		7.12%	0.81[0.33,1.97]
HERS 1998	139/1380	129/1383		92.18%	1.08[0.86,1.36]
WAVE 2002	2/43	1/45		0.7%	2.09[0.2,22.25]
Subtotal (95% CI)	**1527**	**1533**		**100%**	**1.07[0.86,1.33]**
Total events: 149 (Treatment), 140 (Control)					
Heterogeneity: Tau^2=0; Chi^2=0.7, df=2(P=0.7); I^2=0%					
Test for overall effect: Z=0.58(P=0.56)					
2.14.4 CEE 0.625 mg + MPA 2.5 mg for median 4.1 yrs					
HERS 1998	179/1380	182/1383		100%	0.99[0.81,1.19]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**0.99[0.81,1.19]**
Total events: 179 (Treatment), 182 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.15(P=0.88)					
2.14.5 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					
HERS 1998	111/1156	111/1165		100%	1.01[0.78,1.29]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**1.01[0.78,1.29]**
Total events: 111 (Treatment), 111 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.06(P=0.95)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	

Analysis 2.15. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 15 Stroke (first or recurrent): Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.15.1 CEE 0.625 daily for 2.8 yrs					
WAVE 2002	2/60	3/62		100%	0.69[0.12,3.98]
Subtotal (95% CI)	**60**	**62**		**100%**	**0.69[0.12,3.98]**
Total events: 2 (Treatment), 3 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.42(P=0.68)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

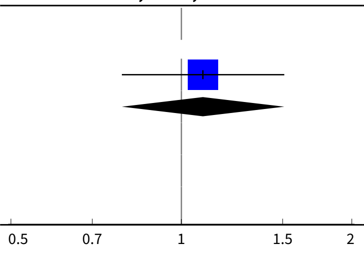
Analysis 2.16. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 16 Stroke (first or recurrent): Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.16.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	10/513	6/504		100%	1.64[0.6,4.47]
Subtotal (95% CI)	**513**	**504**		**100%**	**1.64[0.6,4.47]**
Total events: 10 (Treatment), 6 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.96(P=0.34)					



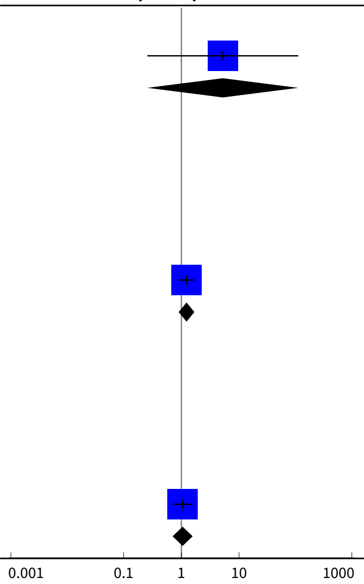
Analysis 2.17. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 17 Stroke (first or recurrent): Oestrogen-only HT (mod dose) if no uterus, plus annual MPA if uterus intact.

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.17.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5mgs for 12 days a year (if uterus intact) for 2.8 yrs					
WEST 2001	63/337	56/327		100%	1.09[0.79,1.51]
Subtotal (95% CI)	**337**	**327**		**100%**	**1.09[0.79,1.51]**
Total events: 63 (Treatment), 56 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.53(P=0.6)					



Analysis 2.18. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 18 Stroke (first or recurrent): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.18.1 CEE 0.625 mg + MPA for 2.8 yrs					
WAVE 2002	2/43	0/45		100%	5.23[0.26,105.85]
Subtotal (95% CI)	**43**	**45**		**100%**	**5.23[0.26,105.85]**
Total events: 2 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.08(P=0.28)					
2.18.2 CEE 0.625 mg + MPA 2.5 mg for median 4.1 yrs					
HERS 1998	82/1380	67/1383		100%	1.23[0.9,1.68]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.23[0.9,1.68]**
Total events: 82 (Treatment), 67 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.27(P=0.2)					
2.18.3 CEE 0.625 mg + MPA for 4-6.8 yrs UNBLINDED					
HERS 1998	47/1156	45/1165		100%	1.05[0.71,1.57]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**1.05[0.71,1.57]**



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Total events: 47 (Treatment), 45 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.25(P=0.8)					
		Favours treatment	0.001 0.1 1 10 1000	Favours control	

Analysis 2.19. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 19 Stroke (first or recurrent): Combined continuous HT (mod/high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.19.1 Continuous oestradiol 2 mg + norethisterone acetate 1 mg for 1.3 yrs					
EVTET 2000	0/71	1/69		100%	0.32[0.01,7.82]
Subtotal (95% CI)	**71**	**69**		**100%**	**0.32[0.01,7.82]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.69(P=0.49)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	

Analysis 2.20. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 20 Transient ischaemic attack: Oestrogen-only HT (mod/high dose).

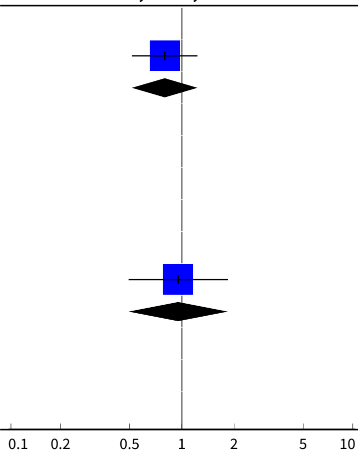
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.20.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	15/513	13/504		100%	1.13[0.54,2.36]
Subtotal (95% CI)	**513**	**504**		**100%**	**1.13[0.54,2.36]**
Total events: 15 (Treatment), 13 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.34(P=0.74)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

Analysis 2.21. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 21 Transient ischaemic attack: Oestrogen-only HT (mod dose) if no uterus, plus sequential MPA if uterus intact.

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.21.1 Oestradiol 1 mg daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 yrs					
WEST 2001	30/337	25/327		100%	1.16[0.7,1.94]
Subtotal (95% CI)	**337**	**327**		**100%**	**1.16[0.7,1.94]**
Total events: 30 (Treatment), 25 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.59(P=0.56)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

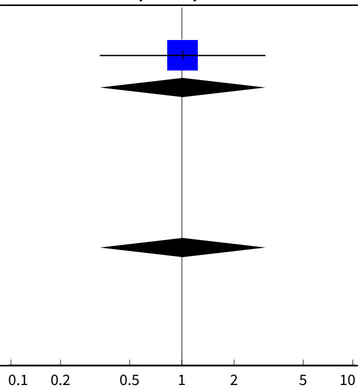
Analysis 2.22. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 22 Transient ischaemic attack: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.22.1 CEE 0.625 mg + MPA 2.5 mg for 4 yrs					
HERS 1998	35/1380	44/1383		100%	0.8[0.51,1.23]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**0.8[0.51,1.23]**
Total events: 35 (Treatment), 44 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.02(P=0.31)					
2.22.2 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					
HERS 1998	17/1156	18/1165		100%	0.95[0.49,1.84]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**0.95[0.49,1.84]**
Total events: 17 (Treatment), 18 (Control)					
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%					
Test for overall effect: Z=0.15(P=0.88)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	



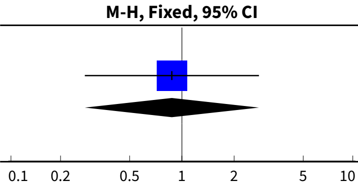
Analysis 2.23. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 23 Stroke or transient ischaemic attack: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.23.1 CEE 0.625 mg + MPA 2.5 mg for 3.2 yrs					
ERA 2000	6/104	6/105		100%	1.01[0.34,3.03]
Subtotal (95% CI)	**104**	**105**		**100%**	**1.01[0.34,3.03]**
Total events: 6 (Treatment), 6 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.02(P=0.99)					
Total (95% CI)	**104**	**105**		**100%**	**1.01[0.34,3.03]**
Total events: 6 (Treatment), 6 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.02(P=0.99)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	



Analysis 2.24. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 24 Stroke or transient ischaemic attack : Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.24.1 CEE 0.625 daily for 3.2 yrs					
ERA 2000	5/100	6/105		100%	0.88[0.28,2.78]
Subtotal (95% CI)	**100**	**105**		**100%**	**0.88[0.28,2.78]**
Total events: 5 (Treatment), 6 (Control)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Heterogeneity: Not applicable					
Test for overall effect: Z=0.23(P=0.82)					
Total (95% CI)	**100**	**105**		**100%**	**0.88[0.28,2.78]**
Total events: 5 (Treatment), 6 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.23(P=0.82)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

Analysis 2.25. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 25 VTE (first or recurrent PE or DVT): Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.25.1 CEE 0.625 daily for 2.8 - 3.2 yrs					
ERA 2000	5/100	1/105		28.4%	5.25[0.62,44.16]
WAVE 2002	0/60	2/62		71.6%	0.21[0.01,4.22]
Subtotal (95% CI)	**160**	**167**		**100%**	**1.64[0.44,6.17]**
Total events: 5 (Treatment), 3 (Control)					
Heterogeneity: Tau²=0; Chi²=2.96, df=1(P=0.09); I²=66.22%					
Test for overall effect: Z=0.73(P=0.47)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	

Analysis 2.26. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 26 VTE (first or recurrent PE or DVT): Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.26.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	5/513	4/504		100%	1.23[0.33,4.55]
Subtotal (95% CI)	**513**	**504**		**100%**	**1.23[0.33,4.55]**
Total events: 5 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.31(P=0.76)					
		Favours treatment	0.1 0.2 0.5 1 2 5 10	Favours control	

Analysis 2.27. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 27 VTE (first or recurrent PE or DVT): Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.27.1 CEE 0.625 mg + MPA 2.5 mg for 1 yr					
HERS 1998	13/1380	4/1383		100%	3.26[1.06,9.96]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**3.26[1.06,9.96]**
		Favours treatment	0.001 0.1 1 10 1000	Favours control	

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Total events: 13 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.07(P=0.04)					
2.27.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
HERS 1998	21/1380	6/1383		100%	3.51[1.42,8.66]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**3.51[1.42,8.66]**
Total events: 21 (Treatment), 6 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.72(P=0.01)					
2.27.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs (2.8 - 3.2)					
ERA 2000	2/104	1/105		9.5%	2.02[0.19,21.93]
HERS 1998	28/1380	9/1383		85.83%	3.12[1.48,6.58]
WAVE 2002	1/43	0/45		4.67%	3.14[0.13,74.95]
Subtotal (95% CI)	**1527**	**1533**		**100%**	**3.01[1.5,6.04]**
Total events: 31 (Treatment), 10 (Control)					
Heterogeneity: Tau²=0; Chi²=0.12, df=2(P=0.94); I²=0%					
Test for overall effect: Z=3.11(P=0)					
2.27.4 CEE 0.625 mg + MPA 2.5 mg for median 4.1 years					
HERS 1998	34/1380	13/1383		100%	2.62[1.39,4.94]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**2.62[1.39,4.94]**
Total events: 34 (Treatment), 13 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.98(P=0)					
2.27.5 CEE 0.625 mg + MPA 2.5 mg for 4-7 yrs UNBLINDED					
HERS 1998	15/1156	11/1165		100%	1.37[0.63,2.98]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**1.37[0.63,2.98]**
Total events: 15 (Treatment), 11 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.81(P=0.42)					
		Favours treatment	0.001 0.1 1 10 1000	Favours control	

Analysis 2.28. Comparison 2 Women with cardiovascular disease (Selected outcomes: death, CVD, cognition, QOL), Outcome 28 VTE (first or recurrent PE or DVT): Combined continuous HT (mod/high dose oestrogen).

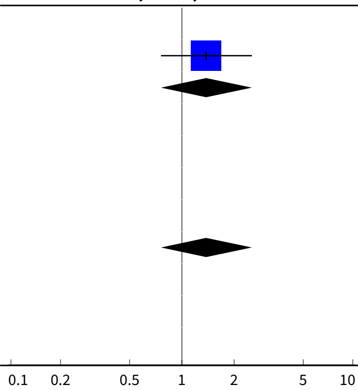
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
2.28.1 Continuous oestradiol 2 mg + norethisterone acetate 1 mg for 1.3 yrs					
EVTET 2000	7/71	1/69		100%	6.8[0.86,53.85]
Subtotal (95% CI)	**71**	**69**		**100%**	**6.8[0.86,53.85]**
Total events: 7 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.82(P=0.07)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	

Comparison 3. Women post surgery for early-stage endometrial cancer (Selected outcomes: death, recurrence)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Death from any cause: Oestrogen-only HT	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.38 [0.76, 2.53]
1.1 CEE 0.625 mg daily for median 3 yrs	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.38 [0.76, 2.53]
2 Death from endometrial cancer: Oestrogen-only HT (moderate dose)	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.25 [0.33, 4.68]
2.1 CEE 0.625 mg daily for median 3 yrs	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.25 [0.33, 4.68]
3 Death from CHD: Oestrogen-only HT (moderate dose)	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.25 [0.33, 4.68]
3.1 CEE 0.625 mg daily for median 3 yrs	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.25 [0.33, 4.68]
4 Recurrence of endometrial cancer: Oestrogen-only HT (moderate dose)	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.17 [0.54, 2.55]
4.1 CEE 0.625 mg daily for median 3 yrs	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.17 [0.54, 2.55]

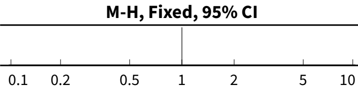
Analysis 3.1. Comparison 3 Women post surgery for early-stage endometrial cancer (Selected outcomes: death, recurrence), Outcome 1 Death from any cause: Oestrogen-only HT.

Study or subgroup	Treatment n/N	Control n/N	Odds Ratio M-H, Fixed, 95% CI	Weight	Odds Ratio M-H, Fixed, 95% CI
3.1.1 CEE 0.625 mg daily for median 3 yrs					
Barakat 2006	26/618	19/618		100%	1.38[0.76,2.53]
Subtotal (95% CI)	**618**	**618**		**100%**	**1.38[0.76,2.53]**
Total events: 26 (Treatment), 19 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.06(P=0.29)					
Total (95% CI)	**618**	**618**		**100%**	**1.38[0.76,2.53]**
Total events: 26 (Treatment), 19 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.06(P=0.29)					



Analysis 3.2. Comparison 3 Women post surgery for early-stage endometrial cancer (Selected outcomes: death, recurrence), Outcome 2 Death from endometrial cancer: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Odds Ratio M-H, Fixed, 95% CI	Weight	Odds Ratio M-H, Fixed, 95% CI
3.2.1 CEE 0.625 mg daily for median 3 yrs					



Study or subgroup	Treatment	Control	Odds Ratio	Weight	Odds Ratio
	n/N	n/N	M-H, Fixed, 95% CI		M-H, Fixed, 95% CI
Barakat 2006	5/618	4/618		100%	1.25[0.33,4.68]
Subtotal (95% CI)	**618**	**618**		**100%**	**1.25[0.33,4.68]**
Total events: 5 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.33(P=0.74)					
Total (95% CI)	**618**	**618**		**100%**	**1.25[0.33,4.68]**
Total events: 5 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.33(P=0.74)					

Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Analysis 3.3. Comparison 3 Women post surgery for early-stage endometrial cancer (Selected outcomes: death, recurrence), Outcome 3 Death from CHD: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment	Control	Odds Ratio	Weight	Odds Ratio
	n/N	n/N	M-H, Fixed, 95% CI		M-H, Fixed, 95% CI
3.3.1 CEE 0.625 mg daily for median 3 yrs					
Barakat 2006	5/618	4/618		100%	1.25[0.33,4.68]
Subtotal (95% CI)	**618**	**618**		**100%**	**1.25[0.33,4.68]**
Total events: 5 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.33(P=0.74)					
Total (95% CI)	**618**	**618**		**100%**	**1.25[0.33,4.68]**
Total events: 5 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.33(P=0.74)					

Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Analysis 3.4. Comparison 3 Women post surgery for early-stage endometrial cancer (Selected outcomes: death, recurrence), Outcome 4 Recurrence of endometrial cancer: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment	Control	Odds Ratio	Weight	Odds Ratio
	n/N	n/N	M-H, Fixed, 95% CI		M-H, Fixed, 95% CI
3.4.1 CEE 0.625 mg daily for median 3 yrs					
Barakat 2006	14/618	12/618		100%	1.17[0.54,2.55]
Subtotal (95% CI)	**618**	**618**		**100%**	**1.17[0.54,2.55]**
Total events: 14 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.4(P=0.69)					
Total (95% CI)	**618**	**618**		**100%**	**1.17[0.54,2.55]**
Total events: 14 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.4(P=0.69)					

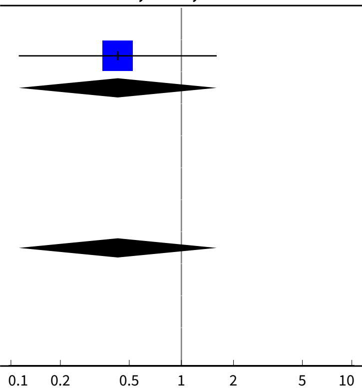
Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Comparison 4. Women hospitalised with chronic illness (Selected outcomes: death, CVD, VTE)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 All cause death: Combined sequential HRT (high dose oestrogen)	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.43 [0.11, 1.60]
1.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.43 [0.11, 1.60]
2 Myocardial infarction: Combined sequential HRT (high dose oestrogen)	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.04, 3.14]
2.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.04, 3.14]
3 Venous thrombo-embolism (DVT or PE): Combined sequential HRT (high dose oestrogen)	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.07]
3.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.07]

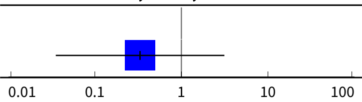
Analysis 4.1. Comparison 4 Women hospitalised with chronic illness (Selected outcomes: death, CVD, VTE), Outcome 1 All cause death: Combined sequential HRT (high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
4.1.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle				
Nachtigall 1979	3/84	7/84	100%	0.43[0.11,1.6]
Subtotal (95% CI)	**84**	**84**	**100%**	**0.43[0.11,1.6]**
Total events: 3 (Treatment), 7 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=1.26(P=0.21)				
Total (95% CI)	**84**	**84**	**100%**	**0.43[0.11,1.6]**
Total events: 3 (Treatment), 7 (Control)				
Heterogeneity: Not applicable				
Test for overall effect: Z=1.26(P=0.21)				

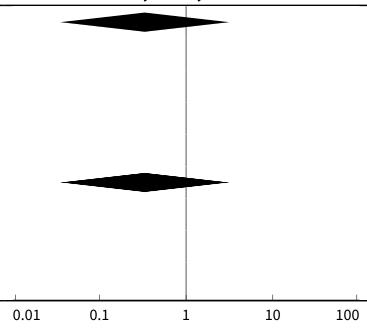


Analysis 4.2. Comparison 4 Women hospitalised with chronic illness (Selected outcomes: death, CVD, VTE), Outcome 2 Myocardial infarction: Combined sequential HRT (high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Weight	Risk Ratio M-H, Fixed, 95% CI
4.2.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle				
Nachtigall 1979	1/84	3/84	100%	0.33[0.04,3.14]

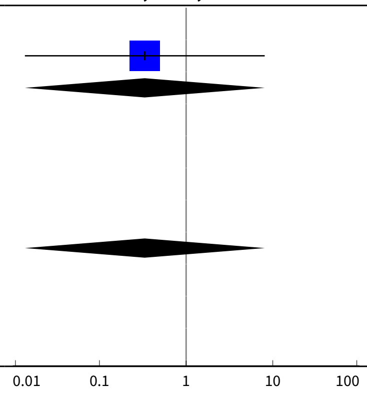


Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Subtotal (95% CI)	**84**	**84**		**100%**	**0.33[0.04,3.14]**
Total events: 1 (Treatment), 3 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.96(P=0.34)					
Total (95% CI)	**84**	**84**		**100%**	**0.33[0.04,3.14]**
Total events: 1 (Treatment), 3 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.96(P=0.34)					



Analysis 4.3. Comparison 4 Women hospitalised with chronic illness (Selected outcomes: death, CVD, VTE), Outcome 3 Venous thrombo-embolism (DVT or PE): Combined sequential HRT (high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
4.3.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle					
Nachtigall 1979	0/84	1/84		100%	0.33[0.01,8.07]
Subtotal (95% CI)	**84**	**84**		**100%**	**0.33[0.01,8.07]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
Total (95% CI)	**84**	**84**		**100%**	**0.33[0.01,8.07]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					

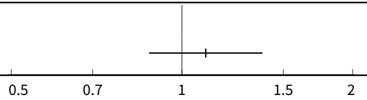


Comparison 5. Women with dementia

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Worsening of dementia on treatment (by ADCS-CGIC score): Oestrogen-only HRT (mod and high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Totals not selected
1.1 Unopposed CEE 0.625 mg or 1.25 mg daily for 1 yr	1		Risk Ratio (M-H, Fixed, 95% CI)	0.0 [0.0, 0.0]

Analysis 5.1. Comparison 5 Women with dementia, Outcome 1 Worsening of dementia on treatment (by ADCS-CGIC score): Oestrogen-only HRT (mod and high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Risk Ratio M-H, Fixed, 95% CI
5.1.1 Unopposed CEE 0.625 mg or 1.25 mg daily for 1 yr				
Mulnard 2000	64/81	28/39		1.1[0.88,1.38]



Comparison 6. All women (Selected outcomes: cancer, cholecystic disease, fractures)

Outcome or subgroup title	No. of studies	No. of partici-pants	Statistical method	Effect size
1 Breast cancer: Oestrogen-only HT (low dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.1 Oestrogen only HRT patch 0.025 mg daily (2 yrs)	1	176	Risk Ratio (M-H, Fixed, 95% CI)	2.93 [0.12, 71.04]
2 Breast cancer: Oestrogen-only HT (moderate dose)	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.1 Oestradiol 1 mg for 2 yrs	1	222	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.10]
2.2 CEE 0.625 for 2.8 - 3.2 yrs	3	676	Risk Ratio (M-H, Fixed, 95% CI)	2.05 [0.38, 11.04]
2.3 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.65, 1.03]
3 Breast cancer: Oestrogen-only HT (mod/high dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.1 Oestradiol patch 0.075 mg for 2 yrs	1	176	Risk Ratio (M-H, Fixed, 95% CI)	2.93 [0.12, 71.04]
3.2 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	0.98 [0.25, 3.91]
4 Breast cancer: Combined continuous HT (moderate dose oestrogen)	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.1 CEE 0.625 mg + MPA 2.5 mg for mean/medi-an 1 yr	2	23182	Risk Ratio (M-H, Fixed, 95% CI)	0.52 [0.28, 0.96]
4.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.71 [0.47, 1.08]
4.3 CEE 0.625 mg + MPA 2.5 mg for 2.8 - 3.4 yrs	3	17733	Risk Ratio (M-H, Fixed, 95% CI)	0.86 [0.62, 1.20]
4.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.36 [0.82, 2.27]
4.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.26 [1.02, 1.56]
4.6 CEE 0.625 mg + MPA 2.5 mg for 4-7 yrs UN-BLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.08 [0.52, 2.23]
5 Breast cancer: Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs	1	348	Risk Ratio (M-H, Fixed, 95% CI)	2.0 [0.18, 21.85]
5.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	352	Risk Ratio (M-H, Fixed, 95% CI)	3.91 [0.44, 34.64]
6 Breast cancer: Combined sequential HT (high dose oestrogen)	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.11 [0.01, 2.03]
6.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.11 [0.01, 2.03]
7 Colorectal cancer: Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
7.1 CEE 0.625 mg for 3 yrs	1	349	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.08]
7.2 CEE 0.625 mg for 6.8 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	1.08 [0.75, 1.54]
8 Colorectal cancer: Combined continuous HT (moderate dose oestrogen)	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
8.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr	2	20993	Risk Ratio (M-H, Fixed, 95% CI)	0.68 [0.32, 1.42]
8.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.83 [0.46, 1.50]
8.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs	2	16956	Risk Ratio (M-H, Fixed, 95% CI)	0.81 [0.49, 1.34]
8.4 CEE 0.625 mg + 2.5 mg MPA for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.69 [0.32, 1.48]
8.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.62 [0.43, 0.89]
8.6 CEE 0.625 mg + 2.5 mg MPA for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	0.81 [0.46, 1.44]
9 Colorectal cancer: Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
9.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs	1	348	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.13]
9.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	352	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 7.95]
10 Colorectal cancer: Combined sequential HT (high dose oestrogen)	1	168	Risk Ratio (M-H, Fixed, 95% CI)	1.0 [0.06, 15.73]
10.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle	1	168	Risk Ratio (M-H, Fixed, 95% CI)	1.0 [0.06, 15.73]
11 Endometrial cancer: Oestrogen-only HT (moderate dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only

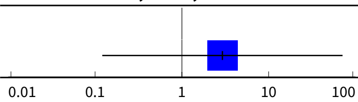
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
11.1 CEE 0.625 mg for 3 - 3.2 yrs	1	238	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.10]
12 Endometrial cancer: Oestrogen only HT (mod/high dose)	1	245	Odds Ratio (M-H, Fixed, 95% CI)	0.0 [0.0, 0.0]
12.1 Oestradiol valerate 2 mg for 2 yrs	1	245	Odds Ratio (M-H, Fixed, 95% CI)	0.0 [0.0, 0.0]
13 Recurrent endometrial cancer:Oestrogen-only HT	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.17 [0.54, 2.55]
13.1 Oestrogen (type and dose not stated) for median 3 yrs	1	1236	Odds Ratio (M-H, Fixed, 95% CI)	1.17 [0.54, 2.55]
14 Endometrial cancer: Combined continuous HT (moderate dose oestrogen)	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
14.1 CEE 0.625 mg + MPA 2.5 mg for 1 yr	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.13, 6.76]
14.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.31, 2.95]
14.3 CEE 0.625 mg + MPA 2.5 mg for 3 - 3.2 yrs	3	17056	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.35, 1.82]
14.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.40 [0.08, 2.06]
14.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.83 [0.50, 1.39]
14.6 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	0.14 [0.01, 2.78]
15 Endometrial cancer: Combined sequential HT (moderate dose oestrogen)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
15.1 Combined sequential: 17B estradiol 1mg + dydrogesterone 5 mg days 14-28 for 2 yrs	1	163	Risk Ratio (M-H, Fixed, 95% CI)	1.90 [0.08, 45.95]
15.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	239	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.03]
16 Endometrial cancer: Combined sequential HT (high dose oestrogen)	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.07]
16.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle	1	168	Risk Ratio (M-H, Fixed, 95% CI)	0.33 [0.01, 8.07]
17 Endometrial cancer: Combined sequential HT (mod/high dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
17.1 Oestradiol 2 mg + dydrogesterone 20 mg	1	159	Risk Ratio (M-H, Fixed, 95% CI)	3.30 [0.16, 67.59]
18 Ovarian cancer: Combined continuous HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
18.1 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	1.59 [0.78, 3.25]
19 Hip fractures: Oestrogen-only HT (moderate dose)	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.64 [0.45, 0.93]
19.1 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.64 [0.45, 0.93]
20 Hip fractures: Combined continuous HT (moderate dose oestrogen)	3		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
20.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr	2	20993	Risk Ratio (M-H, Fixed, 95% CI)	0.64 [0.26, 1.57]
20.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.61 [0.31, 1.18]
20.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.70 [0.42, 1.17]
20.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.55, 2.42]
20.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.68 [0.48, 0.97]
20.6 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	2.10 [1.06, 4.16]
21 Vertebral fractures: Oestrogen-only HT (moderate dose)	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.62 [0.42, 0.93]
21.1 CEE 0.625 mg for 6.8 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.62 [0.42, 0.93]
22 Vertebral fractures: Combined continuous HT (moderate dose oestrogen)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
22.1 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.65 [0.44, 0.97]
22.2 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.74 [0.37, 1.47]
22.3 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.10 [0.49, 2.48]
23 All clinical fractures: Oestrogen-only HT (moderate dose)	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
23.1 CEE 0.625 daily for 3.2 yrs	1	205	Risk Ratio (M-H, Fixed, 95% CI)	0.42 [0.17, 1.04]
23.2 CEE 0.625 mg for 7.1 yrs	1	10739	Risk Ratio (M-H, Fixed, 95% CI)	0.73 [0.65, 0.80]
24 All clinical fractures: Oestrogen-only HT (mod/high dose)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
24.1 Oestradiol valerate 2 mg for 2 yrs	1	1017	Risk Ratio (M-H, Fixed, 95% CI)	0.60 [0.29, 1.26]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
25 All clinical fractures: Combined continuous HT (moderate dose oestrogen)	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
25.1 CEE 0.625 mg + MPA 2.5 mg for median one year	1	4385	Risk Ratio (M-H, Fixed, 95% CI)	0.69 [0.46, 1.02]
25.2 CEE 0.625 mg + MPA 2.5 mg for 3.2-3.4 yrs	2	986	Risk Ratio (M-H, Fixed, 95% CI)	0.52 [0.32, 0.87]
25.3 CEE 0.625 mg + MPA 2.5 mg for median 5.6 yrs	1	16608	Risk Ratio (M-H, Fixed, 95% CI)	0.78 [0.71, 0.85]
25.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs	1	2763	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.76, 1.18]
25.5 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED	1	2321	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.91, 1.65]
26 Gallbladder disease requiring surgery: Oestrogen-only HT (moderate dose)	3	8930	Risk Ratio (M-H, Fixed, 95% CI)	1.75 [1.40, 2.19]
26.1 CEE 0.625 mg for 3 - 3.2 yrs	2	554	Risk Ratio (M-H, Fixed, 95% CI)	0.77 [0.17, 3.39]
26.2 CEE O.625 mg for 7.1 yrs	1	8376	Risk Ratio (M-H, Fixed, 95% CI)	1.78 [1.42, 2.24]
27 Gallbladder disease requiring surgery: Combined continuous HT (moderate dose oestrogen)	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
27.1 CEE 0.625 mg + 2.5 mg MPA for 3 yrs	2	557	Risk Ratio (M-H, Fixed, 95% CI)	2.01 [0.61, 6.59]
27.2 CEE 0.625 mg + 2.5 mg MPA for 4 yrs	1	2253	Risk Ratio (M-H, Fixed, 95% CI)	1.35 [0.98, 1.85]
27.3 CEE 0.625 mg + 2.5 mg MPA for 5.6 yrs	1	14203	Risk Ratio (M-H, Fixed, 95% CI)	1.64 [1.30, 2.06]
28 Gallbladder disease requiring surgery: Combined sequential HT (moderate dose oestrogen)	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
28.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs	1	348	Risk Ratio (M-H, Fixed, 95% CI)	2.0 [0.37, 10.78]
28.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs	1	352	Risk Ratio (M-H, Fixed, 95% CI)	1.47 [0.25, 8.67]

Analysis 6.1. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 1 Breast cancer: Oestrogen-only HT (low dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.1.1 Oestrogen only HRT patch 0.025 mg daily (2 yrs)					
Notelovitz 2002	1/89	0/87		100%	2.93[0.12,71.04]



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Subtotal (95% CI)	**89**	**87**		**100%**	**2.93[0.12,71.04]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.66(P=0.51)					

Favours treatment 0.01 0.1 1 10 100 Favours control

Analysis 6.2. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 2 Breast cancer: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.2.1 Oestradiol 1 mg for 2 yrs					
EPAT 2001	0/111	1/111		100%	0.33[0.01,8.1]
Subtotal (95% CI)	**111**	**111**		**100%**	**0.33[0.01,8.1]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
6.2.2 CEE 0.625 for 2.8 - 3.2 yrs					
ERA 2000	1/100	0/105		24.61%	3.15[0.13,76.39]
PEPI 1995	1/175	1/174		50.58%	0.99[0.06,15.77]
WAVE 2002	1/60	0/62		24.81%	3.1[0.13,74.59]
Subtotal (95% CI)	**335**	**341**		**100%**	**2.05[0.38,11.04]**
Total events: 3 (Treatment), 1 (Control)					
Heterogeneity: $Tau^2=0$; $Chi^2=0.4$, df=2(P=0.82); $I^2=0\%$					
Test for overall effect: Z=0.83(P=0.4)					
6.2.3 CEE 0.625 mg for 7.1 yrs					
WHI 1998	129/5310	161/5429		100%	0.82[0.65,1.03]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**0.82[0.65,1.03]**
Total events: 129 (Treatment), 161 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.71(P=0.09)					

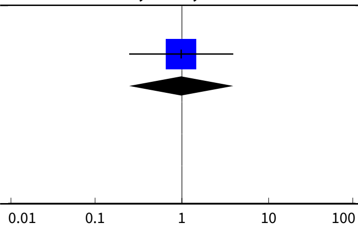
Favours treatment 0.01 0.1 1 10 100 Favours control

Analysis 6.3. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 3 Breast cancer: Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.3.1 Oestradiol patch 0.075 mg for 2 yrs					
Notelovitz 2002	1/89	0/87		100%	2.93[0.12,71.04]
Subtotal (95% CI)	**89**	**87**		**100%**	**2.93[0.12,71.04]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.66(P=0.51)					

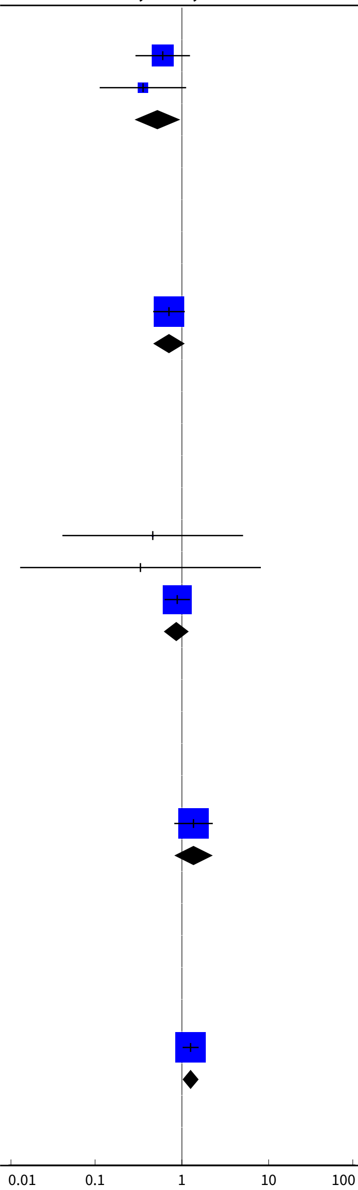
Favours treatment 0.01 0.1 1 10 100 Favours control

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.3.2 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	4/513	4/504		100%	0.98[0.25,3.91]
Subtotal (95% CI)	**513**	**504**		**100%**	**0.98[0.25,3.91]**
Total events: 4 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.03(P=0.98)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	



Analysis 6.4. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 4 Breast cancer: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.4.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr					
WHI 1998	12/8506	19/8102		67.58%	0.6[0.29,1.24]
WISDOM 2007	5/4385	7/2189		32.42%	0.36[0.11,1.12]
Subtotal (95% CI)	**12891**	**10291**		**100%**	**0.52[0.28,0.96]**
Total events: 17 (Treatment), 26 (Control)					
Heterogeneity: Tau²=0; Chi²=0.57, df=1(P=0.45); I²=0%					
Test for overall effect: Z=2.1(P=0.04)					
6.4.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	38/8506	51/8102		100%	0.71[0.47,1.08]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.71[0.47,1.08]**
Total events: 38 (Treatment), 51 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.6(P=0.11)					
6.4.3 CEE 0.625 mg + MPA 2.5 mg for 2.8 - 3.4 yrs					
EPHT 2006	1/404	2/373		2.76%	0.46[0.04,5.07]
PEPI 1995	0/174	1/174		1.99%	0.33[0.01,8.13]
WHI 1998	65/8506	70/8102		95.24%	0.88[0.63,1.24]
Subtotal (95% CI)	**9084**	**8649**		**100%**	**0.86[0.62,1.2]**
Total events: 66 (Treatment), 73 (Control)					
Heterogeneity: Tau²=0; Chi²=0.62, df=2(P=0.73); I²=0%					
Test for overall effect: Z=0.88(P=0.38)					
6.4.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs					
HERS 1998	34/1380	25/1383		100%	1.36[0.82,2.27]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.36[0.82,2.27]**
Total events: 34 (Treatment), 25 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.19(P=0.23)					
6.4.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	199/8506	150/8102		100%	1.26[1.02,1.56]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.26[1.02,1.56]**
Total events: 199 (Treatment), 150 (Control)					
Heterogeneity: Not applicable					
		Favours treatment	0.01 0.1 1 10 100	Favours control	



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Test for overall effect: Z=2.19(P=0.03)					
6.4.6 CEE 0.625 mg + MPA 2.5 mg for 4-7 yrs UNBLINDED					
HERS 1998	15/1156	14/1165		100%	1.08[0.52,2.23]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**1.08[0.52,2.23]**
Total events: 15 (Treatment), 14 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.21(P=0.84)					
	Favours treatment		0.01 0.1 1 10 100	Favours control	

Analysis 6.5. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 5 Breast cancer: Combined sequential HT (moderate dose oestrogen).

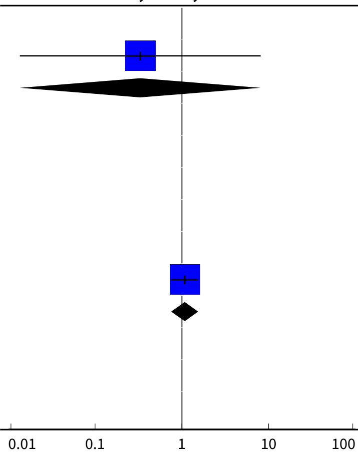
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.5.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs					
PEPI 1995	2/174	1/174		100%	2[0.18,21.85]
Subtotal (95% CI)	**174**	**174**		**100%**	**2[0.18,21.85]**
Total events: 2 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.57(P=0.57)					
6.5.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	4/178	1/174		100%	3.91[0.44,34.64]
Subtotal (95% CI)	**178**	**174**		**100%**	**3.91[0.44,34.64]**
Total events: 4 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.23(P=0.22)					
	Favours treatment		0.001 0.1 1 10 1000	Favours control	

Analysis 6.6. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 6 Breast cancer: Combined sequential HT (high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.6.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle					
Nachtigall 1979	0/84	4/84		100%	0.11[0.01,2.03]
Subtotal (95% CI)	**84**	**84**		**100%**	**0.11[0.01,2.03]**
Total events: 0 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.48(P=0.14)					
Total (95% CI)	**84**	**84**		**100%**	**0.11[0.01,2.03]**
Total events: 0 (Treatment), 4 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.48(P=0.14)					
	Favours treatment		0.001 0.1 1 10 1000	Favours control	

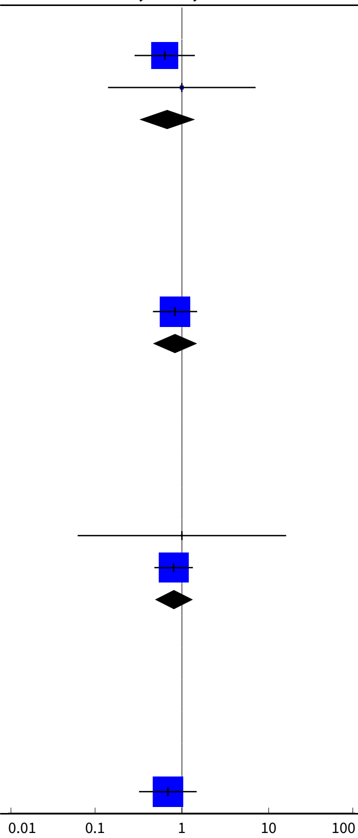
Analysis 6.7. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 7 Colorectal cancer: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.7.1 CEE 0.625 mg for 3 yrs					
PEPI 1995	0/175	1/174		100%	0.33[0.01,8.08]
Subtotal (95% CI)	**175**	**174**		**100%**	**0.33[0.01,8.08]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
6.7.2 CEE 0.625 mg for 6.8 yrs					
WHI 1998	61/5310	58/5429		100%	1.08[0.75,1.54]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**1.08[0.75,1.54]**
Total events: 61 (Treatment), 58 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.4(P=0.69)					



Analysis 6.8. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 8 Colorectal cancer: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.8.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr					
WHI 1998	10/8506	15/8102		88.47%	0.64[0.29,1.41]
WISDOM 2007	2/2196	2/2189		11.53%	1[0.14,7.07]
Subtotal (95% CI)	**10702**	**10291**		**100%**	**0.68[0.32,1.42]**
Total events: 12 (Treatment), 17 (Control)					
Heterogeneity: Tau²=0; Chi²=0.17, df=1(P=0.68); I²=0%					
Test for overall effect: Z=1.04(P=0.3)					
6.8.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	21/8506	24/8102		100%	0.83[0.46,1.5]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.83[0.46,1.5]**
Total events: 21 (Treatment), 24 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.61(P=0.54)					
6.8.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs					
PEPI 1995	1/174	1/174		2.96%	1[0.06,15.86]
WHI 1998	27/8506	32/8102		97.04%	0.8[0.48,1.34]
Subtotal (95% CI)	**8680**	**8276**		**100%**	**0.81[0.49,1.34]**
Total events: 28 (Treatment), 33 (Control)					
Heterogeneity: Tau²=0; Chi²=0.02, df=1(P=0.88); I²=0%					
Test for overall effect: Z=0.82(P=0.41)					
6.8.4 CEE 0.625 mg + 2.5 mg MPA for 4 yrs					
HERS 1998	11/1380	16/1383		100%	0.69[0.32,1.48]



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Subtotal (95% CI)	**1380**	**1383**		**100%**	**0.69[0.32,1.48]**
Total events: 11 (Treatment), 16 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.96(P=0.34)					
6.8.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	48/8506	74/8102		100%	0.62[0.43,0.89]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.62[0.43,0.89]**
Total events: 48 (Treatment), 74 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.61(P=0.01)					
6.8.6 CEE 0.625 mg + 2.5 mg MPA for 4-6.8 yrs UNBLINDED					
HERS 1998	21/1156	26/1165		100%	0.81[0.46,1.44]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**0.81[0.46,1.44]**
Total events: 21 (Treatment), 26 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.71(P=0.48)					
		Favours treatment	0.01 0.1 1 10 100	Favours control	

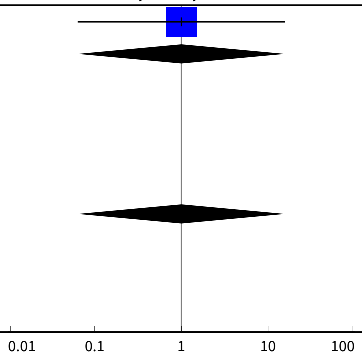
Analysis 6.9. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 9 Colorectal cancer: Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.9.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs					
PEPI 1995	0/174	1/174		100%	0.33[0.01,8.13]
Subtotal (95% CI)	**174**	**174**		**100%**	**0.33[0.01,8.13]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.67(P=0.5)					
6.9.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	0/178	1/174		100%	0.33[0.01,7.95]
Subtotal (95% CI)	**178**	**174**		**100%**	**0.33[0.01,7.95]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.69(P=0.49)					
		Favours treatment	0.001 0.1 1 10 1000	Favours control	

Analysis 6.10. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 10 Colorectal cancer: Combined sequential HT (high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.10.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle					
		Favours treatment	0.01 0.1 1 10 100	Favours control	

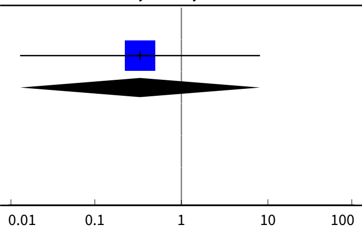
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Nachtigall 1979	1/84	1/84		100%	1[0.06,15.73]
Subtotal (95% CI)	**84**	**84**		**100%**	**1[0.06,15.73]**
Total events: 1 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Not applicable					
Total (95% CI)	**84**	**84**		**100%**	**1[0.06,15.73]**
Total events: 1 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Not applicable					



Favours treatment — Favours control

Analysis 6.11. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 11 Endometrial cancer: Oestrogen-only HT (moderate dose).

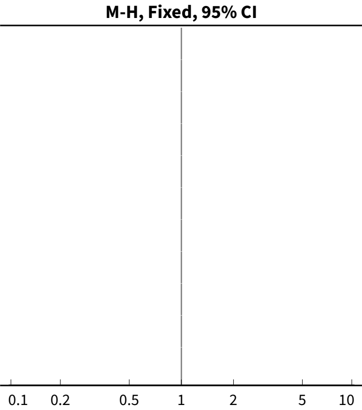
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.11.1 CEE 0.625 mg for 3 - 3.2 yrs					
PEPI 1995	0/119	1/119		100%	0.33[0.01,8.1]
Subtotal (95% CI)	**119**	**119**		**100%**	**0.33[0.01,8.1]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.67(P=0.5)					



Favours treatment — Favours control

Analysis 6.12. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 12 Endometrial cancer: Oestrogen only HT (mod/high dose).

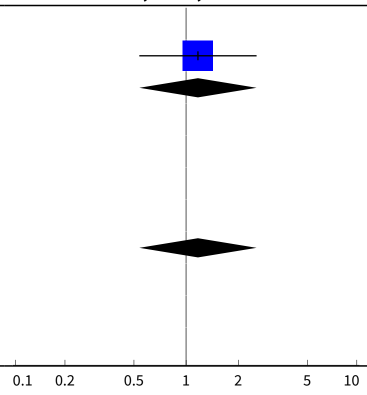
Study or subgroup	Treatment n/N	Control n/N	Odds Ratio M-H, Fixed, 95% CI	Weight	Odds Ratio M-H, Fixed, 95% CI
6.12.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	0/140	0/105			Not estimable
Subtotal (95% CI)	**140**	**105**			**Not estimable**
Total events: 0 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Not applicable					
Total (95% CI)	**140**	**105**			**Not estimable**
Total events: 0 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Not applicable					



Favours treatment — Favours control

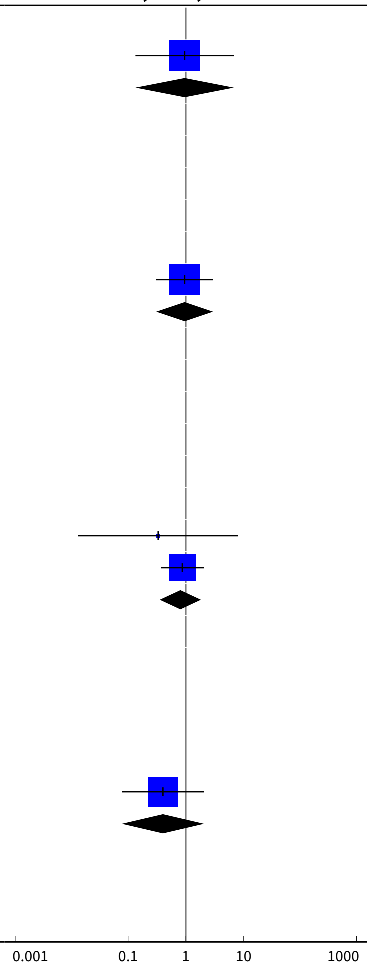
Analysis 6.13. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 13 Recurrent endometrial cancer:Oestrogen-only HT.

Study or subgroup	Treatment n/N	Control n/N	Odds Ratio M-H, Fixed, 95% CI	Weight	Odds Ratio M-H, Fixed, 95% CI
6.13.1 Oestrogen (type and dose not stated) for median 3 yrs					
Barakat 2006	14/618	12/618		100%	1.17[0.54,2.55]
Subtotal (95% CI)	**618**	**618**		**100%**	**1.17[0.54,2.55]**
Total events: 14 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.4(P=0.69)					
Total (95% CI)	**618**	**618**		**100%**	**1.17[0.54,2.55]**
Total events: 14 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.4(P=0.69)					



Analysis 6.14. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 14 Endometrial cancer: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.14.1 CEE 0.625 mg + MPA 2.5 mg for 1 yr					
WHI 1998	2/8506	2/8102		100%	0.95[0.13,6.76]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.95[0.13,6.76]**
Total events: 2 (Treatment), 2 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.05(P=0.96)					
6.14.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	6/8506	6/8102		100%	0.95[0.31,2.95]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.95[0.31,2.95]**
Total events: 6 (Treatment), 6 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.08(P=0.93)					
6.14.3 CEE 0.625 mg + MPA 2.5 mg for 3 - 3.2 yrs					
ERA 2000	0/104	0/105			Not estimable
PEPI 1995	0/120	1/119		11.79%	0.33[0.01,8.03]
WHI 1998	10/8506	11/8102		88.21%	0.87[0.37,2.04]
Subtotal (95% CI)	**8730**	**8326**		**100%**	**0.8[0.35,1.82]**
Total events: 10 (Treatment), 12 (Control)					
Heterogeneity: Tau²=0; Chi²=0.33, df=1(P=0.57); I²=0%					
Test for overall effect: Z=0.52(P=0.6)					
6.14.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs					
HERS 1998	2/1380	5/1383		100%	0.4[0.08,2.06]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**0.4[0.08,2.06]**
Total events: 2 (Treatment), 5 (Control)					
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%					
Test for overall effect: Z=1.09(P=0.27)					



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.14.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	27/8506	31/8102		100%	0.83[0.5,1.39]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.83[0.5,1.39]**
Total events: 27 (Treatment), 31 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.71(P=0.48)					
6.14.6 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					
HERS 1998	0/1156	3/1165		100%	0.14[0.01,2.78]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**0.14[0.01,2.78]**
Total events: 0 (Treatment), 3 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.28(P=0.2)					

Favours treatment 0.001 0.1 1 10 1000 Favours control

Analysis 6.15. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 15 Endometrial cancer: Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.15.1 Combined sequential: 17B estradiol 1mg + dydrogesterone 5 mg days 14-28 for 2 yrs					
Ferenczy 2002	1/100	0/63		100%	1.9[0.08,45.95]
Subtotal (95% CI)	**100**	**63**		**100%**	**1.9[0.08,45.95]**
Total events: 1 (Treatment), 0 (Control)					
Heterogeneity: Tau^2=0; Chi^2=0, df=0(P<0.0001); I^2=100%					
Test for overall effect: Z=0.4(P=0.69)					
6.15.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	0/120	1/119		100%	0.33[0.01,8.03]
Subtotal (95% CI)	**120**	**119**		**100%**	**0.33[0.01,8.03]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					

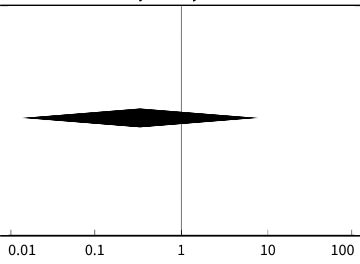
Favours treatment 0.001 0.1 1 10 1000 Favours control

Analysis 6.16. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 16 Endometrial cancer: Combined sequential HT (high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.16.1 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle					
Nachtigall 1979	0/84	1/84		100%	0.33[0.01,8.07]
Subtotal (95% CI)	**84**	**84**		**100%**	**0.33[0.01,8.07]**
Total events: 0 (Treatment), 1 (Control)					

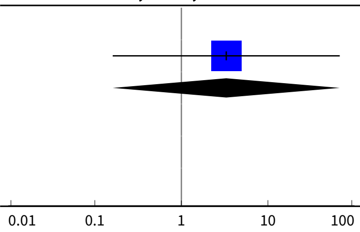
Favours treatment 0.01 0.1 1 10 100 Favours control

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					
Total (95% CI)	**84**	**84**		**100%**	**0.33[0.01,8.07]**
Total events: 0 (Treatment), 1 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.68(P=0.5)					



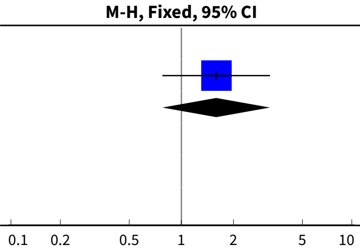
Analysis 6.17. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 17 Endometrial cancer: Combined sequential HT (mod/high dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.17.1 Oestradiol 2 mg + dydrogesterone 20 mg					
Ferenczy 2002	2/96	0/63		100%	3.3[0.16,67.59]
Subtotal (95% CI)	**96**	**63**		**100%**	**3.3[0.16,67.59]**
Total events: 2 (Treatment), 0 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.77(P=0.44)					



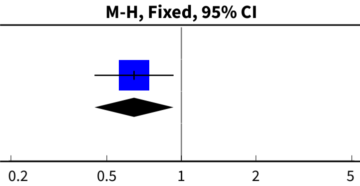
Analysis 6.18. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 18 Ovarian cancer: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.18.1 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	20/8506	12/8102		100%	1.59[0.78,3.25]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**1.59[0.78,3.25]**
Total events: 20 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.27(P=0.21)					



Analysis 6.19. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 19 Hip fractures: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.19.1 CEE 0.625 mg for 7.1 yrs					
WHI 1998	46/5310	73/5429		100%	0.64[0.45,0.93]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**0.64[0.45,0.93]**
Total events: 46 (Treatment), 73 (Control)					



Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Heterogeneity: Not applicable					
Test for overall effect: Z=2.35(P=0.02)					
Total (95% CI)	**5310**	**5429**		**100%**	**0.64[0.45,0.93]**
Total events: 46 (Treatment), 73 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.35(P=0.02)					
	Favours treatment		0.2 0.5 1 2 5	Favours control	

Analysis 6.20. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 20 Hip fractures: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.20.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 yr					
WHI 1998	6/8506	9/8102		75.42%	0.64[0.23,1.78]
WISDOM 2007	2/2196	3/2189		24.58%	0.66[0.11,3.97]
Subtotal (95% CI)	**10702**	**10291**		**100%**	**0.64[0.26,1.57]**
Total events: 8 (Treatment), 12 (Control)					
Heterogeneity: Tau²=0; Chi²=0, df=1(P=0.97); I²=0%					
Test for overall effect: Z=0.97(P=0.33)					
6.20.2 CEE 0.625 mg + MPA 2.5 mg for 2 yrs					
WHI 1998	14/8506	22/8102		100%	0.61[0.31,1.18]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.61[0.31,1.18]**
Total events: 14 (Treatment), 22 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.47(P=0.14)					
6.20.3 CEE 0.625 mg + MPA 2.5 mg for 3 yrs					
WHI 1998	25/8506	34/8102		100%	0.7[0.42,1.17]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.7[0.42,1.17]**
Total events: 25 (Treatment), 34 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.35(P=0.18)					
6.20.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs					
HERS 1998	15/1380	13/1383		100%	1.16[0.55,2.42]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**1.16[0.55,2.42]**
Total events: 15 (Treatment), 13 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.39(P=0.7)					
6.20.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	52/8506	73/8102		100%	0.68[0.48,0.97]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.68[0.48,0.97]**
Total events: 52 (Treatment), 73 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.15(P=0.03)					
	Favours treatment		0.1 0.2 0.5 1 2 5 10	Favours control	

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.20.6 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					
HERS 1998	25/1156	12/1165		100%	2.1[1.06,4.16]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**2.1[1.06,4.16]**
Total events: 25 (Treatment), 12 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.13(P=0.03)					

Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Analysis 6.21. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 21 Vertebral fractures: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.21.1 CEE 0.625 mg for 6.8 yrs					
WHI 1998	39/5310	64/5429		100%	0.62[0.42,0.93]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**0.62[0.42,0.93]**
Total events: 39 (Treatment), 64 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.34(P=0.02)					
Total (95% CI)	**5310**	**5429**		**100%**	**0.62[0.42,0.93]**
Total events: 39 (Treatment), 64 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.34(P=0.02)					

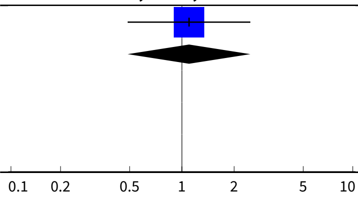
Favours treatment 0.2 0.5 1 2 5 Favours control

Analysis 6.22. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 22 Vertebral fractures: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.22.1 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 yrs					
WHI 1998	41/8506	60/8102		100%	0.65[0.44,0.97]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.65[0.44,0.97]**
Total events: 41 (Treatment), 60 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.13(P=0.03)					
6.22.2 CEE 0.625 mg + MPA 2.5 mg for 4 yrs					
HERS 1998	14/1380	19/1383		100%	0.74[0.37,1.47]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**0.74[0.37,1.47]**
Total events: 14 (Treatment), 19 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.87(P=0.39)					
6.22.3 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					

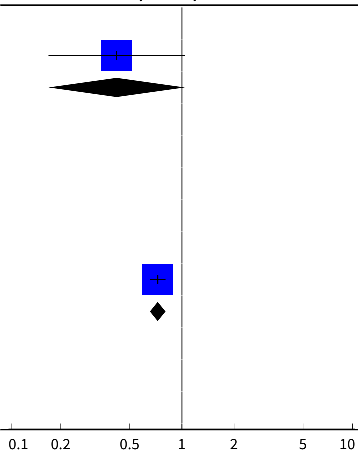
Favours treatment 0.1 0.2 0.5 1 2 5 10 Favours control

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
HERS 1998	12/1156	11/1165		100%	1.1[0.49,2.48]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**1.1[0.49,2.48]**
Total events: 12 (Treatment), 11 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.23(P=0.82)					



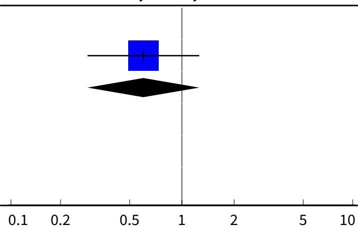
Analysis 6.23. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 23 All clinical fractures: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.23.1 CEE 0.625 daily for 3.2 yrs					
ERA 2000	6/100	15/105		100%	0.42[0.17,1.04]
Subtotal (95% CI)	**100**	**105**		**100%**	**0.42[0.17,1.04]**
Total events: 6 (Treatment), 15 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.88(P=0.06)					
6.23.2 CEE 0.625 mg for 7.1 yrs					
WHI 1998	540/5310	761/5429		100%	0.73[0.65,0.8]
Subtotal (95% CI)	**5310**	**5429**		**100%**	**0.73[0.65,0.8]**
Total events: 540 (Treatment), 761 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=6.07(P<0.0001)					



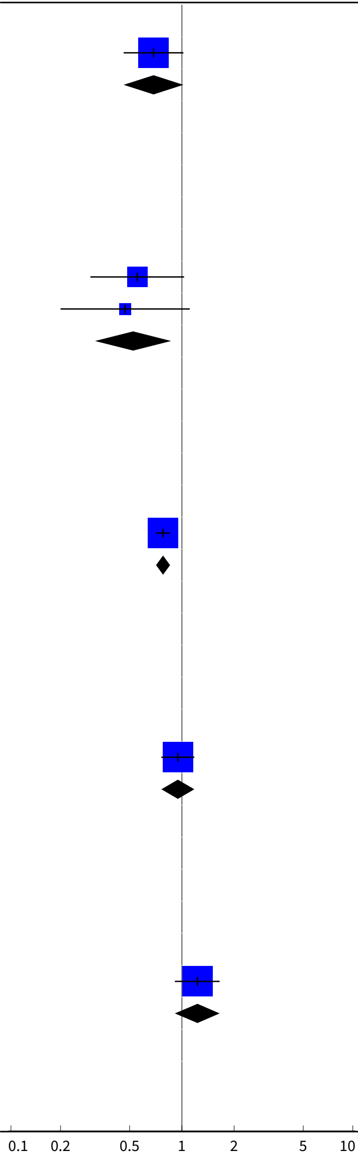
Analysis 6.24. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 24 All clinical fractures: Oestrogen-only HT (mod/high dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.24.1 Oestradiol valerate 2 mg for 2 yrs					
ESPRIT 2002	11/513	18/504		100%	0.6[0.29,1.26]
Subtotal (95% CI)	**513**	**504**		**100%**	**0.6[0.29,1.26]**
Total events: 11 (Treatment), 18 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.35(P=0.18)					



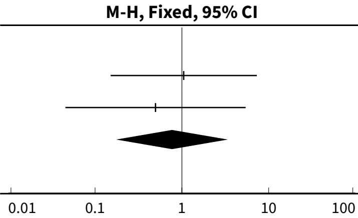
Analysis 6.25. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 25 All clinical fractures: Combined continuous HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.25.1 CEE 0.625 mg + MPA 2.5 mg for median one year					
WISDOM 2007	40/2196	58/2189		100%	0.69[0.46,1.02]
Subtotal (95% CI)	**2196**	**2189**		**100%**	**0.69[0.46,1.02]**
Total events: 40 (Treatment), 58 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.84(P=0.07)					
6.25.2 CEE 0.625 mg + MPA 2.5 mg for 3.2-3.4 yrs					
EPHT 2006	15/404	25/373		63.52%	0.55[0.3,1.03]
ERA 2000	7/104	15/105		36.48%	0.47[0.2,1.11]
Subtotal (95% CI)	**508**	**478**		**100%**	**0.52[0.32,0.87]**
Total events: 22 (Treatment), 40 (Control)					
Heterogeneity: Tau²=0; Chi²=0.09, df=1(P=0.76); I²=0%					
Test for overall effect: Z=2.52(P=0.01)					
6.25.3 CEE 0.625 mg + MPA 2.5 mg for median 5.6 yrs					
WHI 1998	733/8506	896/8102		100%	0.78[0.71,0.85]
Subtotal (95% CI)	**8506**	**8102**		**100%**	**0.78[0.71,0.85]**
Total events: 733 (Treatment), 896 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=5.27(P<0.0001)					
6.25.4 CEE 0.625 mg + MPA 2.5 mg for 4 yrs					
HERS 1998	140/1380	148/1383		100%	0.95[0.76,1.18]
Subtotal (95% CI)	**1380**	**1383**		**100%**	**0.95[0.76,1.18]**
Total events: 140 (Treatment), 148 (Control)					
Heterogeneity: Tau²=0; Chi²=0, df=0(P<0.0001); I²=100%					
Test for overall effect: Z=0.48(P=0.63)					
6.25.5 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 yrs UNBLINDED					
HERS 1998	90/1156	74/1165		100%	1.23[0.91,1.65]
Subtotal (95% CI)	**1156**	**1165**		**100%**	**1.23[0.91,1.65]**
Total events: 90 (Treatment), 74 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.34(P=0.18)					

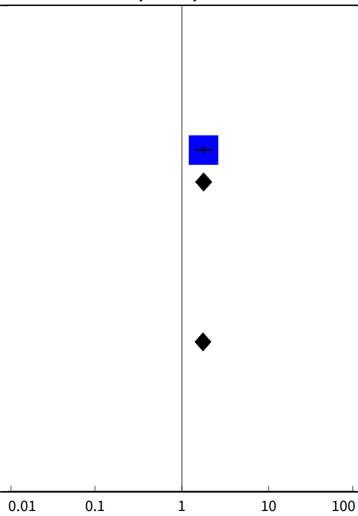


Analysis 6.26. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 26 Gallbladder disease requiring surgery: Oestrogen-only HT (moderate dose).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.26.1 CEE 0.625 mg for 3 - 3.2 yrs					
ERA 2000	2/100	2/105		1.69%	1.05[0.15,7.31]
PEPI 1995	1/175	2/174		1.73%	0.5[0.05,5.43]
Subtotal (95% CI)	**275**	**279**		**3.42%**	**0.77[0.17,3.39]**
Total events: 3 (Treatment), 4 (Control)					



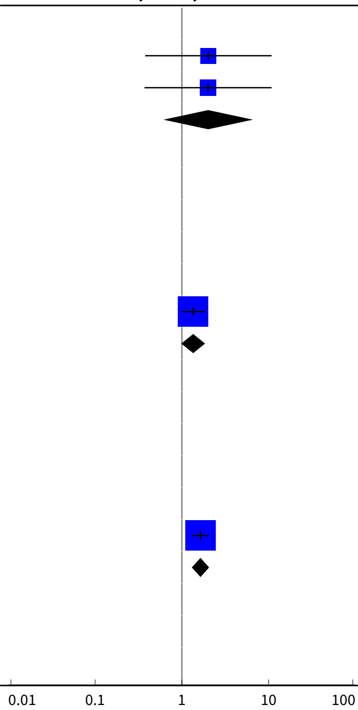
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Heterogeneity: Tau²=0; Chi²=0.23, df=1(P=0.63); I²=0%					
Test for overall effect: Z=0.35(P=0.73)					
6.26.2 CEE 0.625 mg for 7.1 yrs					
WHI 1998	197/4141	113/4235		96.58%	1.78[1.42,2.24]
Subtotal (95% CI)	**4141**	**4235**		**96.58%**	**1.78[1.42,2.24]**
Total events: 197 (Treatment), 113 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=4.99(P<0.0001)					
Total (95% CI)	**4416**	**4514**		**100%**	**1.75[1.4,2.19]**
Total events: 200 (Treatment), 117 (Control)					
Heterogeneity: Tau²=0; Chi²=1.36, df=2(P=0.51); I²=0%					
Test for overall effect: Z=4.88(P<0.0001)					
Test for subgroup differences: Not applicable					



Favours treatment | Favours control

Analysis 6.27. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 27 Gallbladder disease requiring surgery: Combined continuous HT (moderate dose oestrogen).

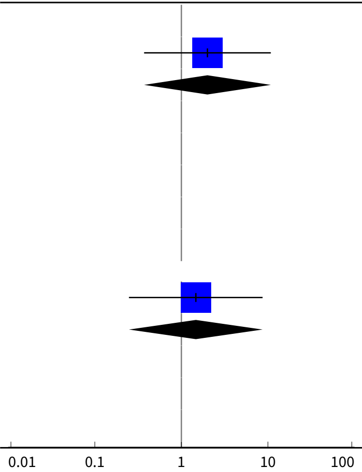
Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.27.1 CEE 0.625 mg + 2.5 mg MPA for 3 yrs					
ERA 2000	4/104	2/105		49.88%	2.02[0.38,10.79]
PEPI 1995	4/174	2/174		50.12%	2[0.37,10.78]
Subtotal (95% CI)	**278**	**279**		**100%**	**2.01[0.61,6.59]**
Total events: 8 (Treatment), 4 (Control)					
Heterogeneity: Tau²=0; Chi²=0, df=1(P=0.99); I²=0%					
Test for overall effect: Z=1.15(P=0.25)					
6.27.2 CEE 0.625 mg + 2.5 mg MPA for 4 yrs					
HERS 1998	85/1135	62/1118		100%	1.35[0.98,1.85]
Subtotal (95% CI)	**1135**	**1118**		**100%**	**1.35[0.98,1.85]**
Total events: 85 (Treatment), 62 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.86(P=0.06)					
6.27.3 CEE 0.625 mg + 2.5 mg MPA for 5.6 yrs					
WHI 1998	196/7308	113/6895		100%	1.64[1.3,2.06]
Subtotal (95% CI)	**7308**	**6895**		**100%**	**1.64[1.3,2.06]**
Total events: 196 (Treatment), 113 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=4.21(P<0.0001)					



Favours treatment | Favours control

Analysis 6.28. Comparison 6 All women (Selected outcomes: cancer, cholecystic disease, fractures), Outcome 28 Gallbladder disease requiring surgery: Combined sequential HT (moderate dose oestrogen).

Study or subgroup	Treatment n/N	Control n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.28.1 CEE 0.625 mg daily + MPA 10 mg days 1-12 for 3 yrs					
PEPI 1995	4/174	2/174		100%	2[0.37,10.78]
Subtotal (95% CI)	**174**	**174**		**100%**	**2[0.37,10.78]**
Total events: 4 (Treatment), 2 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.81(P=0.42)					
6.28.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1-12 for 3 yrs					
PEPI 1995	3/178	2/174		100%	1.47[0.25,8.67]
Subtotal (95% CI)	**178**	**174**		**100%**	**1.47[0.25,8.67]**
Total events: 3 (Treatment), 2 (Control)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.42(P=0.67)					



ADDITIONAL TABLES

Table 1. Adherence to treatment

Study	How defined	Assessment	HT group	Placebo group	Note
Barakat 2006	Discontinuation of therapy for more than a month (or use of HT in placebo group)	Not stated	41.1% compliant for whole follow-up period (median 3 yrs)	50.1% compliant for whole follow-up period (median 3 yrs)	
EPHT	> 80% of prescribed treatment taken	Number of collected and returned drugs and clinic reports	< 40% compliant at 3 yrs (estimated from graph)	< 30% compliant at 3 yrs (estimated from graph)	
ERA 2000	Percentage of study medication taken	Pill counts	Level of adherence at 3.2 years: Women on unopposed oestrogen, measured in 79% of participants only: 74% Women on combined HRT, measured in 82% of participants only: 84%	Level of adherence at 3.2 years: measured in 80% of participants only: 86% 5 women initiated treatment outside study	
EVTET 2000	Adherence not described				
EPAT 2001	Percentage of study medication consumed	Pill counts	Level of adherence 95% in the 87% of participants evaluated	Level of adherence 92% in the 92% of participants evaluated	
Haines 2003	Taking at least 80% of study medication overall, and not	Self-report Pill count	At least 80% compliance women analysed (91% of women in study overall)	At least 80% compliance women analysed (91% of women in study overall)	

Table 1. Adherence to treatment *(Continued)*

	missing more than 4 tablets in any one month				
ESPRIT 2002	"Regular tablet use"	Self-report to family doctor. Self-report to study nurse at 6 weeks and whenever in contact with trial staff	Number non-adherent: 51% at 12 months 57% at 24 months	Number non-adherent: 31% at 12 months 337% at 24 months	Triallists attribute higher non-compliance in HRT group to prevalence of vaginal bleeding (reported by 56% in HRT group, 7% in controls)
Mulnard 2000	Taking at least 80% of study medication	Plasma oestradiol level evaluation at each visit Pill counts at each visit	No information given in publication		
Ferenczy 2002	Adherence not described				
Notelovitz 2002	Adherence not described				
HERS 1998	Taking at least 80% of study medication	Pill counts	79% adherent at 1 year 70% adherent at 3 yrs 3% initiated treatment outside study About 50% continued to use open-label HRT during unblinded follow up (4.2 - 6.8 yrs)	91% adherent at 1 yr 81% non-adherent at 3 yrs Under 10% used HRT during unblinded follow-up (4.2 - 6.8 yrs)	Proportion of women who reported taking study medication at one year: HRT group: 82% Placebo group: 91%
PEPI 1995	Taking at least 80% of study medication	Study diary reviewed at clinic visits Pill counts	Number adherent at 36 months: Women without uterus: 80-89% at 36 months Women with uterus: 1. On unopposed CEE: 44% 2. On combined therapy: 80%	Number adherent at 36 months: Women without uterus: 67% Women with uterus: 76%	
Nachtigall 1979	Adherence not described				
WEST 1998	Percentage of study medication taken	Self-report to study nurse 3 monthly	At 2.8 yrs: Mean adherence including drop-outs: 70%	At 2.8 yrs: Mean adherence including dropouts: 74% over 2.8 yrs	

Table 1. Adherence to treatment *(Continued)*

		Computer chip in medication bottle records opening date and time Pill counts	Mean adherence excluding dropouts: 90% 35% discontinued medication by 2.8 yrs, of whom 1% initiated treatment outside study	Mean adherence excluding dropouts: 90% 24% discontinued medication 2% initiiated treatment outside study	
Obel 1993	Adherence not described				
WHI 1998 (unopposed oestrogen arm)	Taking at least 80% of study medication. Temporary discontinuation (e.g. during surgery) permitted	Weighing of returned medication bottles	At 6.8 years, about 53.8% of women were non-adherent In addition 5.7% of women had initiated hormone use through their own physician	At 6.8 years, about 53.8% of women were non-adherent In addition 9.1% of women had initiated hormone use through their own physician	
WAVE 2002	Percentage of study medication taken	Pill counts	At 2.8 yrs: Adherence 67% in the 78% of women analysed	At 2.8 yrs: Adherence 70% in the 81% of women analysed	
WHI 1998 (combined arm)	Taking at least 80% of study medication. Temporary discontinuation (e.g. during surgery) permitted	Weighing of returned medication bottles	42% non-adherent by 5.2 yrs Of these 6.2% initiated HRT outside study	10.7% crossed to active treatment by 5.2 yrs	Analyses censoring events 6 months after non-adherence increased effect sizes
WISDOM 2007	Supply of study medication	Time at risk minus temporary interruptions and time after withdrawal from treatment	73% of time	86% of time	Women had a 3 month run-in period on placebo. Only women who took 80% of tablets were randomised
Yaffe 2006	Supply of study medication	Patch counts: 75% use over 2 years counted as compliance	84%	84% of time	Women had a 1 week run-in period . Only compliant women were randomised

APPENDICES

Appendix 1. MEDLINE search strategy

1. randomised controlled trial.pt.

2. controlled clinical trial.pt.
3. Randomized controlled trials/
4. random allocation/
5. double-blind method/
6. single-blind method/
7. or/1-6
8. clinical trial.pt.
9. exp clinical trials/
10. (clin$ adj25 trial$).ti,ab,sh.
11. ((singl$ or doubl$ or tripl$ or trebl$) adj25 (blind$ or mask$)).ti,ab,sh.
12. placebos/
13. placebo$.ti,ab,sh.
14. random$.ti,ab,sh.
15. Research design/
16. or/8-15
17. animal/ not (human/ and animal/)
18. 7 or 16
19. 18 not 17
20. hormone therapy.mp. [mp=title, abstract, cas registry/ec number word, mesh subject heading]
21. HT.mp. [mp=title, abstract, cas registry/ec number word, mesh subject heading]
22. (estrogen or oestrogen or estradiol or oestradiol).mp. [mp=title, abstract, cas registry/ec number word, mesh subject heading]
23. (Alora or Climara or Estraderm or Estrofem or Femtran or Cliane or Estrapak or Kliogest or Kliovance or Trisequens or Progynova or Nuvelle or Ovestin or Premarin or Prempro or Progynova or Menoprem or Premia).mp. [mp=title, abstract, cas registry/ec number word, mesh subject heading]
24. (menopaus$ or postmenopaus$ or perimenopaus$).mp. [mp=title, abstract, cas registry/ec number word, mesh subject heading]
25. (22 or 23) and 24
26. 20 or 21 or 25
27. 19 and 26

Appendix 2. EMBASE

1. Controlled study/ or randomized controlled trial/
2. double blind procedure/
3. single blind procedure/
4. crossover procedure/
5. drug comparison/
6. placebo/
7. random$.ti,ab,hw,tn,mf.
8. latin square.ti,ab,hw,tn,mf.
9. crossover.ti,ab,hw,tn,mf.
10. cross-over.ti,ab,hw,tn,mf.
11. placebo$.ti,ab,hw,tn,mf.
12. ((doubl$ or singl$ or tripl$ or treb$) adj5 (blind$ or mask$)).ti,ab,hw,tn,mf.
13. (clinical adj5 trial$).ti,ab,hw,tn,mf.
14. research design/
15. or/1-14
16. nonhuman/
17. animal/ not (human/ and animal/)
18. or/16-17
19. 15 not 18
20. Hormone Substitution/
21. hormone replacement therapy.mp. [mp=title, original title, abstract, name of substance word, subject heading word]
22. HRT.mp. [mp=title, original title, abstract, name of substance word, subject heading word]
23. (estrogen or oestrogen or estradiol or oestradiol).mp. [mp=title, original title, abstract, name of substance word, subject heading word]
24. (Alora or Climara or Estraderm or Estrofem or Femtran or Cliane or Estrapak or Kliogest or Kliovance or Trisequens or Progynova or Nuvelle or Ovestin or Premarin or Prempro or Progynova or Menoprem or Premia).mp. [mp=title, original title, abstract, name of substance word, subject heading word]
25. (Menopaus$ or postmenopaus$ or perimenopaus$).mp. [mp=title, original title, abstract, name of substance word, subject heading word]
26. (23 or 24) and 25
27. (or/20 -22) or 26
21. 19 and 27

Appendix 3. BIO Abstracts

BIOLOGICAL ABSTRACTS
1. hormone replacement therapy.mp. [mp=abstract, biosystematic codes, original language book title (non-english), book title (english), chemicals & biochemicals, concept codes, diseases, geopolitical locations, gene name, major concepts, miscellaneous descriptors, methods & equipment, organisms, parts,
structures & systems of organisms, sequence data, super taxa, title, time, taxa notes]
2. hormone therapy.mp. [mp=abstract, biosystematic codes, original language book title (non-english), book title (english), chemicals & biochemicals, concept codes, diseases, geopolitical locations, gene name, major concepts, miscellaneous descriptors, methods & equipment, organisms, parts, structures & systems of organisms, sequence data, super taxa, title, time, taxa notes]
3. 1 or 2
4. (random$ adj2 trial$).mp. [mp=abstract, biosystematic codes, original language book title (non-english), book title (english), chemicals & biochemicals, concept codes, diseases, geopolitical locations, gene name, major concepts, miscellaneous descriptors, methods & equipment, organisms, parts, structures & systems of organisms, sequence data, super taxa, title, time, taxa notes]
5. 3 and 4
6. limit 5 to human

WHAT'S NEW

Date	Event	Description
31 May 2008	New search has been performed	The following new studies have been added: Barakat 2006, EPHT 2006, WISDOM 2007, Yaffe 2006. <u>WHI 1998 oestrogen-only arm, data added:</u> Venous thrombo-embolism at 2 & 7.1 years of follow up. Coronary heart disease, venous thromboembolism, stroke, breast cancer, fractures and quality of life at 7.1 years of follow up. <u>WHI 1998 combined arm, data added:</u> Subgroup analysis of breast cancer risk by prior hormone summarised in the text (Discussion section). Data on main outcomes after 3 years post-intervention (Discussion section). Results from WHI 1998 (WHISCA) on specific cognitive functions in older women added in the text. There have been no substantial changes to the overall findings of this review. Statistically significant risk of venous thromboembolism for WHI oestrogen-only arm now evident at follow up of "up to 2 years".
18 May 2008	Amended	Converted to new review format

HISTORY

Protocol first published: Issue 2, 2003
Review first published: Issue 3, 2005

Date	Event	Description
20 November 2003	New citation required and conclusions have changed	Substantive amendment

CONTRIBUTIONS OF AUTHORS

C Farquhar and A Lethaby developed the protocol and circulated it to members of the Cochrane HRT Study Group for comment. The following people contributed specifically to the protocol: Professor Shah Ebrahim, Dr Peter Tugwell, Teresa Moore and Maria Judd.

For the original version of the review, Jane Marjoribanks and Jane Suckling searched for relevant studies and selected studies for inclusion based on the protocol criteria. Jane Marjoribanks extracted and entered data, wrote all sections of the review, circulated it to other members of the Cochrane HRT Study Group for comment and edited the draft. Quirine Lamberts checked all the data extraction.

Jane Marjoribanks updated the review.

The following members of the Group commented on the draft.

- Breast Cancer Group: Sue Carrick Sue Lockwood (Editor)
- Dementia and Cognitive Improvement Group: Professor Leon Flicker (Editor), Professor Lon Schneider (Editor)
- Heart Group: Lee Hooper (Editor), Theresa Moore (Review Group Coordinator)
- Menstrual Disorders and Subfertility Group: Cindy Farquhar (Co-ordinating Editor), Anne Lethaby (Editor)
- Stroke Group: Professor Ale Agra (Editor), Steff Lewis (Statistical Editor)

DECLARATIONS OF INTEREST

None known

SOURCES OF SUPPORT

Internal sources

- University of Auckland, New Zealand.

External sources

- No sources of support supplied

INDEX TERMS

Medical Subject Headings (MeSH)

*Perimenopause; *Postmenopause; Cardiovascular Diseases [chemically induced] [mortality]; Cause of Death; Estrogen Replacement Therapy [*adverse effects] [methods]; Estrogens [*adverse effects] [therapeutic use]; Hot Flashes [drug therapy]; Neoplasms [chemically induced] [mortality]; Progesterone [*adverse effects] [therapeutic use]; Randomized Controlled Trials as Topic; Venous Thromboembolism [chemically induced]

MeSH check words

Adult; Aged; Aged, 80 and over; Female; Humans; Middle Aged